Risk aversion

More subtle is the property of *risk aversion*. First we need a piece of notation. For a lottery p, let Ep represent the expected value of p, or $Ep = \sum_x xp(x)$.[b]

Proposition 3.3. *Suppose that for all lotteries p, $\delta_{Ep} \succeq p$. This is true if and only if the utility function u is concave.*

A consumer who prefers to get the expected value of a gamble for sure instead of taking the risky gamble, (and whose utility function, in consequence, is concave) is said to be *risk averse*. We could also define a *risk-seeking* consumer as one for whom $p \succeq \delta_{Ep}$ for all p; this sort of behavior goes with a convex utility function u. And a consumer is *risk neutral* if $p \sim \delta_{Ep}$, which goes with a linear utility function. In economic theory, risk aversion, which includes risk neutrality as a special case, is typically assumed.

The proof of proposition 3.3 is left as an exercise. We draw a picture that indicates what is going on. Consider figure 3.4. We have graphed there a concave utility function u, and we've joined by a line segment two points $(x, u(x))$ and $(x', u(x'))$. Take any $\alpha \in (0,1)$, say $\alpha = 2/3$, and consider the lottery $p = \alpha\delta_x + (1-\alpha)\delta_{x'}$; that is, an α chance at x and a $1-\alpha$ chance at x'. The expected value of p is precisely $x'' = \alpha x + (1-\alpha)x'$.[1] Which does our consumer prefer, $\delta_{x''}$ or the lottery p? We can answer this by comparing the expected utilities of the two. For $\delta_{x''}$, we have utility $u(\alpha x + (1-\alpha)x')$, while the expected utility of p is $\alpha u(x) + (1-\alpha)u(x')$. By concavity, the former is at least as large as the latter, which is what we want. Of course, the property of risk aversion is meant to hold for all lotteries and not just those with supports of size two. But concavity of u is just what is needed, in general.

We have defined risk aversion in terms of comparisons of a lottery with its expected value. We can generalize this sort of thing as follows. It is possible to partially order lotteries according to their riskiness. That is, for *some* pairs of lotteries p and p', it makes sense to say that p is *riskier than* p'. For example, suppose that p' gives prizes 10 and 20 with probabilities 2/3 and 1/3, and p gives prizes 5, 15, and 30 with probabilities 1/3, 5/9, and 1/9, respectively. I claim that p is riskier than p'. Among the many ways to justify this claim, one is that p is p' "plus noise." Imagine conducting p'. If the outcome is 10, then either take 5 away (leaving 5) or give 5 more (leaving 15), each with probability 1/2. If the outcome is 20, either take away 5 with probability 2/3 (leaving 15) or give 10 more with probability 1/3. Think of this as a compound lottery, and you will see that first I do p' and then, conditional on the outcome, I conduct a lottery with expected value zero. But this compound lottery reduces to p. Since p is gotten from p' in this fashion, if a consumer was risk averse, she would (weakly) prefer p' to p. (Why?) In general, if p and p' are related in this fashion (p is p' plus zero-conditional-mean noise), then any risk averse decision maker will (weakly) prefer p' to p. This will be true also if p is p_1 plus zero-conditional-mean noise, p_1 is p_2 plus such noise, …, and p' is p_n plus such noise.

As something of a converse, we can define "riskier than" by p is riskier than p' if p and p' have the same expected value and *every* risk averse consumer (weakly) prefers p' to p. Then an interesting question is, What is the relationship between this definition of "riskier than" and the characterizations that involve zero-conditional-mean noise? The general notion of more or less risky gambles is an important one in the literature of the economics of uncertainty, although we will not develop it here; see Rothschild and Stiglitz (1970, 1971, 1973), Diamond and Stiglitz (1974), or Machina (forthcoming).

[b] We later will write $E\theta$ for the expected value of a random variable θ. No confusion should result.

[1] Careful! The lottery $p = \alpha\delta_x + (1-\alpha)\delta_{x'}$ denotes a probability distribution with two possible outcomes, whereas the number $x'' = \alpha x + (1-\alpha)x'$ is the convex combination of two numbers. And, appearing momentarily, $\delta_{x''}$ is a probability distribution with one possible outcome, namely x''.

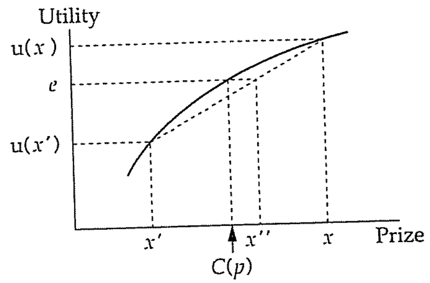

Figure 3.4. Concavity, risk aversion and certainty equivalents.
For a lottery p that gives prize x with probability 2/3 and x' with probability 1/3, we mark the expected value of the lottery as x'', the expected utility of the lottery as e, and the certainty equivalent of the lottery as $C(p)$.

Certainty equivalents

Because the utility function u in figure 3.4 is continuous, we know (from the intermediate value theorem of calculus) that for every $\alpha \in [0,1]$ there is some value x^* with $u(x^*) = \alpha u(x) + (1-\alpha)u(x')$. For any such x^*, we know from the expected utility representation that $\delta_{x^*} \sim p$. Such an x^* is called a *certainty equivalent* of p. In general,

Definition. *A **certainty equivalent** for a lottery p is any prize x such that $\delta_x \sim p$.*

Proposition 3.3. *If X is an interval of R and u is continuous, then every lottery p has at least one certainty equivalent. If u is strictly increasing, every p has at most one certainty equivalent.*

(The proof is left as an exercise.) We henceforth assume that the utility function under consideration is continuous, strictly increasing, and concave, the latter two reflecting increasing and risk averse preferences. Hence every p has a unique certainty equivalent, which we denote $C(p)$. Note that risk aversion, in this setting, can be characterized by $C(p) \leq Ep$.

What right have we to assume that u is continuous? It turns out that concavity of u implies that u is continuous on the interior of the interval X, although perhaps not at any endpoints of the interval. (Can you show this? Does the assumption that u is strictly increasing help about the endpoints in any way?) But we will want u to be continuous even at the endpoints, and we might want u to be continuous even if u didn't represent risk averse preferences. One can prove that u must be continuous if and only if preferences are continuous in the weak topology, relative to the space of simple probability distributions. So if you know about the (relative) weak topology, you know what is necessary and sufficient for this often very useful assumption about u.

Absolute (and relative) risk aversion

Assume henceforth that any von Neumann–Morgenstern utility function u we consider is strictly increasing, concave, and has at least two continuous derivatives.[c] This implies, of course, that u is continuous, and so each lottery p has a unique certainty equivalent $C(p)$ that is less than or equal to Ep. We will call the difference $Ep - C(p)$ the *risk premium* of p, written $R(p)$.

Now consider a lottery p and a (dollar) amount z. Write $p \oplus z$ for the lottery that gives prize $x+z$ with probability $p(x)$. That is, $p \oplus z$ is just the

[c] What justification is there for the assumption that u is twice differentiable? We can justify continuity and concavity. This then implies that u will have left- and right-hand derivatives everywhere. The properties of nondecreasing or nonincreasing risk aversion are then sufficient to show that u is continuously differentiable. (This makes a good homework exercise.) Concavity again implies that u' is decreasing, and we could, by complicating matters substantially, get away without assuming a continuous second derivative. But unless one is being very much more exact about this than we are, it isn't worth the difficulty entailed.

lottery constructed from p by increasing each prize of p by the amount z. By thinking of these prizes as the "after the gamble" wealth level of our consumer, as we increase z in $p \oplus z$, we increase the consumer's general wealth level. It seems somewhat natural to suppose that as someone becomes richer, she cares less and less about risks that she takes in given gambles. In symbols, this would say that as z increases, $R(p \oplus z)$ should not increase; the consumer's risk premium for a fixed gamble should not increase as the consumer becomes wealthier. We formalize this notion and two related to it as follows:

Definition. *For a fixed consumer with utility function u, if $R(p \oplus z)$ is nonincreasing in z, the consumer is said to be **nonincreasingly risk averse.**[d] If $R(p \oplus z)$ is constant in z, we say that the consumer has **constant risk aversion.** If $R(p \oplus z)$ is nondecreasing in z, then we say the consumer is **nondecreasingly risk averse.***

Our assertion about what seems "somewhat natural" is that consumers are nonincreasingly or perhaps constantly risk averse.

What does all this portend for the utility function u? We give a definition and then the result.

Definition. *Given a (twice continuously differentiable, concave, strictly increasing) utility function u, let $\lambda(x) = -u''(x)/u'(x)$, and call the function λ the **coefficient of absolute risk aversion** for the consumer.*

Since u is concave and strictly increasing, $\lambda(x) \geq 0$.

Proposition 3.4. *A consumer is nonincreasingly risk averse if and only if λ (defined from her von Neumann–Morgenstern utility function u) is a nonincreasing function of x. The consumer is nondecreasingly risk averse if and only if λ is a nondecreasing function of x. And the consumer has constant risk aversion if and only if λ is a constant function, in which case the utility function u is a positive affine translate of the utility function $-e^{-\lambda x}$. (If λ is the constant zero, then $u(x)$ is a positive affine translate of the function x; the consumer is risk neutral.)*

We do not attempt to prove this or the subsequent proposition in this subsection. References will be supplied at the end of the chapter.

[d] Sloppy terminology is sometimes used, and this property is called decreasing risk aversion.

We can use this general concept of the coefficient of risk aversion to compare the risk averseness of two different consumers. Imagine we have two consumers who conform to the model we are investigating. Let u be the first consumer's von Neumann–Morgenstern utility function, with $\lambda(x) = -u''(x)/u(x)$, and let v be the second consumer's von Neumann–Morgenstern utility function, with $\eta(x) = -v''(x)/v'(x)$. We wish to capture formally the idea that the first consumer is *at least as risk averse* as the second. A natural definition[e] is:

Definition. *The first consumer is at* ***least as risk averse*** *as the second if, for every lottery p and sure thing x such that the first consumer weakly prefers the lottery p to x for sure, the second consumer prefers the lottery as well.*

Put another way, any time the at-least-as-risk-averse consumer is willing to take the risk, so is the at-most-as-risk-averse consumer.

Proposition 3.5. *The first consumer is at least as risk averse as the second if and only if $\lambda(x) \geq \eta(x)$ for every x, which is equivalent to the statement that $v(\cdot) \equiv f(u(\cdot))$ for some strictly increasing, concave function f.*

Another variation sometimes played on this theme concerns how a single consumer responds to proportional gambles with her wealth. We suppose, for purposes of this discussion, that $X = (0, \infty)$, and the consumer's von Neumann–Morgenstern utility function u is concave, strictly increasing, and twice continuously differentiable. We imagine that the consumer has at her disposal an amount of wealth x, all of which she stakes in a gamble that pays a random *gross return*. Such a random gross return is specified by a simple probability distribution p with domain X, where our consumer's wealth after the gamble is θx (for $\theta \in \text{supp}(p)$) with probability $p(\theta)$. We can define a *certainty equivalent rate of return*, $CRR(p;x)$, which is that number $\hat{\theta}$ such that our consumer is indifferent between staking her wealth according to the gross return distribution p or taking $\hat{\theta}x$ for sure. And we can ask how $CRR(p;x)$ changes with changes in the consumer's initial level of wealth x. It is generally considered somewhat natural to suppose that $CRR(p;x)$ is nonincreasing in x — the richer our consumer is, the more conservative she becomes in staking all of her wealth. If we define $\mu(x) = -xu''(x)/u'(x)$, we can get results such as $CRR(p;x)$ is nonincreasing in x if and only if $\mu(x)$ is nonincreasing in x. The function μ is called the *coefficient of relative risk aversion*, as distinguished from $\lambda \equiv -u''/u'$, the coefficient of *absolute* risk aversion. For more on this, consult one of the references at the end of the chapter.

[e] But see Ross (1981) for a critique of this definition.

3.3. Applications to market demand

Many applications can be given for the model developed above, and many books are filled with those applications. What follows is a very haphazard selection of a few applications. The first is intended to tie up a loose end from the previous section; the second two are there to whet your appetite for more.

Induced preferences for income

In section 3.2 we looked at the special case where the prize space X is single dimensional. The obvious application is to the case of money prizes. But consumers don't eat money — money is useful for the commodities one can buy with it. So a fairly obvious question is: If a consumer has von Neumann–Morgenstern preferences for consumption bundles and if her preferences for money arise entirely from the purchasing power of that money, what conditions on her preferences for consumption bundles will translate into the sorts of properties we discussed in section 3.2?

To explore this question, we take the following setting. There are k commodities, and the consumption space X for our consumer is the positive orthant in R^k. We assume that the consumer, considering simple lotteries over the consumption bundles x that she might consume, conforms to the three assumptions of section 3.1. Thus there is a function $u : X \to R$, which is her von Neumann–Morgenstern utility function for simple lotteries on X. Note well, u is a perfectly good ordinal representation for our consumer's preferences on sure-thing consumption bundles. That is, u could be used in all of the developments of chapter 2; in particular, corresponding to u is an indirect utility function $\nu(p, Y)$ which gives the amount of utility (on the scale of u) our consumer derives from income Y at prices p. But u is more than just any ordinal numerical representation for preferences on X. We can use u (and positive affine transformations of u) to compute expected utilities for simple lotteries on X to rank-order our consumer's preferences for those simple lotteries.

We also assume for the time being that prices are fixed and given by some price vector p. (We will explore uncertain prices later.)

The notation we used earlier in this chapter conflicts with the notation from chapter 2, and we will proceed conforming to the notation in chapter 2. We will use Y to denote a level of income, so that simple lotteries on income have prizes denoted by Y. The space of income levels will be taken to be $[0, \infty)$. A simple probability distribution or lottery on the space of possible income levels will be denoted by π.

Now we pose the basic issue: Suppose our consumer attempts to compare two simple probability distributions on her income, say π and π'. *Suppose that our consumer learns how much income she has to spend before she makes any purchases of consumption bundles. Then it is natural to assume that, if she has income Y to spend, she purchases $x(p, Y)$ (her Marshallian demand) which gives her utility $\nu(p, Y)$. Accordingly, the expected utility she achieves from π is*

$$\sum_{Y \in \text{supp}(\pi)} \nu(p, Y)\pi(Y),$$

so that $\pi \succ \pi'$ if and only if

$$\sum_{Y \in \text{supp}(\pi)} \nu(p, Y)\pi(Y) > \sum_{Y \in \text{supp}(\pi')} \nu(p, Y)\pi'(Y).$$

That is to say, $\nu(p, Y)$, viewed as a function of Y for the fixed prices p, is a von Neumann–Morgenstern utility function for our consumer when she examines her preferences for lotteries over levels of income.[f]

The italic in the previous paragraph is there because what is asserted is more subtle than may be apparent. In the first place, the assumption that all uncertainty about income resolves before any consumption decisions are made is crucial. See the remarks about temporal resolution of uncertainty in section 3.5. Secondly, here, for the first time in this book, is a dynamic structure to choice. The consumer is (presumably) choosing some lottery on income that she will spend for consumption purposes. That uncertainty resolves, and *then* she chooses how to spend that income. The phrase *it is natural* in the previous paragraph assumes that when our consumer chooses her income lottery, she thinks ahead to how she will spend the income she might have available, and her tastes over income levels today are completely consonant with her later tastes for consumption bundles. All of chapter 4 is given over to exploring just how natural this really is. You should grant that this is natural for the sake of argument now, but don't be too convinced by it at least until after finishing chapter 4.

So questions about properties of the consumer's utility function for income become questions about properties of the indirect utility function.

[f] We have used the Marshallian demand function $x(p, Y)$ for purposes of exposition only. It should be clear that we don't require the consumer's problem (CP) to have a unique solution.

We give a few basic results in the following proposition. We begin by summarizing the basic state of affairs.

Proposition 3.6. *Suppose the consumer has preferences over lotteries of consumption bundles that satisfy the three assumptions of section 3.1. Let u be the resulting von Neumann–Morgenstern utility function on consumption bundles and let $\nu(p, Y)$ be the corresponding indirect utility function. Then the consumer's preferences over lotteries pertaining to income satisfy the three von Neumann–Morgenstern assumptions, and $Y \to \nu(p, Y)$ is a von Neumann–Morgenstern utility function for income. Moreover:*

(a) Assuming the consumer is locally insatiable, $\nu(p, Y)$ is strictly increasing in Y. If u is continuous, $\nu(p, Y)$ is continuous in Y.

(b) If u is a concave function, then $\nu(p, Y)$ is a concave function in Y; that is, our consumer is risk averse.

Proof. To begin, note that the preamble asserts that the consumer's preferences on income lotteries satisfy the three von Neumann–Morgenstern assumptions. We know this *because* her preferences over income lotteries have an expected utility representation; note that proposition 3.1 states that the three assumptions are necessary and sufficient for expected utility. Part (a) follows from proposition 2.13(b,c). As for (b), take any two income levels Y and Y' and $a \in (0, 1)$. Let x solve (CP) for (p, Y), and let x' solve (CP) for (p, Y'). Then $ax + (1-a)x'$ is feasible at prices p with income $aY + (1-a)Y'$, and

$$\begin{aligned} a\nu(p, Y) + (1-a)\nu(p, Y') &= au(x) + (1-a)u(x') \\ &\leq u(ax + (1-a)x') \leq \nu(p, aY + (1-a)Y'), \end{aligned}$$

where the first inequality follows from the concavity of u, and the second follows from the definition of ν as the most utility that can be obtained at prices p with a given income level.

We had rather a lot to say in chapter 2 about concavity of utility functions on consumption bundles; essentially we said there is no particular fundamental reason to think it is a valid assumption. Now it would seem we need this assumption. But now we can motivate this assumption by considerations that were not available to us in chapter 2. Now we can ask our consumer: For any two bundles x and x', would you rather have the bundle $.5x + .5x'$ (where we are taking the convex combination of the bundles) or a lottery where you get x with probability $1/2$ and x' with probability $1/2$? If our consumer always (weakly) prefers the sure thing, and if she conforms to the assumptions of the von Neumann–Morgenstern theory, then her von Neumann–Morgenstern utility function, which is a perfectly good representation of her (ordinal) preferences on X, will be concave.[g]

We leave for the reader the interesting task of exploring what conditions on u give properties such as constant absolute risk aversion for lotteries in Y.

In the derivation above, we assume that prices are certain, given by a fixed p. What if there is uncertainty about prices? Specifically, imagine that our consumer enters into a lottery that determines her income Y, and at the same time prices are determined by a lottery ρ. All uncertainty resolves, and then our consumer chooses what to consume.

If we want to speak of the consumer's preferences over lotteries in her income in this setting, we must assume that the uncertainty affecting her income is statistically independent of the uncertainty affecting prices. A simple example will illustrate the point. Consider a consumer who is evaluating an income lottery that gives her \$10,000 with probability 1/2 and \$20,000 with probability 1/2. Suppose that prices will be either p or $2p$, each with probability 1/2. If our consumer's income level is perfectly positively correlated with the price level, she faces no real uncertainty; her real purchasing power is unchanged. If, on the other hand, her income level is perfectly negatively correlated with the price level, she faces rather a lot of uncertainty in her purchasing power. It will be a rare consumer indeed who is indifferent between these two situations, even though in terms of the lotteries on (nominal) income, the two situations are identical. Either we must consider how the consumer feels about lotteries jointly on prices and income (in which case all the developments of section 3.2, which depend on one-dimensional prizes, go by the board), or we have to make some assumption that prevents this sort of problem. Independence is one such assumption.

Assuming the price and income level are independent and letting ρ be the probability distribution on prices, the induced expected utility our consumer obtains from the probability distribution π on Y is given by

$$\sum_{Y \in \mathrm{supp}(\pi)} \sum_{p \in \mathrm{supp}(\rho)} \nu(p, Y)\rho(p)\pi(Y).$$

Thus the consumer's von Neumann–Morgenstern utility function for income, which we will now write $v(Y)$, is

$$v(Y) = \sum_{p \in \mathrm{supp}(\rho)} \nu(p, Y)\rho(p).$$

[g] Assume that her preferences are continuous in the weak topology, so checking for 50-50 convex combinations is sufficient to obtain concavity. Alternatively, ask the question for general $a \in (0, 1)$.

You can quickly check that in this setting proposition 3.6 holds without any change except for this redefinition of the utility function on income.

We can ask here a different question. How does the consumer respond to uncertainty in the price level? Imagine that we fix the consumer's income Y and we let the consumer select the economy in which to buy, where different economies have different distributions over prices. To put it simply, imagine we ask this consumer: Given two price vectors p and p', would you rather be in an economy where prices are certain to be given by $.5p + .5p'$ or in an economy where prices are either p or p', each with probability 1/2? There is no clear answer to this question; see problem 3.

Demand for insurance

One of the most important markets in which uncertainty plays a role is the insurance market. We will return to this market in part IV of the book, when we examine some advanced issues pertaining to insurance. But for now, let us establish some simple basic properties about the demand for insurance.

Imagine a consumer whose income level is subject to some uncertainty. Specifically, her income will be Y with probability π and Y' with probability $1 - \pi$, where $Y > Y'$. Think of the difference $\Delta = Y - Y'$ as some loss the consumer might sustain, either because of an accident, or ill health, or theft, or some other misfortune. An insurance company is willing to insure against this loss; if the consumer will pay a premium of δ, the insurance company is prepared to pay Δ back to the consumer if she sustains this loss. The consumer may buy partial coverage; if she pays $a\delta$, she gets back $a\Delta$ if she sustains the loss. We do not restrict a to $[0, 1]$. There are good reasons why such a restriction or one even more severe might be in place, but they have to do with matters we take up later in the book.

We assume that this consumer satisfies the three assumptions of the von Neumann–Morgenstern theory concerning lotteries in her final level of income, net of any payments to/from the insurance company, and her utility function v is strictly increasing, concave, and differentiable. Then the consumer's problem concerning how much insurance to buy can be written

$$\max_a \pi v(Y - a\delta) + (1 - \pi)v(Y' + a\Delta - a\delta).$$

Rewrite $Y' + a\Delta$ as $Y - (1 - a)\Delta$, and the first-order condition on this problem becomes

$$\pi\delta v'(Y - a\delta) = (1 - \pi)(\Delta - \delta)v'(Y - (1 - a)\Delta - a\delta).$$

Note that since u is concave and a is unconstrained, the first-order condition is necessary and sufficient for a solution.

The insurance contract is said to be *actuarially fair* if the expected payout $(1-\pi)\Delta$ equals the premium δ. This equality can be rewritten as $\pi\delta = (1-\pi)(\Delta - \delta)$; so if the contract is actuarially fair, the first-order condition becomes

$$v'(Y - a\delta) = v'(Y - (1-a)\Delta - a\delta),$$

and we see that this equation holds when $a = 1$.

The contract is *actuarially unfair* if the expected payout is less than the premium. Let $\beta = \pi\delta/((1-\pi)(\Delta-\delta))$, so that for an actuarially unfair contract, $\beta > 1$. The first-order conditions can be written

$$\beta v'(Y - a\delta) = v'(Y - (1-a)\Delta - a\delta),$$

so at any solution to the first-order equation, we will need

$$v'(Y - a\delta) < v'(Y - (1-a)\Delta - a\delta).$$

Since v is assumed to be concave, its derivative is decreasing, and so we will find a solution to the first-order equation at a point where

$$Y - a\delta > Y - (1-a)\Delta - a\delta,$$

which is at some $a < 1$.

We conclude that *if the insurance contract is actuarially fair, the consumer will purchase full insurance. If the insurance contract is actuarially unfair, the consumer will purchase only partial insurance.* (If the contract were actuarially *over*fair, the consumer would overinsure.) These are somewhat rough statements because we need to worry about whether the first-order condition and the problem have a solution at all.[2] And we must worry whether this problem has multiple solutions.[3] But except for these worries, these are precise statements, and they can be generalized substantially. The basic idea is that insofar as insurance insures dollar for dollar against a loss, consumers will wish to smooth their income. If the insurance is actuarially

[2] What would happen if the consumer was risk neutral and the contract was actuarially unfair? What would happen if we added in the constraint that $a \geq 0$?

[3] Suppose the consumer is risk neutral and the contract is actuarially fair. What is the set of all solutions?

unfair, there is a cost to do this, and consumers do not completely smooth their income. Instead, in this case they choose to bear *some* of the risk. (But see problems 4 and 5.)

Demand for financial assets

Imagine a consumer who will do all her consuming one year hence. This consumer has W dollars to invest, and the income out of which she will consume next year will consist of the proceeds from her investments. Letting Y denote the proceeds from her investments, we assume that her preferences over lotteries on Y satisfy the von Neumann–Morgenstern assumptions, with v her von Neumann–Morgenstern utility function on Y, which is strictly increasing, concave, and differentiable.

Assume that our consumer can invest her money in one of two assets. The first of these is a riskless asset — for every dollar invested, our consumer gets back $r > 1$ dollars next year. The second asset is risky. Its gross return, denoted by θ, has a simple probability distribution π, so that $\pi(\theta)$ represents the probability that the risky asset has a gross return of θ. By gross return we mean a dollar invested in the risky asset returns θ dollars next year.

The consumer's problem, then, is to decide how much money to invest in each of the two assets. Since every dollar invested in the second asset is precisely one dollar less invested in the first, we can write her decision problem as a problem in one variable α, the amount of money she invests in the second, risky asset. Now if θ is the gross return on the risky asset, investing α in this asset means she will have

$$Y = \theta\alpha + r(W - \alpha) = \alpha(\theta - r) + rW$$

dollars to spend on consumption next period. So her problem is to

$$\max_{\alpha} \sum_{\theta \in \text{supp}(\pi)} v(\alpha(\theta - r) + rW)\pi(\theta).$$

Should we constrain α? That depends on the ability of the consumer to *short-sell* either or both of the assets. If she can borrow as much money as she pleases at the riskless rate of r to invest in the risky asset, then we would not constrain α above. (If, as is more realistic, she can borrow money but at a rate of interest exceeding what she may lend at, the problem becomes more difficult but not intractable.) If she can short-sell the risky asset, which means she is given the purchase price of the asset today (to invest in the riskless asset) but must make good on the return of the asset in a year, then we would not constrain α below. We will proceed assuming that she cannot short-sell the asset but that she can borrow money at the riskless rate; you may wish to try other variations on your own.

Hence we add to this problem the constraint that $\alpha \geq 0$. Since v is concave, we find the solution to the problem by looking for a solution to the first-order condition

$$\sum_{\theta \in \mathrm{supp}(\pi)} (\theta - r)v'(\alpha(\theta - r) + rW)\pi(\theta) \leq 0,$$

where the inequality must be an equality if $\alpha > 0$.

What can we say about solutions to this problem? A few obvious remarks are

(a) If $\theta > r$ for all $\theta \in \mathrm{supp}(\pi)$, then there can be no solution. This is so mathematically because v' is always strictly positive. (That is, v is strictly increasing.) The economic intuition behind this is straightforward. If $\theta > r$ for all possible values of θ, then the consumer can make money for sure by borrowing money (selling the riskless asset) and buying the risky asset. Since we have not constrained borrowing, the consumer will wish to do this on an infinite scale, reaping infinite profits. Such a thing is called an *arbitrage opportunity*, and one presumes that arbitrage opportunities are rarely to be found. (Note that if $\theta \geq r$ with strict inequality for at least one $\theta \in \mathrm{supp}(\pi)$, we have exactly the same problem.)

(b) Suppose the consumer is risk neutral. That is, $v(Y) = aY + b$ for $a > 0$. The first-order condition is then $\sum_{\theta \in \mathrm{supp}(\pi)} (\theta - r)a\pi(\theta) \leq 0$. If we write $E\theta$ for the expected gross return on the risky asset, or $E\theta = \sum_{\theta \in \mathrm{supp}(\pi)} \theta\pi(\theta)$, the first-order condition becomes $E\theta \leq r$, with equality if $\alpha > 0$. Since α is entirely removed from the first-order condition, there are three possibilities: If $E\theta < r$, the consumer chooses $\alpha = 0$. If $E\theta = r$, any α is a solution. And if $E\theta > r$, there is no solution. The economic intuition is: If the consumer is risk neutral, she compares assets according to their expected gross return only. If the expected gross return on the risky asset equals that of the riskless asset (the middle case), she is happy to invest in any fashion. If the expected gross return on the risky asset exceeds that of the riskless asset, she wishes to borrow at the riskless rate in infinite amounts and invest in the risky asset. If the expected gross return on the risky asset is less than that of the riskless asset, she wants none of the risky asset and, in fact, she would wish to short-sell the risky asset in infinite amounts if she were allowed to do so.

(c) Now return to the case where the consumer is strictly risk averse. Assume that there is a solution to the first-order conditions. (See problem 6 for a case in which this can be guaranteed.) Suppose that $E\theta < r$. Then we assert that the only possible solution of the first-order conditions is $\alpha = 0$. That is, if the expected return of the risky asset is less than that of the riskless asset, a risk averse consumer will necessarily buy none of it.[h] Constructing a precise and detailed argument that this is so is left as an exercise. But the basic idea is easily sketched. Recall that if v is concave, v' will be decreasing. Suppose that $\alpha > 0$. In the first-order condition, we multiply $(\theta - r)$ by $v'(\cdots)\pi(\theta)$. The stuff inside the v' increases with increasing θ and so multiplying by v' (which decreases as the stuff inside increases) puts relatively more weight on $(\theta - r)$ for smaller values of θ. But if we look at the weighted sum of $\theta - r$ weighted by the probabilities $\pi(\theta)$ alone, we would get something strictly negative. Putting more weight on the smaller terms makes the weighted sum even smaller, so for any $\alpha > 0$, the sum is strictly negative. Only $\alpha = 0$ can be a solution.[i]

(d) Finally, suppose that $E\theta > r$. Then the only solutions of the first-order conditions have $\alpha > 0$. It is easy to see this mathematically; plug $\alpha = 0$ into the first-order condition, and you will find that the sum term is $v'(Wr)(E\theta - r)$. In terms of economic intuition, for the first little bit of the risky asset the consumer buys, she is approximately risk neutral. If $E\theta > r$, the risky asset is a good deal for a risk neutral investor. It is only when the scale of her investment in the risky asset grows to some significant size that her risk aversion begins to take over, and then she may find that, on the margin, it is no longer worth purchasing still more of the risky asset.

In fact, the intuition of this problem is entirely captured by the previous sentence. Think of beginning in a position where the investor has none of the risky asset. Then for small changes in her holdings, she is risk neutral, and the direction she wishes to move in depends on the sign of $E\theta - r$. If this sign is positive and she increases her holdings of the risky asset, the marginal impact on her expected utility of still more risky asset is given by the sum term in the first-order conditions. As noted above, this puts more and more relative weight on the smaller values of $\theta - r$ as

[h] Before getting too excited about this, see later discussion concerning more than one risky asset.

[i] We can shortcut this argument if we assume that v is strictly concave. Using the strict concavity, we can prove rather simply that the problem has a unique solution in α as long as the risky asset is risky. And verifying that the first-order conditions are satisfied at $\alpha = 0$ is easy.

α grows. Now if all the terms are positive, there is no end to this process (which is [a]). And if the shift in weight is never sufficient to make the marginal impact on her expected utility nonpositive, this process never stops (which includes [b]). Of course, if we start out with the sign being negative, we never get started at all (which is [c]). And if the sign is positive, we will have to move some distance away from $\alpha = 0$ before coming to the level where the marginal value of the risky asset is zero (which is [d]). Once you understand intuitively why the sum term is the marginal value of the risky asset and you understand how, when u is concave, the relative weights shift as they do as a function of α, the intuition of this situation should be crystal clear.

Conclusions (c) and (d) are no longer valid when there is more than one risky asset, unless the returns on risky assets are statistically independent of one another. We will illustrate the basic problem in a setting with two risky assets. As before, we imagine that there is a riskless asset with gross rate of return r. The two risky assets will be indexed by $i = 1, 2$, with θ^i representing the gross return on asset i. We do not assume that the two assets returns are statistically independent; instead we take as given the joint distribution of their gross returns. This is given by a probability distribution π which, for a finite number of two dimensional vectors (θ^1, θ^2), specifies that $\pi(\theta^1, \theta^2)$ is the joint probability that the first asset has gross return θ^1 and the second has gross return θ^2. (You will recall, I hope, that the two random returns are statistically independent if and only if there are one-dimensional probability distributions π^1 and π^2 such that $\pi(\theta^1, \theta^2) = \pi^1(\theta^1)\pi^2(\theta^2)$.) The expected values of the two assets' gross returns will be written $E\theta^i$ for $i = 1, 2$.

Our consumer's problem is to decide how much to invest in each of the two risky assets, with anything left over going into the riskless asset. We let α^i be the amount invested in asset i. As in the previous subsection, we work with the case in which the risky assets can't be sold short but the riskless asset can be (that is, the consumer can borrow at the riskless rate), so the consumer's problem is

$$\max_{\alpha^1, \alpha^2} \sum_{(\theta^1, \theta^2)} v\Big(\alpha^1(\theta^1 - r) + \alpha^2(\theta^2 - r) + Wr\Big)\pi(\theta^1, \theta^2)$$

subject to $\alpha^1 \geq 0, \alpha^2 \geq 0$. (I will be sloppy about the supports of distributions throughout this discussion, on the presumption that, if you've survived this far, you can stand the sloppiness.) The first-order condition for α^i is

$$\sum_{(\theta^1, \theta^2)} (\theta^i - r)v'\Big(\alpha^1(\theta^1 - r) + \alpha^2(\theta^2 - r) + Wr\Big)\pi(\theta^1, \theta^2) \leq 0,$$

with equality if $\alpha^i > 0$.

It shouldn't be hard to see that these first-order conditions can have no solution if either (a') $\theta^i > r$ with probability one for one of the two assets, or (b') the consumer is risk neutral and $E\theta^i > r$ for one of the two assets. Those conclusions carry over intact. In fact, we can sharpen (a') a bit: It must not be possible to find nonnegative β^1 and β^2 such that $\beta^1 + \beta^2 = 1$ and $\beta^1\theta^1 + \beta^2\theta^2 \geq r$ with probability one, with a strict inequality with positive probability. If we could find such β^1 and β^2, then arbitrage would be possible. (Prove this!)

However, in contrast to the case of one risky asset, we assert that it is possible that (c') $\alpha^1 > 0$ when $E\theta^1 < r$, and (d') $\alpha^1 = 0$ when $E\theta^1 > r$.[j] The "problem" is that we don't know, at a particular α^1 and α^2, how the term $v'(\cdots)$ is shifting weight on the $\theta^1 - r$ terms. For example, if both α^1 and α^2 are positive and θ^1 and θ^2 are negatively correlated, then at "large" values of θ^1 the stuff inside v' could tend to be small because of the $\alpha^2(\theta^2 - r)$ part, which would make v' *large* when θ^1 is large. This, then, could give (c'). Alternatively, if the two assets' gross returns are positively correlated, even at $\alpha^1 = 0$, we get from the $\alpha^2(\theta^2 - r)$ term some relative shift of weight towards lower values of θ^1, which would permit (d').

That, to be sure, is a fairly opaque way of demonstrating the possibilities suggested in (c') and (d'). In fact, this is no demonstration at all. A proper demonstration would consist of a pair of concrete (and, one hopes, simple) examples. This task is left to you; see problem 8 for some hints. Moreover, we asserted at the outset that, if assets 1 and 2 had gross return distributions that were statistically independent, everything would work as in the previous subsection. This is left to you as well.

The first-order conditions in our problem with two risky assets tell us that the marginal effect of putting money into risky asset i when one is holding a particular portfolio of assets depends upon how the return of asset i "covaries" with the marginal utility of consumption at the given portfolio. Very roughly put, the smaller the "covariance," the worse it is for putting additional funds into asset i, since this means weight is shifted to the smaller terms in the first-order conditions. The intuition behind this is not too difficult. If asset i pays off relatively well when times are good (when the marginal utility of consumption is low) and relatively poorly when times are bad, holding more of asset i compounds the risk the investor faces. But if asset i pays off well when times are bad and (relatively) poorly when times are good overall, then asset i serves as insurance against the other assets being held. And as we saw in the subsection on insurance, one may be willing to take up some insurance, even if it is actuarially unfair.

Moreover, since the marginal utility of consumption at a given portfolio is a decreasing function of the overall return on one's portfolio, this intuition suggests that the "marginal value" of an asset in a given portfolio has to do with the "covariance" between the returns of the asset and the returns of the portfolio; everything else held equal, the marginal value of an asset is smaller the larger is its "covariance" with the overall return on the portfolio.

[j] Assertion (c') shouldn't come as a complete surprise. Remember that there can be demand for insurance, even when the insurance is actuarially unfair. But the point is a bit more subtle here, since the risk against which one buys insurance is given exogenously, and here all risks that the consumer faces are chosen by her.

This intuition can be made quite precise (which is to say, we can remove the quote marks from the word covariance) if we assume that our consumer evaluates her possible portfolios in terms of the portfolios' means and variances. Holding variance fixed, a higher mean return is better, and holding mean fixed, a lower variance is better. *If* our consumer's preferences over portfolios could be reduced to something this simple, then we would have available a very nice theory. Looking at any finite set of risky assets (and, for simplicity, one riskless asset), all we would need to compute the mean and variance of a given portfolio of assets is the vector of expected returns and the variance-covariance matrix of those returns.

We will not develop the theory of mean-variance efficient portfolios here; this is one of the starting points of the modern theory of finance, and you may consult one of the many textbooks on the subject. (See the end of this chapter for one recommendation.) But I did wish to suggest this last step and to make one comment which good textbooks in finance will amplify.

Moving from von Neumann–Morgenstern preferences to mean-variance driven preferences is not a trivial step. A consumer who maximizes the expectation of some utility function v defined on the gross income from her investments does not, in general, have preferences that can be reduced to a comparison of the mean–variance pairs of various portfolios she might hold. *If* her utility function is quadratic, $v(Y) = a + bY - cY^2$ (for $b, c > 0$), then this works. (Why?) But quadratic utility functions are not so nice; they are concave, but for $Y \geq b/(2c)$ they are decreasing. Alternatively, we could restrict attention to assets whose distributions are such that risk averse expected utility maximizers always prefer higher expected returns holding variance constant and lower variance holding expected return constant. For example, if assets have a joint Normal (multivariate) distribution, then any portfolio of the assets has a Normal distribution. And for any risk averse expected utility maximizer (whose utility function is increasing), two Normal distributions with the same variance are ranked according to their means (higher is better), and two Normal distributions with the same mean are ranked according to their variances (lower is better). The reader who knows a bit of mathematics and who knows about multivariate Normals may wish to construct a proof.

3.4. States of nature and subjective probability

In the preceding development, the objects of choice were probability distributions over prizes. That is, the probabilities — numerical expressions of likelihood — were given exogenously. As discussed in section 3.1, this would be inadequate for modeling certain situations. In some contexts, the odds of various outcomes are not at all clear, and what a consumer chooses depends very critically on what she subjectively assesses as the odds of the outcomes. To take a very simple example, imagine you are offered your choice of two bets: In the first, you win \$100 if a Latin American team wins the next World Cup soccer tournament, and zero otherwise. In the second, you win \$100 if

a European team wins the next World Cup. If we tried to apply the von Neumann–Morgenstern model to this problem, things would be very trivial. The von Neumann–Morgenstern model would regard the two gambles as lotteries with objectively specified probabilities. And then our consumer would pick whichever has the better chance of getting for her the \$100 prize. Any (reasonable) consumer would bet the same way. But what makes this sort of bet interesting are the odds one consumer or another assesses. We don't expect all consumers to bet the same way on this (although see the final subsection of this section). And so we need a model of choice under uncertainty that develops within itself the *subjective* probabilities that are assessed by a consumer faced with this sort of uncertainty. (To distinguish between situations where the probabilities are objective and those where probabilities are subjective, one sometimes encounters the following terminology: The former situations are cases where there is *risk*, while the latter are cases of *uncertainty*.)

The framework

The classic formal treatment of choice where there is subjective uncertainty is that of Savage (1954). The framework employed by Savage runs as follows. We have a set of prizes X, just as before. And we have a set of *states of nature*, S. Each $s \in S$ is a description of the resolution of any (relevant) uncertainty. The states in S should be mutually exclusive and exhaustive. You might think in terms of a horse race (or, if you prefer, the next World Cup competition). Each s will describe the order of finish of all the horses in the race, and S will be the set of all such orders of finish.

Since the formality may be a bit forbidding, let us carry along a simple example. Imagine a two-horse race between horses named Secretariat and Kelso. There are three possible outcomes of the race: Kelso wins, Secretariat wins, or the race is a dead heat. We will label these three s_1, s_2 and s_3, respectively.

We next form from X and S the set of all *horse race lotteries*, H. A horse race lottery h will describe for each state s a prize x that is won if s is the outcome of the race. Formally, H is the set of all functions from S to X; in very formal notation, $H = X^S$. We write $h(s)$ for the prize won at the horse race lottery h if the state is s.

So, for example, imagine that for \$2 you can purchase a betting ticket that pays off \$3.85 if Secretariat wins the race and \$0 if Kelso wins. If the race is a dead heat, then you get your \$2 back. Net of the cost of the ticket, this gamble is represented by the function h, which is given

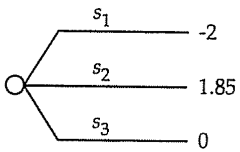

Figure 3.5. A horse race chance node.

by $h(s_1) = -2$, $h(s_2) = 1.85$, and $h(s_3) = 0$. We will depict horse race gambles by "chance nodes"; in figure 3.5, you see this particular horse race gamble.

If we were being less fanciful — thinking of S as the set of all states of the world, and X as a space of commodity bundles — then standard terminology would style H as the set of *state contingent commodity bundles* or *state contingent claims.*

In the classic theory due to Savage, H is the set of objects of choice. (He calls them *acts* instead of horse race lotteries.) The consumer's preferences are given by $\succ$ defined on H; certain properties of $\succ$ are assumed and a representation of $\succ$ is derived.

The representation and a weaker alternative

What sort of representation is sought? Savage seeks (and most economists employ) a representation of the following form. There exist a probability distribution π over S and a utility function $u : X \to R$ such that

$$h \succ h' \text{ if and only if } \sum_{s \in S} u(h(s))\pi(s) > \sum_{s \in S} u(h'(s))\pi(s). \qquad (\star)$$

(This representation has been written under the presumption that S is a finite set. In fact, Savage's treatment requires that S be infinite, and the summations just given become appropriately defined integrals.) This is just like von Neumann–Morgenstern expected utility, except that the probabilities of the various prizes are obtained by a two-step process: Probabilities are *subjectively* assessed for the various states of nature. The probability of getting a particular prize x if horse race lottery h is chosen is just the sum of the probabilities of those states $s \in S$ such that $h(s) = x$.

If we are able to obtain representation $(\star)$, we have done quite well. To understand how well, consider the following two cases in which $(\star)$ is inappropriate:

(1) Imagine you are marketing a particular product and trying to decide how much advertising to do. To keep things simple, imagine that the product will either sell 1,000 units or 10,000. If it sells 1,000, you will lose \$1,000, not including the cost of advertising. If it sells 10,000, you will make \$3,000. You can either advertise a lot or not at all. Not advertising costs you nothing, while advertising a lot costs you \$1,000.

Create the following model. The prize is your net profit. Possible values are $\$-2,000$, $\$-1,000$, $\$2,000$, and $\$3,000$, so these four dollar amounts constitute X. The states are your level of sales: You sell 1,000 units (state s_1) or 10,000 (state s_2). Hence there are two states in S. The two acts you can consider are: "advertise," which we denote by h, where $h(s_1) = \$-2,000$ and $h(s_2) = \$2,000$; and "don't advertise," which we denote by h', where $h'(s_1) = \$-1,000$ and $h'(s_2) = \$3,000$.

As long as your utility function u is increasing in increasing profits, if the representation $(\star)$ holds you never choose to advertise. Whatever probabilities you assess for s_1 and s_2, h' always gives a better outcome (by the \$1,000 cost of advertising) in both states. The problem is that your choice of an act presumably influences how many units you sell. *In Savage's representation* $(\star)$, *the probabilities of the states do not depend on the act chosen.* That is, we don't write $\pi(s;h)$ or anything like that. So in cases where acts influence state probabilities, we couldn't expect $(\star)$ to hold.

(In this example, a cure is very simple. The problem is that our state space is deficient. We really need three states: s_1, which is the state that the product sells 1,000 units whether you advertise or not; s_2, which is the state that the product sells 1,000 units if you don't advertise and 10,000 units if you do; and s_3, which is the state that the product sells 10,000 whether you advertise or not.[4] The question of whether to advertise becomes one of how likely is the state s_2, in which you increase your profits by \$3,000 if you do advertise, compared to the "lost" \$1,000 in advertising fees in states s_1 and s_3.)

(2) Imagine that you are thinking of undertaking one of two gambles. In the first, you will win \$100 if the yen:dollar exchange rate (i.e., the number of yen that can be bought with \$1) increases over the next month and otherwise you lose \$100. In the second, you win \$100 if the yen:dollar ratio doesn't increase and lose \$100 if it does. We create a model where the states of natures ($s \in S$) are the various possible yen:dollar exchange rates and where the prizes ($x \in X$) are $\$-100$ and \$100. It should be easy for you to see (given today's yen:dollar exchange rate) how to model the lotteries from which you are meant to choose.

[4] If there is a chance that advertising will anger potential customers, we would add a fourth state.

Do we expect the representation (⋆) to hold in this case? Probably not. *In the representation (⋆) the value of any prize $u(x)$ is independent of the state of nature in which the prize is won.* But winning \$100 when the yen:dollar exchange rate is low is worse than winning \$100 when the yen:dollar exchange rate is high. This should be very clear if you live in Japan. But, if you think about it, it is probably true if you live in some country other than Japan.[5] Put another way, suppose, for the sake of argument, that you think it as likely that the yen:dollar ratio will go up as that it will go down. Then according to the representation (⋆), you should be indifferent between the two gambles. But because \$100 is worth more when the yen:dollar ratio is high and less when the ratio is low, the first of these two gambles is (pretty clearly) the better of the two.

What can we do to salvage the representation (⋆) in cases such as this? One thing to try is to redefine prizes X in a way that makes the value of a given prize independent of the state in which it is won. Instead of making the prize a dollar amount, we might consider recording prizes in terms of the purchasing power the prize represents, using some standard market bundle of goods. But this can get very messy, very quickly. And so, as an alternative, we might consider giving up on (⋆) and trying for a weaker representation. We might ask for a probability distribution π on S and a "state-dependent utility function" $u : X \times S \to R$ such that

$$h \succ h' \text{ if and only if } \sum_{s\in S} u(h(s), s)\pi(s) > \sum_{s\in S} u(h'(s), s)\pi(s).$$

If this is what we are after, we can simplify one level further. Given π and u, define $v : X \times S \to R$ by $v(x, s) = u(x, s)\pi(s)$. Then the representation just given becomes

$$h \succ h' \text{ if and only if } \sum_{s\in S} v(h(s), s) > \sum_{s\in S} v(h'(s), s). \qquad (\star\star)$$

Since this form conveys what is important in this weaker type of representation, we will use it. Whereas representation (⋆) is styled as a *subjective expected utility* representation, the weaker alternative (⋆⋆) is styled as an *additive across states* representation.[k]

[5] Suppose you live in the United States. The "market bundle" of goods that \$100 will buy at 150 yen to the dollar is better than the bundle that \$100 will buy at 125 yen to the dollar, insofar as the lower exchange rate means that Japanese goods in the market bundle are more expensive.

[k] Does the probability distribution π convey any information if we allow u to be state dependent? See problem 10 for more on this.

Savage's sure thing principle

Now that we know what sort of representation we are after, we can see how it might be obtained. The classic derivation of the representation (⋆), as we said before, is due to Savage (1954). Savage begins with preferences $\succ$ defined on H and develops a number of properties of $\succ$ that lead to the representation (⋆). Some of the properties are those you've seen before; $\succ$ is asymmetric and negatively transitive.[l] Other properties exploit the particular structure we have here. Although we won't go through the Savage treatment here, it may help (to give a flavor of the subject) to give the "most important" of the other properties that he invokes, which is called the *independence axiom* or the *sure thing principle.* (This comes the closest to our substitution axiom in the theory of section 3.1.) It runs as follows:

Savage's sure thing principle. *Suppose we are comparing two horse race lotteries h and g. Suppose moreover that the set of states S contains a subset T on which h and g are identical; for every $s \in T$, $h(s) = g(s)$. Then how the consumer feels about h compared to g depends only on how h and g compare on states that are not in T. Formally, if $h \succ g$, and if h' and g' are two other horse race lotteries such that (a) h is identical with h' and g is identical with g' on $S \setminus T$, and (b) h' and g' are identical on T, then $h' \succ g'$ must follow.*

This is easier than it seems. The idea is that the ranking of h and g doesn't depend on the specific way that they agree on T — that they agree is enough. Figure 3.6 shows this idea in pictorial form.

You should have little difficulty seeing that Savage's sure thing principle is implied by the representation (⋆) and even by the representation (⋆⋆). If we write, in the spirit of (⋆⋆), $U(h) = \sum_{s\in S} v(h(s), s) = \sum_{s\in T} v(h(s), s) + \sum_{s\in S\setminus T} v(h(s), s)$, and write a similar expression for g, then as $h(s) = g(s)$ for $s \in T$, a comparison of $U(h)$ and $U(g)$ depends on how $\sum_{s\in S\setminus T} v(h(s), s)$ compares with $\sum_{s\in S\setminus T} v(g(s), s)$. This comparison is entirely independent of what h and g are on T.

The theory of Anscombe and Aumann

Savage's treatment goes on from the sure thing principle to other assumptions that are more complex, and then develops these assumptions mathematically in very complex fashion. His treatment is a bit hard for

[l] These would have to be entailed, since we obtain a numerical representation.

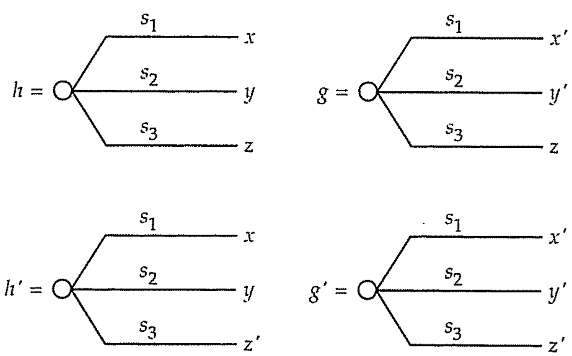

Figure 3.6. Savage's independence axiom in action.
Since h and g give the same prize in s_3, a comparison between them depends only on what they give on the other states. The prize they give on s_3 is irrelevant to the comparison. Hence, however h and g are ranked, h' and g' rank the same way.

a first pass on this subject, so we will continue with a different, easier formalization, taken from Anscombe and Aumann (1963).

Anscombe and Aumann avoid the complexities that Savage encounters by enriching the space over which preferences are defined. Like Savage, they take as given a prize space X and a state space S. We will assume that S is a finite set.

Their innovation comes at this point. Let P denote the set of all simple probability distributions on X, and redefine H so that it is the set of all functions from S to P. In comparison, Savage has H as the set of functions from S to X. A story that might be told about the Anscombe–Aumann construction runs as follows: Imagine that we have at our disposal a set of "objective" randomizing devices — coins we can flip, dice we can roll, roulette wheels we can spin — that allow us to construct any "objective" probability distribution that we wish. That is, given any $p \in P$, we can use the devices at our disposal to stage the gamble p in a way such that every consumer we might meet agrees to the objective probabilities in p. For example, if we wish to create a lottery that gives the prize $10 with probability 4/5 and $0 with probability 1/5, we might roll a symmetrical die, paying $10 if it comes up with one through four spots up, paying $0 if it comes up with five spots up, and rolling again (until we get an outcome between one and five) if it comes up with six spots up. It is assumed that every consumer we might encounter regards this as an objective probability lottery with prizes $10 and $0 and probabilities 4/5 and 1/5, respectively.

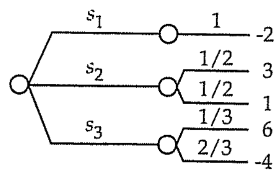

Figure 3.7. A horse race lottery with roulette wheel lotteries for prizes.

We'll call such objective gambles *roulette wheel gambles* or *lotteries*. The next step is to construct the set H of all functions from S to P. In words, an h from this H is a betting ticket that specifies, for each possible outcome of the horse race, a roulette wheel lottery that is won by the holder of the betting ticket. Embedded in this new, fancy H is the old (Savage) H, where the prize contingent on each state s is a degenerate roulette wheel lottery that gives some prize in X for sure. But this new H is bigger, because the set of possible prizes is larger; a prize in the horse race lottery is now a roulette lottery.

To give an example, imagine a (very complex) betting ticket that pays the following: In state s_1 (Kelso wins), you get $\$-2$ for sure. (That is, you lose $2 for sure.) In state s_2 (Secretariat wins), you get $3 with probability 1/2 and $1 with probability 1/2. In state s_3 (a dead heat), you get $6 with probability 1/3 and you lose $4 with probability 2/3. This sort of creature is depicted by a compound chance node, as in figure 3.7, where the compounding is now an important part of the mathematical formalism.

And (the punchline) Anscombe and Aumann (and we) will assume that the consumer has preferences given by $\succ$ defined on this fancy H.

Before talking about those preferences, we introduce a bit of notation. First, if $h \in H$ is a horse race lottery in the new, expanded sense, we write $h(\cdot|s)$ for that element of P that is the prize under h in state s, and we write $h(x|s)$ for the objective probability of winning $x \in X$ under h, contingent or conditional on the state s. Second, suppose h and g are two horse race lotteries (in the new, expanded sense), and α is a number between zero and one. Define a new horse race lottery, denoted $\alpha h+(1-\alpha)g$, as follows: For each state s, the new horse race lottery gives as prize the roulette wheel lottery $\alpha h(\cdot|s) + (1-\alpha)g(\cdot|s)$, where this last object is defined as in section 3.1. Figure 3.8 gives an example of this sort of operation.

Now we can start suggesting properties for the consumer's preferences $\succ$.

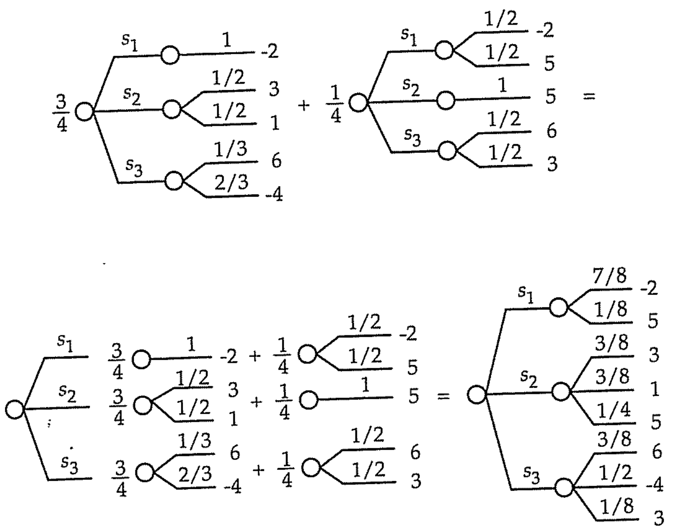

Figure 3.8. Taking mixtures of horse race lotteries.
When mixing horse race lotteries, you mix the prizes for each outcome of the horse race.

Assumption 4. *(a) $\succ$ is asymmetric.*

(b) $\succ$ is negatively transitive.

(c) $\succ$ satisfies the substitution axiom of section 3.1.

(d) $\succ$ satisfies the Archimedean axiom of section 3.1.

The first two parts of this assumption should come as no surprise, but the latter two properties may take some explanation. Look back at the substitution and Archimedean axioms in section 3.1. They don't depend on the fact that p, q, and r are probability distributions — only that convex combinations of the objects of choice can be taken.[m] And in the preceding paragraph you were told how to take convex combinations of horse race lotteries. So the latter two properties make mathematical sense.

Do they make sense as properties of consumer choice? That is a harder question to answer, of course. This is left to your own judgment (although read section 3.5 if you are too optimistic or too pessimistic) with the (obvious) remark that buying these properties in this setting takes somewhat more courage than buying them in the context of section 3.1.

[m] In more advanced texts these properties are stated for algebraic objects called *mixture spaces*, and the conclusions of these properties for general mixture spaces, which is something just short of the von Neumann–Morgenstern expected utility representation, is given by the so-called *Mixture Space Theorem*. See any of the advanced texts referenced in section 3.7.

Whether you find them sensible or not, they do produce a rather nice result.

Proposition 3.7. *Suppose that S is finite. Then the four properties in assumption 4 are necessary and sufficient for $\succ$ to have a numerical representation U of the following form. There exists for each state s a function u_s from X to the real line such that for $h \in H$,*

$$U(h) = \sum_{s \in S} \sum_{x} u_s(x) h(x|s).$$

Some hard staring is required to comprehend this result. Of course, when we say that U is a numerical representation of $\succ$, we mean that

$$h \succ g \quad \text{if and only if} \quad U(h) > U(g).$$

The result here establishes the existence of such a numerical representation *and* tells us something about the structure we can assume for the function U: For each state, there is a state-dependent utility function u_s. To evaluate h, first, for each state s, compute the expected utility of the roulette gamble $h(\cdot|s)$ using the utility function for state s. This is the term $\sum_x u_s(x)h(x|s)$. Then sum these expected utilities, summing over the states s.

Example: Suppose that your utility functions, depending on the outcome of the horse race, are: In state s_1, $u_{s_1}(x) = x/2$; in state s_2, $u_{s_2}(x) = -2e^{-.2x}$; in state s_3, $u_{s_3}(x) = -.01e^{-.1x}$. Then the overall utility of the gamble depicted in figure 3.7 is computed as follows. In state s_1, you have \$$-2$ for sure, so your state-dependent expected utility in state s_1 is $-2/2 = -1$. In state s_2, your state-dependent expected utility is $(1/2)(-2e^{-.2\times 3}) + (1/2)(-2e^{-.2\times 1}) = -1.3675$. And in state s_3, your state-dependent expected utility is $(1/3)(-.01e^{-.1\times 6}) + (2/3)(-.01e^{-.1\times -4}) = -.0118$. Hence the overall index (utility) of this gamble is $(-1) + (-1.3675) + (-.0118) = -2.3793$.

What is nice here is that (a) we get additive separability over outcomes of the horse race and (b) for each outcome of the horse race, we compute expected utility as in section 3.1. What is less nice is that we only get additive separability over horse race outcomes; the state-dependent utility functions needn't bear any particular relationship to one another. Put another way, this representation is very much like the representation (⋆⋆), where the state dependent utility function $v(h(\cdot|s), s)$ defined on $P \times S$ has the form, for each s, of von Neumann–Morgenstern expected utility. Now we could stop here (and go on to see what consequences this representation has for demand behavior — the consequences are substantial); or we could add properties for each state-dependent utility function (such as decreasing risk aversion, etc., if the prizes are money amounts). But instead we will add another property for $\succ$ that will tie together the various u_s.

This requires a bit of notation. For $p \in P$ (a roulette lottery), write p as an element of H (a horse race lottery) meaning the horse race lottery that gives, regardless of the outcome of the horse race, the prize p. With this, we can give the new property:

Assumption 5. *Fix any p and q from P. Fix any state s^*. Construct horse race lotteries h and g as follows: Pick (arbitrarily) any $r \in P$, and let $h(s) = r$ if $s \neq s^*$, $h(s^*) = p$, $g(s) = r$ if $s \neq s^*$, and $g(s^*) = q$. Then $p \succ q$ if and only if $h \succ g$.*

What does this mean? First, understand what h and g are. Both h and g give prize r if the horse race outcome is anything other than s^*; h has prize p if the outcome is s^*; and g has prize q if the outcome is s^*. Note that as long as the other four properties hold, the choice of r is (therefore) irrelevant to the preference relation between h and g; the difference in $U(h)$ and $U(g)$ is the difference between

$$\sum_x u_{s^*}(x)p(x) \quad \text{and} \quad \sum_x u_{s^*}(x)q(x).$$

With this, we can reinterpret the property given: It says that the sign of the difference between the two terms in the display just above is independent of the particular state s^*. If the difference runs one way for one s^*, it runs the same way for all. (This isn't quite what the property says, but you should be able to convince yourself that it is equivalent.) In other words, the preference that the consumer has between a pair of roulette lotteries is independent of the state in which the two are compared. See figure 3.9 for an illustration.

And the consequence is

Proposition 3.8. *For finite S, $\succ$ satisfies assumptions 4 and 5 if and only if we have a representation U of the following type. There is a function $u : X \to R$*

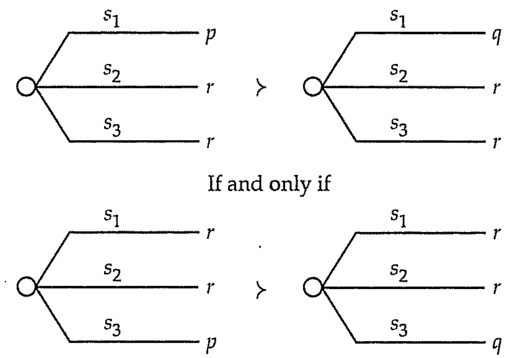

Figure 3.9. An equivalent formulation of the extra property.
The extra property is equivalent to the assumption that the consumer's preference over roulette wheel lotteries at any given outcome of the horse race doesn't depend on outcome. (We find preference over roulette wheel lotteries at a given outcome by holding constant the prizes won at all other outcomes.)

and a strictly positive probability distribution π on S such that

$$U(h) = \sum_{s \in S} \pi(s) \sum_x u(x)h(x|s).$$

In this representation, π is unique and u is unique up to positive affine transformations.

The interpretation is that our consumer assesses subjective probability $\pi(s)$ that the outcome of the horse race will be s, and he has a utility function for prizes u that is (otherwise) independent of the state. In words, a horse race gamble is as good as its *subjective expected utility*. Except for the objective lottery part of these gambles, we have Savage's representation (⋆).

For example, in our horse race example, a "typical" representation might be that $u(x) = -e^{-x}$, $\pi(s_1) = .6$, $\pi(s_2) = .399$, and $\pi(s_3) = .001$. We would interpret this as *the consumer has constant absolute risk aversion* (recall the previous section) *and assesses probability* .6 *that Kelso will win, probability* .399 *that Secretariat will win, and probability* .001 *that the race will end in a dead heat.*

If you like a challenge, you should be able to prove this proposition from earlier propositions without difficulty; remember the uniqueness part of the proposition in section 3.1. In this version we require that the subjective probability of every state s is strictly positive. This we did for convenience; it is easy to dispense with. Finally, all this requires a finite state space S. Infinite state spaces are significantly harder to handle.

The Anscombe–Aumann treatment, and especially the result of proposition 3.7, is more subtle than it may seem at first. Let me support that contention with one that is more concrete: It is *crucial* for this theory that the horses run before the roulette wheel is spun. That is, one could imagine beginning with $H = X^S$, as in Savage's treatment, and then letting $\succ$ be defined on all simple probability distributions on H. Posing assumption 4 on this space, where, more or less, we build compound lotteries with the objective uncertainty coming first, would lead to a result far weaker than the representation in proposition 3.7. Until you understand why this is, you miss an implicit assumption that is crucial to the Anscombe–Aumann representation. I will not attempt to explain this here but send you instead to Kreps (1988,105-8).

The Harsanyi doctrine

There is a final philosophical point to make about subjective expected utility and subjective probability. In economics, one usually takes the tastes and preferences of the consumer as exogenous data. This isn't to say, for example, that you can't present a consumer with information that will cause the consumer to realize a particular product is a good one he will enjoy; but, in the end, there is no arguing about tastes. Is the same true of subjective probability assessments? Are subjective probability assessments as subjective and personal as tastes?

To be very pedantic about this, return to the question of betting on the next World Cup. We will look at gambles whose outcomes depend only on whether the winner of the next Cup comes from Europe (including the Soviet Union) or from Latin America or from the rest of the world, so we take a three-element state space $S = \{s_1, s_2, s_3\}$ where s_1 is the state that a European team wins the next World Cup, s_2 is the state that a Latin American team wins, and s_3 is the state that some other team wins. We will take the prize space X to be $[0,100]$, with numerical amounts representing dollars won. Now imagine that we give some consumer her choice of the three gambles (from the Anscombe–Aumann version of H) depicted in figure 3.10. Suppose she ranks the three as $h \succ h' \succ h''$. Then if her preferences conform to the Anscombe–Aumann axioms (and if she prefers more money to less), we interpret this as: Our consumer assesses probability greater than .48 that a Latin American team will win and probability less than .48 that a European team will win. Indeed, by varying the probability in h' and looking for a value where she is

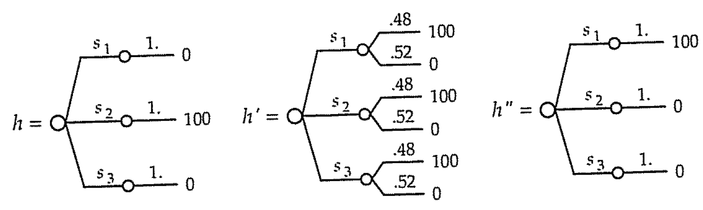

Figure 3.10. Three compound lotteries.

indifferent between h and h' (as modified), we will find the subjective probability that she assesses that a Latin American team will win. All this talk about subjective probabilities that she assesses is entirely a construct of the representation that is guaranteed by the Anscombe–Aumann axioms, and it arises entirely from our consumer's preferences for lotteries like those shown. A different consumer might express the preferences $h'' \succ h' \succ h$, and we would, therefore, know that these consumers' preferences are represented by different subjective assessments over the state space S, if each set of preferences satisfies the Anscombe–Aumann axioms.

Economists (of a neoclassical stripe) would rarely if ever insist that consumers have the same ordinal preferences over bundles of goods or the same levels of risk aversion or risk tolerence. Individual consumers are allowed to have individual preferences. Seemingly then, one would allow subjective probabilities, as another part of the expression of personal preferences, to vary across individuals. If this seems so to you, and it certainly does to me, you are forewarned that, amongst many microeconomists, it is dogma (philosophy?) that two individuals *having access to the same information* will necessarily come up with the same subjective probability assessments. Any difference in subjective probability assessments must be the result of differences in information. Our two consumers might express the preferences given above, but only if they have been exposed to different pieces of information about the qualities of the various teams, etc. This assumption has very substantial implications for exchange among agents; we will encounter some of these later in the book. I leave it to others to defend this assumption — see, for example, Aumann (1987, section 5) — as I cannot do so. But the reader should be alerted to this modeling assumption, which plays an important role in parts of modern microeconomic theory; it is called both the *common prior assumption* and the *Harsanyi doctrine*.

3.5. Problems with these models

The von Neumann–Morgenstern model, where probabilities are objective, and the Savage (or Anscombe–Aumann) model, where probabilities are subjective, are the chief models of consumer choice under uncertainty in microeconomics. Indeed, "chief" is an understatement; "predominant" is better, but still not strong enough. One wonders, then: (a) Are there theoretical reasons to distrust these models? (b) How good are they empirically? (c) If there are problems with the models, what are the alternatives? In this section we begin to answer these questions.

Theoretical problems

There are a number of theoretical reasons for distrusting these models. These reasons often arise from the way the models are used in applications; only a limited portion of the consumer's overall decision problem is modeled, and it is assumed that the consumer's choice behavior on the limited problem conforms to this sort of model. *Even though the consumer's overall choice behavior might conform to this sort of model, choice behavior can fail to conform on pieces examined in isolation.* Two of the most important reasons for this have to do with portfolio effects and the temporal resolution of uncertainty.

Portfolio effects are, in spirit, the same as complement/substitute effects in the usual theory of consumer choice under certainty. If we wished to model the demand for, say, wheat, it would be important to realize that the price of corn is a key variable. We (probably) wouldn't try to analyze the wheat market in isolation. Or, rather, when we do try to analyze the wheat market in isolation, there is a lot of incanting of ceteris paribus (or, everything else, and in particular the price of corn, held equal). With lotteries, something similar but even more sinister takes place. Suppose we asked the consumer how he felt about lotteries posed by taking positions in the stock of General Motors. According to these models, we would describe the lotteries by probability distributions over the dollar returns from the positions (possibly having the consumer give his subjective assessment of those distributions), and we would expect the consumer's preferences concerning the gambles are given by the expected utilities they engender. But if this consumer already has a position in the stock of General Motors, or a position in the stock of, say, Ford, we would expect his preferences to be affected. Put another way, given the consumer has an initial position in Ford, if we had two gambles with the exact same probability distribution, but one involved the stock of General Motors and the other involved the stock of General Mills, *because of differences in the correlations in returns between Ford and General Motors and between Ford and General Mills*, we expect the consumer to feel differently about them. We would see this if we modeled the entire portfolio choice problem of the individual — including in one model all the separate gambles that make up the one grand lottery that this individual faces. But we cannot look at preferences between gambles involving the stock of General Motors and the stock of General Mills in isolation and expect the model to work. And we can't incant ceteris paribus here with much conviction — holding fixed the consumer's net position in Ford isn't the issue. The issue is instead the differences in correlation between the gambles being investigated and the gambles left out of the model. The (marginal) probability distributions of the gambles under consideration are not sufficient descriptions of the objects under choice; two gambles with the same marginal distributions (but different joint distributions with the consumer's unmodeled portfolio) would rank differently.[n]

As for temporal resolution of uncertainty, the easiest way to introduce this is with an example. Suppose a coin will be flipped (independent of everything else in the world). If it comes up heads, you'll be given a check for $10,000; if it comes up tails, you'll get nothing. Or you can have a check for $3,000. (You may wish to supply a number in place of the $3,000, so that you just slightly prefer the gamble.) One complication arises; the money will be given to you next September 1. This isn't so complicated that we can't gear up the models of choice from the previous sections. Assuming it is a fair coin, this sounds like a gamble with probability .5 of $10,000 and .5 of nothing, against a sure thing of $3,000.

But consider a real complication. We have in mind three gambles that you might be given. The first is just as above, a coin flip gamble, where the coin is flipped today *and you are told the result today*. The second is just as above, a coin flip gamble, where the coin is flipped next September 1 *and you are told the result at that date*. And in the third, we have the same coin flip, flipped today, *and you will be told the result next September 1*. How do you rank these three?

If you conformed to the models above when applied to this choice problem in isolation, then you would rank them identically. They all have the same probability distribution on prizes. (Note also, they are independent of other gambles you may be facing; this isn't a portfolio problem.) But if you are like most people, you like the first gamble better

[n] Readers who followed our earlier discussion of demand for assets with more than one risky asset will recognize that discussion as a more precise version of what has been described here.

than the second two, and you are indifferent between the second two. Two things are at work here:

(a) Getting the information sooner is valuable to you

(b) The important time is when the uncertainty resolves for you, the decision maker, and not when the uncertainty is "physically" resolved

These two statements follow from things we leave out of the model when we think of these gambles simply as probability distributions over some income you might receive. For example, between now and next September 1 you have a number of consumption choices to make — how much to spend on an apartment, on food, on a vacation. What you will do between now and then depends, at least in part, on the resources you have now and the resources you think you will have later. If the uncertainty in this gamble is resolved for you now, you can use that information better to fit your consumption choices between now and September 1 to your (now better-known) resources. If the uncertainty resolves for you next September 1, you will have to temporize in the meantime. Note well that what drives this is the presence of (consumption) decisions you must make that are not analyzed if we look at the gambles in isolation. It is a problem, as before, of a partial or incomplete model of your overall choice problem. (This is why in our discussion of induced preferences for income, induced from preferences for consumption, we said that it was important that the uncertainty about income resolved before any consumption decisions had to be made.)

This shows that timing of resolution of uncertainty can matter.[6] When viewing these gambles in isolation, it isn't enough to describe the possible prizes and their probabilities. What is less obvious (but still true) is that we mightn't expect the usual sort of model to hold in comparing gambles, if we fix all gambles to resolve at a specified later date. (Problem 11 will confirm this.)

What do we do in the face of these problems? One solution is to include in our models all the relevant portions of the consumer's overall choice problem. This is fairly unpalatable; putting everything in makes the model unwieldy and, often, intractable. Or you could restrict attention

[6] The timing of resolution of uncertainty can be important for "psychological" reasons as well, even if the consumer has no particular use for the information in the sense that it could change some decisions he must make. Pending/hanging uncertainty can, in itself, be a good or a bad. The commonplace view is that hanging uncertainty when there is a small chance of very good news and no chance of bad news is typically good, while any hanging uncertainty about a possible bad is in itself a bad. These are important effects, worthy of attention. But they are different from the effects that concern us here.

only to those pieces for which the standard models do apply. This is unpalatable because the conditions required for the standard models to apply are quite stringent (see the next paragraph). A better alternative is to look for models of choice that are correct for pieces of a bigger model. Such models will necessarily have less structure than the standard von Neumann–Morgenstern or Savage models but can still have enough structure to permit some analysis. For example, in dealing with portfolio effects, if you choose a "states of the world" model, then you can have additive separability across states and (even) some comparability of the various state-dependent utility functions. In dealing with problems in temporal resolution of uncertainty, you can have preferences that are "locally linear in probabilities"; recent work by Machina (1982, 1984) shows that this is enough to get back some of the standard results in the economics of uncertainty.

Empirical evidence in simple settings

For the two reasons above (and others besides), one worries about the application of these models of choice to contexts that study an isolated piece of the consumer's overall choice problem. It can be shown, though, that isolating a piece is okay if the consumer obeys the model of choice overall and if the piece has the simplest possible structure; the risk in the gambles is independent of other risks the decision maker might face, and the uncertainty in the gambles resolves immediately. But we can (and do) still wonder how these models perform as *descriptive* models of individual choice under uncertainty, even in settings with this ultra-simple structure.

The typical method for investigating this question is to run experiments on subject populations. The caricature subject population consists of college sophomores who are told in their introductory psych courses that they need to take part in a certain number of such experiments to pass the course. The subjects are asked hypothetical questions: How would you pick if offered a choice between ____ and ____; between ____ and ____; etc.? And we look for violations of the usual models of choice. Such studies are often criticized as being artificial: The subject population isn't used to dealing with the sort of question asked; the subjects aren't given appropriate incentives to take the question seriously (the questions are hypothetical); it isn't clear that the experiments have anything to say about choices made in economically important contexts, and so on. (Since these studies draw so much fire, you can already guess what they will say.) But experimental economists and psychologists have dealt within their limited budgets with these criticisms, and these experiments continue to give over and over a few results that are not in keeping with the models above and that make it incumbent on economists to offer excuses for themselves and their models.

We will discuss three of the most important "effects" suggested by the experimental literature. They all have to do with choice under uncertainty. This isn't to say that the models of choice under certainty haven't been scrutinized in like manner — they have been — and problems arise there as well, along with very nice perceptual and cognitive models concerning how individuals do make choices.

(a) The Allais Paradox. The most famous violation of the von Neumann–Morgenstern expected utility model is named after its discoverer, Maurice Allais. One variation on this runs as follows:

> Choose between two gambles. The first gives a .33 chance of \$27,500, a .66 chance of \$24,000, and a .01 chance of nothing. The second gives \$24,000 for sure.
>
> Choose between two gambles. The first gives a .33 chance of \$27,500 and a .67 chance of nothing. The second gives a .34 chance of \$24,000 and a .66 chance of nothing.

The typical (modal) response pattern is to take the sure thing in the first case, and the first gamble in the second. This particular response pattern violates the substitution axiom.[7]

A number of explanations have been hypothesized for this. The spirit of most of them is that individuals rescale probabilities, with more weight (proportionately) given to small probability events.

(b) The Ellsberg Paradox. Due to Daniel Ellsberg, this runs as follows:

> An urn contains 300 colored marbles; 100 of the marbles are red, and 200 are some mixture of blue and green. We will reach into this urn and select a marble at random:
>
> You receive \$1,000 if the marble selected is of a specified color. Would you rather that color be red or blue?
>
> You receive \$1,000 if the marble selected is not of a specified color. Would you rather that color be blue or red?

The modal responses are red in the first case and red in the second. And the preferences are strict.

[7] Try to prove this yourself, or see the discussion on framing. Alternatively, you can show that no expected utility maximizer could express the two choices above.

This violates the Savage or Anscombe–Aumann axioms. If those axioms are satisfied, then the decision maker chooses as if he assessed subjective probabilities that the marble selected is red, blue, and green. (It seems natural to assess probability 1/3 for red, and it seems natural to me at least to assess 1/3 each for blue and green as well. But the assessment is up to the consumer.) If the consumer strictly prefers red in the first case, he must assess a higher probability for red than for blue. But then he should assess a higher probability for the event *the marble is not blue* than for the event that *the marble is not red.* Hence he should, if he conforms to the model, strictly prefer to answer blue in the second question.

Ellsberg (and others both before and after him) use this sort of behavior to claim that there is a real distinction to be made between situations where there is risk (objective uncertainty) and situations with uncertainty (subjective uncertainty). The story goes: People dislike the ambiguity that comes with choice under uncertainty; they dislike the possibility that they may have the odds wrong and so make a wrong choice. Hence in this example in each case they go with the gamble where they know the odds — betting for or against red.

(c) Framing. Early in chapter 2, we discussed the effect that framing can have on choice. In fact, the example given there about various vaccination programs was an example of choice under uncertainty. In that example, the two different ways of framing the options amounted to a shift of the "status quo" — are the six hundred still alive or already dead? — and experimenters have shown that, in general, choices made under uncertainty can be powerfully influenced by the decision maker's perception of the status quo. (See Kahneman and Tversky, 1979.)

There are many interesting aspects of framing and its impact on choice, many more than we can describe here. But one aspect of framing is especially relevant to our earlier discussion. Consider the two compound gambles in figure 3.11(a) and (b). Which would you choose?

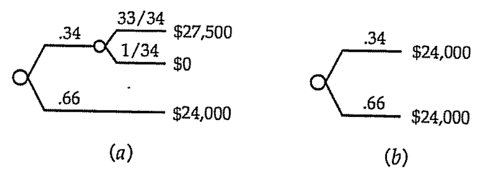

Figure 3.11. Two compound lotteries.

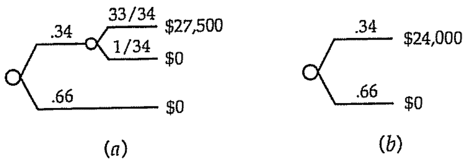

Figure 3.12. Two more compound lotteries.

And how would you choose between the two compound lotteries in figure 3.12?

These gambles are drawn so that the decision maker naturally focuses on the "differences" between the two in each pair — a 33/34 chance of $27,500 (and a 1/34 chance of nothing) in (a) against $24,000 for sure in (b). The incidence of "consistent choices," that is, (a) in both cases or (b) in both cases, is higher than in the Allais paradox.[8] But these compound gambles "are" precisely the gambles of the Allais paradox; that is, they reduce to the single-stage gambles of the Allais paradox if the rules of probability theory are applied.[9] Insofar as framing these gambles as compound lotteries changes choice from when the gambles are framed as single-stage lotteries, the entire structure on which the standard theory is built begins to shake.

Rather a lot of work has been done recently on recasting von Neumann–Morgenstern expected utility theory in ways that allow preferences to accommodate the Allais paradox. Somewhat less, but still some, work has been done on modifying the standard Savage and Anscombe–Aumann models to allow for Ellsberg paradox-like preferences. Machina (forthcoming) contains a good survey of this material, together with an exposition of the standard models and the experimental evidence that is available. Very little indeed has been done (at least, in the literature of economics) to confront generally problems of framing. This is a very important frontier in choice theory.

Validity of the models in complex settings

Another reason to disbelieve these models of choice, especially in how they are applied in economic contexts, concerns the complexity of the decision problems in which they are employed. They are used, for

[8] See, for example, Tversky and Kahneman (1986).

[9] This indicates, of course, how it is that the Allais paradox is a violation of the von Neumann–Morgenstern substitution axiom.

example, to model the consumption-investment budgeting problems of an individual, over the individual's entire life. They are used to model individuals who are assumed to be solving statistical inference problems that the economist who is writing down the model cannot solve. They are used to model extraordinarily complex interactions between individuals, where each individual is (correctly) assessing what others will be doing in response to information that they have, even though the person writing down the model can't make that assessment, and so on. (Lest you think I am sneering at such silly applications, which are quite common among microeconomic theorists, the three sins listed above are my own.)

This problem is sometimes described as one of assumed *unlimited rationality* on the part of individuals in economic models. Individuals are assumed to understand to an amazing extent the environment within which they act, and it is assumed they can perform fantastic calculations to find their own best course of actions at no cost and taking no time. This is, of course, ridiculous. Faced with complexity, individuals resort to rules of thumb, to "back of the envelope" calculations, to satisficing behavior (taking the first "satisfactory" alternative that arises), and the like. It seems patent that such *limitedly rational* behavior would, when placed in a complex economic context, have important implications.

We hinted in chapter 1 at the excuses that are typically made. There is the "positive economics" defense: It is only necessary that behavior is "as if" these calculations were performed. For example (this line of defense often goes), the calculations necessary when driving a car and determining whether it is safe to pass someone ahead are incredibly complex. Yet we see individuals, even those without the benefit of a high school education, succeed in such calculations every day. (We also see them fail occasionally, but there is some chance of a bad outcome any time there is uncertainty.) One might retort that such choices are made, but they are done heuristically, and (probably) with a greater than optimal margin of safety built in. The defender of the faith would respond that this is "as if" the individual were optimizing with a more risk averse utility function.

Moreover (the defender of the faith will say) small variations from the theory on the individual level are not important if they cancel out in aggregate. Of course, this presumes that it is the aggregate outcomes that are of consequence. And it presumes no systematic bias in the deviations in behavior from that which is posited. Any systematic bias would (probably) have aggregate consequences that are important.

The defender of the faith has yet another line of defense that was not mentioned in chapter 1: There are no palatable alternatives. Most alternatives that have been proposed have been criticized as being ad hoc and without any testable restrictions: If we allow any behavior at all, or if the behavior that is allowed is at the whim of the modeler, predictive power is lost. But this rejoinder is something of a non sequitur. *If* the alternatives so far developed are ad hoc (and I don't mean to prejudge this issue at all), this doesn't mean that acceptable alternatives cannot be developed. One always hopes that some good and generally accepted models of limitedly rational behavior will come along. (And there are, occasionally, some attempts in this direction. Indeed, as this is being written, there is rather more activity in this direction than usual.) But until they do....

Finally (and this is the line of defense with which I am personally the most comfortable), you should not rule out the possibility that you can learn things of value by making such heroic assumptions about powers of economic actors. This isn't to say that you can't be misled, or that you can't learn other things with more reasonable assumptions. But neither of these precludes insight coming from models of behavior that are somewhat incredible along this dimension.

3.6. Normative applications of the theory

Another way to view the models of this chapter is to interpret them as normative rather than descriptive theory. The normative view runs:

(0) Suppose you are in a situation in which you must make a particular, fairly complex choice. The choice in question might involve random events without "objective" probabilities, many different outcomes, and so on.

(1) Ask yourself: Can this situation be embedded within a general choice situation that has the structure of one of the models above?

(2) If the answer to the first question is yes, do the "properties of preference" that we have discussed seem sensible in this situation? In other words, is it reasonable to take a choice based on preferences that conform to these properties?

(3) If the answer is yes again, then we know that you want your choice to be based on the sort of criteria that is implied by the particular framework and properties — expected utility maximization or subjective expected utility maximization.

(4) And (the punchline to this) it may be possible to get the data needed for the representation by asking yourself about less complex specific choices and about general properties of your preferences. This will give you the data for the representation — then with that data you *compute* the best option in the complex situation.

For example, suppose you are offered a gamble in which the prizes are \$1,000, \$700, \$470, \$225, \$32, \$−87, \$−385, \$−642, and \$−745. The probabilities depend on the exact closing level of the Dow Jones Industrial Average next Friday afternoon. For example, the gamble might specify that you get \$1,000 if the DJIA is 100 points above its current level, \$700 if it is between 100 and 80 points over its current level, and so on.

(1) Can we formulate this in terms of one of the models above? Quite definitely — the Anscombe–Aumann model applies, where the states of the world are various levels of the DJIA.

(2) Do you subscribe to the appropriate axioms? In this case, you might. The uncertainty resolves quite soon, and you may not have significant asset holdings that are correlated with this uncertainty. (If you did, a more complex normative model could be used.)

(3) Hence your choice whether to take this gamble or not should be made according to the representation of the Anscombe–Aumann framework. We need to get your subjective probability assessment for the future level of the DJIA, and we need to get your utility function for money.

So far, nothing useful has come out of this exercise. Indeed, in (1) and (2) you were asked some hard questions, which (one hopes) you thought hard about. But now:

(4a) Consider first your subjective assessment for the DJIA. A number of procedures help to do this. Some, involving inference from past data, are probably already familiar to you. Others involve framing questions for yourself in a way that is easy to perceive and that eliminates systematic biases. In the end, you will still have to give your subjective assessment, but there are ways to increase your confidence in the "validity" of the assessment you make.

(4b) It is somewhat easier to describe how to simplify getting your utility function. First, it is probably safe to assume that your preferences are increasing in money; your utility function is monotone. Second, as long as you can "afford" to lose the \$745, you can probably be convinced that you are risk averse; your utility function will be concave. Kahneman and Tversky (and others before them) tell us to be careful around the level of zero — you may exhibit a "zero illusion" — putting more significance into the difference between gains and losses on this gamble than you would want, once this is pointed out to you.[o] We need your utility function for a range of \$1,000 to \$ − 745, and we can get this, at least roughly, by asking you relatively simple questions such as: What is your certainty equivalent for a coin flip gamble where the prizes are \$1,000 and \$ − 800? Note what we're doing here. We ask the simplest possible certainty equivalent question, involving a coin flip gamble. With the answers to such relatively easy questions, we can build your utility function for the entire range, and we can put you through consistency checks of the answers you give.

And more besides. Now you know about increasing and decreasing and constant risk aversion. It is likely that you can be convinced in general that you wish to be (weakly) nonincreasingly risk averse. And if the range of prizes isn't too significant for you, as might be the case for this particular problem, then you may even be able to be convinced that you wish to exhibit virtually constant absolute risk aversion over this range. If you accept this, then we are really in business; we have the determination of your utility function down to finding a single constant, which we determine by a single question about a coin flip gamble (although we'll ask several, as a consistency check). The greater is the number of qualitative properties to which you are willing to ascribe in this particular problem (and constant risk aversion over this range of prizes is an easy-to-understand and incredibly powerful property), the easier it is to carry out probability and utility function *assessment*. Once we have your subjective probability assessments and your utility function, we can quickly grind through the complex gamble given to see if you "want" to take it.

3.7. Bibliographic notes

The material of this chapter combines a lot of the theory of choice under uncertainty and a bit of the economics of uncertainty. On the subject of choice under uncertainty, you are directed to three text/reference books: Fishburn (1970), Kreps (1988), and Machina (forthcoming). For getting all the details of the standard model, Fishburn is highly recommended. Kreps (1988) omits some of the details of proofs, but is perhaps more immediately accessible than Fishburn. Machina (forthcoming), besides providing a complete treatment of the standard theory, gives the reader an up-to-date account of recent developments in choice theory that are intended to address Allais and Ellsberg style paradoxes. If you like to consult original sources, see von Neumann and Morgenstern (1944), Savage (1954), and Anscombe and Aumann (1963).

[o] One approach is to frame the same gamble several different ways. For example, we ask you for your certainty equivalent for a coin flip gamble where the prizes are \$ − 1,000 and \$0, and then we ask you for your certainty equivalent for a coin flip gamble where the prizes are \$0 and \$1,000, where you first must pay \$1,000. These are the "same" questions in terms of the final balance of your bank account, and, upon reflection, you are likely to decide that you want to give the same net answer.

The material on utility functions for money (and especially absolute and relative risk aversion) is further developed (in varying degrees) in the three books listed above. The classic references are Arrow (1974) and Pratt (1964) (the coefficient of absolute risk aversion is sometimes called the Arrow-Pratt measure of risk aversion in consequence). We touched very briefly (in smaller type) on the subject of more and less risky gambles; Machina (forthcoming) gives a particularly good treatment of this.

The economics of uncertainty (demand for insurance, demand for risky assets) is developed in many different places. Borch (1968) provides a very readable introduction and Arrow (1974) contains a number of classic developments. Matters concerned with asset markets quickly merge in the academic discipline of finance; a good text on this subject is Huang and Litzenberger (1988).

The papers cited concerning the empirical problems with these models are Allais (1953), Ellsberg (1961), and Kahneman and Tversky (1979). Machina (forthcoming) provides an overview of much of this, with recent theoretical developments that are intended to respond to Allais and Ellsberg style paradoxes.

References

Allais, M. 1953. "Le Comportement de l'Homme Rationnel devant le Risque, Critique des Postulates et Axiomes de l'École Americaine." *Econometrica* 21:503–46.

Anscombe, F., and R. Aumann. 1963. "A Definition of Subjective Probability." *Annals of Mathematical Statistics* 34:199–205.

Arrow, K. 1974. *Essays in the Theory of Risk Bearing*. Amsterdam: North Holland.

Aumann, R. 1987. "Correlated Equilibrium as an Expression of Bayesian Rationality." *Econometrica* 55:1–18.

Borch, K. 1968. *The Economics of Uncertainty*. Princeton, N.J.: Princeton University Press.

Diamond, P., and J. Stiglitz. 1974. "Increases in Risk and in Risk Aversion." *Journal of Economic Theory* 8:337–60.

Ellsberg, D. 1961. "Risk, Ambiguity, and the Savage Axioms." *Quarterly Journal of Economics* 75:643–69.

Fishburn, P. [1970] 1979. *Utility Theory for Decision Making*. New York: John Wiley and Sons. Reprint. Huntington, N.Y.: R. E. Krieger Publishing.

Huang, C-f., and R. Litzenberger. 1988. *Foundations for Financial Economics*. New York: Elsevier Science Publishing Company.

Kahneman, D., and A. Tversky. 1979. "Prospect Theory: An Analysis of Decision Under Risk." *Econometrica* 47:263–91.

Kreps, D. 1988. *Notes on the Theory of Choice*. Boulder, Colo.: Westview Press.

Machina, M. 1982. "'Expected Utility' Analysis without the Independence Axiom." *Econometrica* 50:277–323.

———. 1984. "Temporal Risk and the Nature of Induced Preferences." *Journal of Economic Theory* 33:199–231.

———. (Forthcoming). *The Economic Theory of Individual Choice Under Uncertainty: Theory, Evidence, and New Directions*. Cambridge: Cambridge University Press.

Pratt, J. 1964. "Risk Aversion in the Small and in the Large." *Econometrica* 32:122–36.

Ross, S. 1981. "Some Stronger Measures of Risk Aversion in the Small and in the Large with Applications." *Econometrica* 49:621–38.

Rothschild, M., and J. Stiglitz. 1970. "Increasing Risk. I: A Definition." *Journal of Economic Theory* 2:225–43.

———. 1971. "Increasing Risk. II: Its Economic Consequences." *Journal of Economic Theory* 3:66–84.

———. 1973. "Addendum to Increasing Risk." *Journal of Economic Theory* 5:306.

Savage, L. 1954. *The Foundations of Statistics*. New York: John Wiley and Sons. Rev. and enl. ed. New York: Dover Publications, 1972.

Tversky, A., and D. Kahneman. 1986. "Rational Choice and the Framing of Decisions." *Journal of Business* 59:S251–78.

Von Neumann, J., and O. Morgenstern. 1944. *Theory of Games and Economic Behavior*. Princeton, N.J.: Princeton University Press. (2d edition 1947; 3d edition 1953).

3.8. Problems

■ 1. (a) Prove lemma 1. (Assumption 2 should figure prominently in your proof. Begin with the case $\alpha = 1$.)

(b) Prove lemma 2. (Deal with the cases $p \sim \delta_b$ and $p \sim \delta_w$ first. Then define $\alpha = \inf\{\beta : \beta\delta_b + (1-\beta)\delta_w \succ p\}$. Use the Archimedean property to show that $\alpha\delta_b + (1-\alpha)\delta_w \sim p$.)

(c) Prove lemma 3. (This is harder than lemmas 1 and 2 and is left as something of a challenge. Consider three cases: $p \sim q \sim s$ for all $s \in P$; there is some $s \in P$ with $s \succ p$; and then there is some $s \in P$ with $p \succ s$.)

(d) Give a formal proof of lemma 4. (Use induction on the size of the support of p. Recall here that we assume that all probability distributions in question have finite support.)

(e) Extend the proof of lemma 4 to cases without best and worst prizes. (Suppose there are two prizes with $\delta_b \succ \delta_w$. Arbitrarily set $u(b) = 1$ and $u(0) = 0$. If $\delta_b \succeq \delta_x \succeq \delta_w$, define $u(x)$ as in lemma 2. If $\delta_x \succ \delta_b$, use lemma 2 to show that there is a unique α such that $\alpha\delta_x + (1-\alpha)\delta_w \sim \delta_b$ and define $u(x) = 1/\alpha$. If $\delta_w \succ \delta_x$, use lemma 2 to show that there is a unique α such that $\alpha\delta_b + (1-\alpha)\delta_x \sim \delta_w$, and define $u(x) = -\alpha/(1-\alpha)$. Then show that this u works. Everything should be clear, if a bit tedious, once you figure out why this gives the right definition for u.)

(f) Prove the "uniqueness" part of proposition 3.1. (Hint: Begin your proof as follows. Suppose u and v (with domain X) both give expected utility representations for $\succ$. Suppose it is not the case that $v(\cdot) \equiv au(\cdot) + b$ for constants $a > 0$ and b. Then there exist three members of X such that....)

■ 2. Let p' be a probability distribution giving prizes \$10 and \$20 with probabilities $2/3$ and $1/3$, respectively, and let p be a probability distribution giving prizes \$5, \$15, and \$30 with probabilities $1/3, 5/9$ and $1/9$, respectively. Show that any risk averse expected utility maximizer will (weakly) prefer p' to p. (Hint: Construct p as a compound lottery as discussed at the end of the subsection on risk aversion.) Can you supply a general statement of the principle at work in this specific example?

■ 3. At the end of the first subsection of section 3.3, we asked the question: Fix a consumer with von Neumann–Morgenstern preferences over lotteries for consumption bundles, and fix this consumer's income at some level Y. Given two price vectors p and p', would the consumer rather be in an economy where prices are either p or p', each probability $1/2$, or in an economy where the prices are sure to be $.5p + .5p'$? We asserted that there is no clear answer to that question. In this problem, you are asked to develop two examples that indicate what can happen along these lines.

(a) Imagine that there are two goods, and the consumer's *ordinal* preferences are given by $U(x_1, x_2) = x_1 + x_2$. That is, the consumer's von Neumann–Morgenstern utility function is $u(x_1, x_2) = f(x_1 + x_2)$ for some strictly increasing function f on the real line. Suppose that $p = (1, 3)$ and $p' = (3, 1)$. Show that *regardless of what the function f is*, this consumer prefers to take her chances with the risky prices.

(b) Imagine that there are two goods and the consumer's von Neumann–Morgenstern utility function is $u(x_1, x_2) = f(\min\{x_1, x_2\})$ for some concave, strictly increasing function f on the real line. Assume that $f(0)$ is finite. Now suppose that the risk in prices is entirely risk in the overall price level: $p = (\gamma, \gamma)$ and $p' = (1/\gamma, 1/\gamma)$ for some scalar $\gamma > 1$. Prove that for fixed γ you can always find a function f such that the consumer prefers the certain prices $.5p + .5p'$ to the risky prices. And prove that for every concave, strictly increasing function f with $f(0)$ finite there is a γ sufficiently large so the consumer prefers the risky prices to the certain prices.

■ 4. Recall the analysis of an insurance buying consumer in section 3.3. Suppose this consumer has a concave utility function for net income that is not necessarily differentiable. What changes does this cause for the results given in our discussion?

■ 5. Suppose that an insurance policy compensates a victim for loss, but does so somewhat imperfectly. That is, imagine that in the story given about the insurance buying consumer our consumer's income prior to any insurance is $Y - \Delta$, where Δ is a simple lottery whose support includes 0 and some strictly positive amounts. The available insurance policy pays a flat amount B in the event of any loss, that is, in the event that Δ exceeds zero. The premium is still δ. Let π be the probability that the consumer sustains a loss. The contract is actuarially fair if $\delta = \pi B$. Suppose this is so and that B is the expected amount of the loss, if there is a loss. If the consumer has a concave, differentiable utility function, will she buy full insurance? (Use a simple parameterized example if you can't do this in general.)

Questions 6 through 9 all concern the discussion in section 3.3 of demand for risky assets.

■ 6. Consider the following specialization of the discussion in section 3.3: The consumer has \$$W$ to invest, which he must allocate between two possible investments. The first is a sure thing — put in \$1, and get \$$r >$ \$1 back. The second is risky — for every \$1 invested, it returns a random

amount \$$\theta$, where θ is a simple lottery on $(0, \infty)$ with distribution π. We assume that the expected value of θ is strictly greater than r but that there is positive probability that θ takes on some value that is strictly less than r. We allow this investor to *sell short* the riskless asset but not the risky asset. And we do not worry about the net payoff being negative; the consumer's utility function will be defined for negative arguments.

This investor evaluates his initial choice of portfolio according to the expected utility the portfolio produces for his net payoff. Moreover, this consumer has a constant coefficient of absolute risk aversion $\lambda > 0$. That is, he chooses the amount of his wealth to invest in the risky asset (with the residual invested in the safe asset) to maximize the expectation of $-e^{-\lambda Y}$, where Y is his (random) net payoff from the portfolio he picks. Let us write $\alpha(W, \lambda)$ for the optimal amount of money to invest in the risky asset, as a function of the consumer's initial wealth W and coefficient of risk aversion λ.

(a) Prove that $\alpha(W, \lambda)$ is finite for all W and α. That is, the consumer's investment problem has a well-defined solution.

Even if you can't do part (a), assume its result and go on to parts (b) and (c).

(b) Prove that $\alpha(W, \lambda)$ is independent of W; no matter what his initial wealth, the consumer invests the same amount in the risky asset.

(c) Prove that $\alpha(W, \lambda)$ is nonincreasing in λ; the more risk averse the individual, the less he invests in the risky asset.

■ 7. Consider the general formulation of the consumer's problem with one risky asset. Show that if the risky asset is genuinely risky — that is, it has non-zero variance — and if the consumer's von Neumann–Morgenstern utility function is strictly concave, then the consumer's problem, if it has any solution at all, must have a unique solution.

■ 8. Recall that in the discussion of demand for risky assets with more than one risky asset, we asserted that it was possible that an asset could have an expected return less than r and still be demanded and that an asset could have an expected return greater than r and not be demanded at all. (Recall that we are not allowing short-sales of risky assets.) Produce examples to support these claims. (Hints: For the first example, recall that you want negative correlation between the returns on the two risky assets. Imagine that each asset returns either $\theta = 1$ or $\theta = 5$ and that $r = 2$. You should be able to construct an example from this. The second example is even easier. What happens if θ^1 and θ^2 are perfectly positively correlated and θ^2 always exceeds θ^1?)

■ 9. Prove that if we have two risky assets with independent return distributions, all the results obtained in the text for the case of one risky asset extend.

■ 10. With regard to the *additive across states* representation (★★) in section 3.4, we might wonder whether there is any value in obtaining the representation with subjective probabilities over the states, so that the representation becomes one of ***subjective state dependent** expected utility*. Prove the following: If we have a representation (★★), then for *any* strictly positive probability distribution π on S, there is a state-dependent utility function $u : X \times S \to R$ such that subjective state-dependent expected utility, using this probability distribution π and the utility function u, represents $\succ$. (Since π is completely arbitrary here, up to being strictly positive, the answer to the question posed at the start of this question is there is apparently very little value. Which is why we don't include subjective probabilities in (★★).)

■ 11. Consider a consumer who will live today and tomorrow. On each date, this consumer consumes a single good (manna), whose price on each date is \$100 per unit. We write c_0 for the amount of manna this consumer consumes today and c_1 for the amount he consumes tomorrow. This consumer faces some uncertainty in how much manna he will be able to afford (to be explained in the next paragraph), and his preferences over uncertain consumption pairs (c_0, c_1) are given by the expectation of the von Neumann–Morgenstern utility function $u(c_0, c_1) = \ln(c_0) + \ln(c_1)$, where ln is the logarithm to the base e.

This consumer currently possesses \$100 of wealth. He can spend it all today, or he can put some of it in the bank to take out and use to buy manna tomorrow. This bank pays no interest and has no deposit or withdrawal charges — \$1 put in today will mean \$1 taken out tomorrow. In addition to any savings that he carries forward, the consumer has a random amount of extra income, which he will receive tomorrow. If there is any uncertainty about this income, that uncertainty doesn't resolve until tomorrow — until after today's levels of consumption and savings must be fixed.

(a) Suppose the consumer is sure to receive an extra \$34 tomorrow. What consumption level does he choose today, and what is his overall expected utility?

(b) Suppose that, tomorrow, the consumer will receive an extra \$100 with probability 1/2, and nothing more with probability 1/2. What consumption level does he choose today, and what is his overall expected utility? (To solve this analytically, you will have to solve a quadratic equation.)

(c) Suppose that, tomorrow, the consumer will receive nothing with probability 1/4, \$34 with probability 1/2, and \$100 with probability 1/4. What consumption level does he choose today, and what is his overall expected utility? (A good approximate answer is acceptable here, because to get the exact solution you have to solve a cubic equation. You will need to be quite accurate in your calculations; the third significant digit is important.)

(d) Assuming you get this far, what strikes you about the answers to parts (a), (b), and (c)? In particular, if we asked for this consumer's preferences over lotteries in the "extra income," where it is understood that all uncertainty resolves tomorrow, would this consumer satisfy the von Neumann–Morgenstern axioms?

■ 12. Kahneman and Tversky (1979) give the following example of a violation of von Neumann–Morgenstern theory. Ninety-five subjects were asked:

> Suppose you consider the possibility of insuring some property against damage, e.g., fire or theft. After examining the risks and the premium you find that you have no clear preference between the options of purchasing insurance or leaving the property uninsured.
>
> It is then called to your attention that the insurance company offers a new program called *probabilistic insurance*. In this program you pay half of the regular premium. In case of damage, there is a 50 percent chance that you pay the other half of the premium and the insurance company covers all the losses, and there is a 50 percent chance that you get back your insurance payment and suffer all the losses....
>
> Recall that the premium is such that you find this insurance is barely worth its cost.
>
> Under these circumstances, would you purchase probabilistic insurance?

And 80 percent of the subjects said they wouldn't.

Ignore the "time value of money."[10] Does this provide a violation of the

[10] Because the insurance company gets the premium now, or half now and half later, the interest that the premium might earn can be consequential. I want you to ignore such effects.

von Neumann–Morgenstern model, if we assume (as we usually do) that all expected utility maximizers are risk neutral or risk averse? Is someone who definitely turns down probabilistic insurance exhibiting behavior that is inconsistent with the model (with risk aversion/neutrality a maintained assumption)? You will have gotten the problem half right if you show rigorously that the answer to this question is yes. But you will be getting the entire point of this problem if you can provide a reason why I, an utterly consistent von Neumann–Morgenstern expected utility maximizer, would say no to probabilistic insurance. (Hint: See problem 11.)

■ 13. We earlier showed that the Ellsberg paradox is a violation of the Savage–Anscombe–Aumann theories by referring to their representations of preference. Demonstrate directly that the Ellsberg paradox is a violation of their axioms; that is, show (by example) a specific axiom that is being violated. You may choose either the Savage axioms or the Anscombe–Aumann axioms — only one of the two is necessary. (In the case of Savage, you only know three of the axioms — preference is asymmetric, negatively transitive, and the sure thing principle. One of these three is violated by the Ellsberg example. There are five axioms in Anscombe–Aumann to choose from, so it might be easier to work with the three Savage axioms, which are fewer.)

■ 14. The following bit of nonsense is often heard:

> Suppose I offered you, absolutely for free, a gamble where with probability .4 you win \$1,000 and with probability .6 you lose \$500. You might well choose not to take this gamble (if the alternative is zero) if you are risk averse; although this gamble has a positive expected value ((.4)(1000) + (.6)(−500) = \$100), it also has substantial risk. But if I offered you, say, 100 independent trials of this gamble, then you would certainly wish to take them; the law of large numbers says that you will wind up ahead. That is, risk aversion is perhaps sensible when a single gamble is being contemplated. But it is senseless when we are looking at many independent copies of the same gamble; then the only sensible thing is to go with the long-run averages.

Is this nonsense? Can you produce a particular "consumer" who is rational according to the von Neumann–Morgenstern axioms, and who would turn down all independent copies of this gamble, no matter how many were offered? Or would any von Neumann–Morgenstern expected utility maximizer take these gambles if offered enough independent copies? (Hints: Either answer can be correct, depending on how you interpret the phrase "enough copies of the gamble." The problem is easiest if we read this phrase as: We offer the consumer a number of copies, fixed in advance, but very large. Then you should be able to produce a consumer who will not take any of the gambles. If you are worried about bankruptcy of this consumer, you may take your pick: [1] This consumer is never bankrupt — his utility function is defined for all monetary levels, however positive or negative; [2] this consumer is bankrupt if his wealth, which begins at a level of \$2,000, ever reaches \$0; and we will stop gambling with this consumer the moment he becomes bankrupt. Interpretation [1] is the easier to work with, but either is okay.)

And, if you like challenges, try to prove the following: Suppose that we play according to rule (2): The consumer is offered "up to N gambles" with the proviso that we stop gambling if ever the consumer's wealth falls to \$0. Assume that the consumer has a utility function for final wealth that is strictly increasing and that is finite at zero. (a) If the consumer's utility function is unbounded, then there is an N sufficiently large so that, offered N gambles or more, the consumer will take them. (b) While if the consumer's utility function is bounded above, the result can go either way: The consumer might turn down the gambles, for all sufficiently large N; or the consumer might accept the gambles, for all sufficiently large N. If you can further characterize the two cases in (b), that would be better still.

■ 15. A particular consumer that I know must choose between (1) a sure payment of \$200; (2) a gamble with prizes \$0, \$200, \$450, and \$1,000, with respective probabilities .5, .3, .1, and .1; (3) a gamble with prizes \$0, \$100, \$200, and \$520, each with probability 1/4. This consumer, in the context of these choices, subscribes to the von Neumann–Morgenstern axioms. Moreover, this consumer, upon reflection, is willing to state that her preferences exhibit constant absolute risk aversion over this range of prizes, and her certainty equivalent for a gamble with prizes \$1,000 and \$0, each equally likely, is \$470. Which of the three gambles given is best for this consumer (granting the validity of all her judgments)?

To do this, you could assume that if the insurance company does insure you, the second half of the premium must be increased to account for the interest the company has foregone. While if they do not, when they return the first half premium, they must return it with the interest it has earned. But it is easiest to simply ignore these complications altogether.

chapter four

Dynamic choice

Many important choices made in economic contexts are made through time. The consumer takes some action today, knowing that subsequent choices will be required tomorrow and the day following and so on. And today's choice has impact on either how the consumer views later choices or what choices will later be available or both. We refer to this as a situation of *dynamic choice*, and in this short chapter we discuss how economists model the process of dynamic choice.

Actually, there are two theoretically distinct issues here. When a consumer makes choices today, she presumably does so with some notion of what further choices she plans to be making in the future. We can ask how those plans affect the choice currently being made. That is, what are the consequences for *static choice* of the fact that static choice is part of a larger dynamic choice problem? Then tomorrow comes, and our consumer makes a subsequent choice. We ask how choice on subsequent dates knits together with choice on earlier dates. That is, what is the structure of *dynamic choice*?

In microeconomics, the issue of dynamic choice is usually dealt with by reducing dynamic choice to the static choice of an *optimal dynamic strategy* which is then carried out. We will discuss this standard approach by means of an example in section 4.1. (We have already seen examples in chapter 3, but here we will be more explicit about what is going on.) Then in section 4.2 we will discuss in the context of a different example both the standard approach and a pair of alternatives. General discussion follows in section 4.3.

4.1. Optimal dynamic strategies

A simple example

Imagine a consumer faced with the following problem. This consumer will live for two periods, consuming in each. She consumes only two things: artichokes and broccoli. Hence a full consumption bundle for her is a four-tuple $x = (a_0, b_0, a_1, b_1)$ where a_t is the amount of artichoke she consumes at date t and b_t is the amount of broccoli she consumes at date t, for $t = 0, 1$. We will refer to date zero as *today* and date one as *tomorrow*.

This consumer's preferences over vector consumption bundles x are given by some ordinal utility function $U(a_0, b_0, a_1, b_1)$. (We do not consider her preferences over lotteries of consumption for a while.)

We could imagine that at date zero markets are open in which this consumer can purchase any bundle x that she wishes (and can afford); that is, she can *forward contract* for artichoke and broccoli delivery tomorrow. This would make the choice problem that she faces a simple static problem in the style of chapter 2, and no complications would be encountered. But instead we imagine that today markets are open for current artichokes and broccoli; tomorrow new markets will open for then current artichokes and broccoli; and today the consumer has the ability to save (or borrow) money at a bank that she can withdraw (repay) tomorrow for purposes of subsequent consumption.[a] Using, say, dollars as the unit of account, we assume this consumer has $\$Y_0$ to spend or save today, and she will receive an additional $\$Y_1$ in income tomorrow, to which will be added any savings she may bring forward and from which will be subtracted any loans she must repay. We assume that the dollar prices of artichoke and broccoli are p_0^a and p_0^b today, the prices will be p_1^a and p_1^b tomorrow, and the bank with which our consumer deals will return r dollars tomorrow for every dollar invested today (or will give a dollar today in return for a promise to repay r dollars tomorrow).

Our consumer, knowing all this, thinks things through strategically.

(1) She must decide today how much artichoke and broccoli to consume today and how much to borrow from or lend to the bank. We use a_0 and b_0 for the first two decision variables, and we use z for her net position at the bank today, where $z > 0$ means that she loans z to the bank today and $z < 0$ means that she borrows $-z$ from the bank. We assume that she can consume only nonnegative amounts of vegetables, and so a_0 and b_0 are constrained to be nonnegative. And she has to finance all her purchases and banking activities with her current income Y_0; she faces the budget constraint

$$p_0^a a_0 + p_0^b b_0 + z \le Y_0. \tag{BC1}$$

Note carefully the appearance of z in the budget constraint; if $z > 0$, she is saving money for tomorrow, and this increases her "expenditures" today,

[a] To keep matters simple, we assume that vegetables spoil very quickly, so any vegetables bought today must be consumed today; they cannot be left for consumption tomorrow.

whereas if $z < 0$, she has extra money to spend today, which loosens her budget constraint on vegetables.

(2) Tomorrow she must decide how much artichoke and how much broccoli to consume. Her resources for doing so will consist of her income tomorrow Y_1 and her net proceeds from the bank, rz. So tomorrow she anticipates she will choose a_1 and b_1 subject to the budget constraint

$$p_1^a a_1 + p_1^b b_1 \le Y_1 + rz. \tag{BC2}$$

Note again that the sign of z is correct; if $z > 0$, she saved money in the first period, giving her more money to spend on vegetables tomorrow, whereas if $z < 0$, she borrowed, giving her less.

(3) Anticipating all this, our consumer recognizes that she can choose any bundle $x = (a_0, b_0, a_1, b_1)$ that she wishes, subject to the single budget constraint

$$p_0^a a_0 + p_0^b b_0 + (p_1^a/r)a_1 + (p_1^b/r)b_1 \le Y_0 + (Y_1/r). \tag{BC}$$

We obtain this single, combined budget constraint by moving the z to the right-hand side in (BC1), dividing (BC2) by r, and then adding the two.

The point is that we have turned the consumer's dynamic choice problem into a static choice problem of the sort discussed in chapter 2. We *presume* that our consumer will choose x subject to (BC) (and the implicit nonnegativity constraints) to maximize her utility from consumption. Once we obtain a solution to this static overall choice problem, we can work out how much the consumer has to save or borrow to implement the solution; that is, given the optimal levels of a_0 and b_0, the consumer will set $z = Y_0 - p_0^a a_0 - p_0^b b_0$ today.

This example illustrates the basic approach that is typically taken to dynamic choice. One imagines that the consumer looks ahead to all decisions that she will be called upon to make and considers her choice today as part of an overall strategy for choice. She works out what the consequences of each strategy will be in terms of an overall outcome; she considers her preferences over the possible overall outcomes; and she evaluates strategies accordingly. She chooses the optimal stategy according to her preferences and proceeds to carry out this strategy. The key implicit assumptions are that her preferences are for overall outcomes; these preferences on overall outcomes conform to a standard model of choice of the sort we explored in chapters 2 and 3, and they are stable through time. And the consumer is blessed with a farsighted strategic sense: She can foresee what her options will be in the future; how current decisions will affect later choices; what overall strategies she has available; and what the consequences of her strategic choices will be.

The example complicated with uncertainty

One should not be misled by the example into believing that the consumer's farsighted strategic sense implies that she possesses perfect foresight. There may be things about which she is uncertain, including the consequences of some of her actions. But she accommodates those with a model with uncertainty in the style of chapter 3. We can adapt our basic example to illustrate how this works.

Imagine that the consumer in our example can place her money not only in the bank but also in a risky asset that pays either $\overline{\theta}$ or $\underline{\theta}$ next period, each with probability $1/2$. (We assume $\overline{\theta} > r > \underline{\theta}$.) She cannot sell this risky asset short, although she can borrow money from the bank. Imagine as well that the consumer is uncertain about the price of artichokes tomorrow; the price will be $\overline{p}_1^a$ with probability $1/3$ and $\underline{p}_1^a$ with probability $2/3$. (We assume, to keep the example simple, that there is no uncertainty about the price of broccoli tomorrow.) Finally, the return on the risky asset and the price of artichokes are correlated: The joint probability that the price of artichokes will be $\overline{p}_1^a$ and the return on the risky asset will be $\overline{\theta}$ is $1/4$.

Since there is now uncertainty, we assume that our consumer's preferences over lotteries of consumption bundles x satisfy the assumptions of von Neumann-Morgenstern expected utility theory, and $u(x)$ is the consumer's von Neumann-Morgenstern utility function.

As in the simple example, our consumer thinks through her choice problem strategically.

(1) Today she must decide how much to consume, how much to invest in the risky asset, and how much to borrow or lend to the bank. We denote her consumption decision variables by a_0 and b_0 as before, and we let ζ be the amount of money she invests in the risky asset. All these must be nonnegative. Since our consumer always likes more to eat, we can safely

assume that she will satisfy her budget constraint today with equality. The amount she lends to the bank is, therefore,

$$z = Y_0 - \zeta - p_0^a a_0 - p_0^b b_0.$$

(Note that if $z < 0$, she is borrowing.)

(2) Next period both her assets and the prices of the consumption goods are random. Four "states" are possible: She has at her disposal $Y_1 + rz + \zeta\overline{\theta}$ and the price of artichokes is $\overline{p}_1^a$ with probability $1/4$; she has at her disposal $Y_1 + rz + \zeta\overline{\theta}$ and the price of artichokes is $\underline{p}_1^a$ with probability $1/4$; she has $Y_1 + rz + \zeta\underline{\theta}$ to spend and the price of artichokes is $\overline{p}_1^a$ with probability $1/12$; and she has $Y_1 + rz + \zeta\underline{\theta}$ and the price of artichokes is $\underline{p}_1^a$ with probability $5/12$.[1]

We assume that our consumer will know which of these four situations prevails when she must decide how to allocate her resources between artichokes and broccoli tomorrow. Thus she has eight decision variables tomorrow: How much artichoke and how much broccoli to consume in the first state (as a function of her decisions today, since they determine the resources she has to spend on vegetables tomorrow); how much to consume in the second state; and so on. Since she always likes more vegetable to less, we can reduce the problem to four decision variables for tomorrow. Given the amount of artichoke she purchases in the first state a_1^1, where the subscript pertains to time and the superscript to state one, it can be assumed that her broccoli consumption will be

$$b_1^1 = \frac{Y_1 + rz + \zeta\overline{\theta} - a_1^1\overline{p}_1^a}{p_1^b},$$

and so on for states 2, 3, and 4, making the appropriate substitutions for $\overline{\theta}$ and for $\overline{p}_1^a$.[b]

(3) So we have her decision problem down to a seven variable problem: a_0, b_0, ζ, and a_1^n for $n = 1, 2, 3, 4$. Given values for these seven decision

[1] Where did these probabilities come from? This is the first time in the book that we have a two-by-two outcome space and you are given the two marginal distributions and one joint probability, but it is very far from the last. If you have any problems replicating these probabilities from the data given a couple of paragraphs ago, seek assistance immediately!

[b] Of course, when we eliminate the decision variables b_1^n in this fashion, the nonnegativity constraints on these variables must be incorporated as more constraints on the decision variables that we leave in the problem. For example, the constraint $b_1^1 \geq 0$, which is $[Y_1 + rz + \zeta\overline{\theta} - a_1^1\overline{p}_1^a]/p_1^b \geq 0$, is reexpressed as $Y_1 + rz + \zeta\overline{\theta} \geq a_1^1\overline{p}_1^a$.

variables, we can work out our consumer's expected utility of consumption; just for the record, this is a sum of four terms, the first of which (for the first state) is

$$\frac{1}{4}u\left(a_0, b_0, a_1^1, \frac{Y_1 + r[Y_0 - \zeta - p_0^a a_0 - p_0^b b_0] + \zeta\overline{\theta} - a_1^1\overline{p}_1^a}{p_1^b}\right).$$

(This is the probability 1/4 of the first state times our consumer's von Neumann-Morgenstern utility of consumption in the first state, as a function of her decisions. Be sure you could have come up with this term and you know what the other three are.) We have nonnegativity constraints to watch for, and the nonnegativity constraints on tomorrow's broccoli consumption in each of the four states become nonnegativity constraints on the fourth argument of u above and its mates. But given all this, if we have u, we have a perfectly well-posed static optimization problem to solve. This isn't as simple a problem as in the first example; in particular, we aren't able to reduce the problem to a problem with a single budget constraint. But it is still an optimization problem that can be solved without much difficulty.

As in the simpler version of the problem, we are following the basic pattern of reducing the dynamic problem to a static problem of finding an optimal strategy. A strategy here is a seven-tuple (subject to some constraints); for each strategy we know how to evaluate its consequences in terms of overall expected utility; and we assume that our consumer selects what to consume and how to invest today and tomorrow in a way that is consistent with the optimal overall strategy so determined. This is the standard approach to dynamic choice in microeconomic theory.

At this point we can take two paths in further developments and discussion. The usual path is to begin to think about how we might effectively solve optimization problems such as the one just posed. The mathematical technique of *dynamic programming* is especially useful here. An unusual path is to wonder about alternatives to the standard way of modeling dynamic choice. We will proceed in this chapter along the unusual path, on the assumption that you have either already followed or will follow the former. Appendix 2 gives a brief summary of the parts of dynamic programming that are important in this book; all readers should know about finite horizon dynamic programming, which is discussed in the first two sections of appendix 2. And readers who consume all the small type will need to know a bit about solving problems with infinite horizons, which is covered in the second two sections of appendix 2. In

case you wish to test your knowledge of these topics, problem 3 provides the appropriate diagnostic.

4.2. Menus and meals

A context

Rather than work with the dynamic consumption budgeting problem of the previous section, we will work with a simpler, discrete setting. As in the previous section, we look at a two-period problem, with dates $t = 0$ (today) and $t = 1$ (tomorrow). Tomorrow the consumer will find herself at a restaurant, and she will select some meal from the menu that restaurant offers. Today the consumer selects the restaurant, which (effectively) comes down to choosing the menu from which choice will be made tomorrow. To keep matters simple, we will assume that menu items are completely described by the entrée served, and the same entrée at two different restaurants is the same meal. Atmosphere, the chef's skills, etc., all come to naught; the only difference between menus/restaurants is in the menus of meals that they offer.

Formally, we suppose that the set of all possible entrées is given by some finite set X, and the set of all possible menus is the set of all nonempty subsets of X, which we denote by M. Today our consumer has her choice from some subset of M; we write M' for this set of feasible menus. And if she chooses $m \in M'$ today, then tomorrow she chooses some x from m.

This is, to be sure, a very stylized example. But for the examples we wish to present, it captures the salient aspect of dynamic choice: Choice today constrains the *opportunity set* from which one subsequently chooses. (Another aspect of how choice today affects subsequent choice is by changing one's later tastes. This is something we will miss in our examples but which you will see briefly in problem 5.) If the reader's imagination is fired by the toy examples we give next, more realistic applications will not be hard to imagine.

The standard, strategic approach

To develop alternatives to the standard approach, we first see what the standard approach would look like in this context. It is quite simple. We assume that the consumer has preferences defined on the space of "final consumption" X. These preferences will be assumed to satisfy the standard properties and (since X is finite) to be represented by some (ordinal) utility function $U : X \to R$. The consumer, looking at all the menus in M', considers that a strategy for dynamic choice is to choose first an $m \in M'$ and then some $x \in m$, which has consequences x. Thus she has available to herself any meal $x \in X' = \bigcup_{m \in M'} m$. If we let x^* be the meal in X' that has the greatest utility (according to the index U), our consumer plans to choose x^*, and she begins carrying out this plan by choosing any $m \in M'$ that contains x^*. As we said, this is simplicity itself.

We can take a step in this context that we didn't take in the first section and ask what consequences this model of dynamic choice has for our consumer's static choice behavior today. The consumer's choice behavior today is represented quite simply: Comparing any two menus m and m', she strictly prefers m to m' if and only if the U-best meal in m is strictly better than the U-best meal in m'. That is, if we define a function $V : M \to R$ from U by

$$V(m) = \max_{x \in m} U(x),$$

then V is a numerical representation of our consumer's static preferences over menus.

Note that V so defined has the property that if $V(m) \geq V(m')$, then $V(m \cup m') = V(m)$. In words, if m is at least as good as m', then $m \cup m'$ is indifferent to m. Or, in symbols, if we let $\succ$ denote preferences *defined on* M according to V (and we use $\succeq$ for weak preferences and $\sim$ for indifference), we have

$$m \succeq m' \text{ implies } m \sim m \cup m'. \qquad (\clubsuit)$$

Note well, $\succ$ is preferences defined on M; other, closely related preferences are defined on X and represented by U.

Revealed preference and the standard model

Now we can turn the questions we've been asking somewhat on their heads. Following the standard approach to dynamic choice, we've looked at the problem from the point of view of the consumer; what should she do today and subsequently in view of her overall objectives (where it is a maintained assumption that she has overall objectives). Suppose we observe a consumer making choices today and we ask: Does this choice behavior *reveal* that she is acting according to the standard model, or does it refute that model?

Suppose that we are in the enviable position of having quite a lot of data with which to answer this question. Specifically, suppose we know the consumer's preferences $\succ$ defined on M. Since, in the standard model, $\succ$ has a numerical representation V, we know that it is necessary (if the standard model is to apply) that the consumer's preferences are asymmetric and negatively transitive. Moreover, the previous discussion indicates that ($\clubsuit$) is a necessary condition for the standard model. In fact, these conditions are sufficient.

Proposition 4.1. *A consumer's preferences $\succ$ on M are asymmetric, negatively transitive, and satisfy ($\clubsuit$) if and only if the consumer's preferences arise from some function $U : X \to R$ according to the standard model (as sketched in the previous subsection).*[c]

Our statement of this proposition is no model of mathematical elegance, but its meaning should be clear. The standard model generates very specific "testable restrictions" on static choice behavior on M. When we observe individuals who choose today in ways that are inconsistent with those restrictions, we have individuals whose choice behavior is inconsistent with the standard model.

The weak-willed dieter

With this as prologue, we can give our first example of choice behavior that is inconsistent with the standard model. Imagine that X consists of precisely two meals, *fish* and *pasta*. Hence M contains three elements: $\{fish, pasta\}$, $\{fish\}$, and $\{pasta\}$. Consider a consumer whose static choice behavior is given by the preferences

$$\{fish\} \succ \{fish, pasta\} \sim \{pasta\}.$$

This consumer's preferences (on M) satisfy the standard assumptions on M; that is, her strict preferences are asymmetric and negatively transitive. But she strictly prefers the *fish* only menu to the menu that gives her a choice of *fish* or *pasta*. It is obvious that this doesn't conform to ($\clubsuit$), so the standard model doesn't apply.

But our consumer has a very simple explanation: She is on a diet and is weak-willed. Today, as she chooses the menu from which she will later choose a meal, her will is strong and she knows that she should avoid *pasta*. And she knows that when it comes to the event and she must choose a meal, her will will weaken; if *pasta* is available, she will choose it. Hence

[c] You are asked to prove this in problem 4. It isn't hard.

she prefers to choose today in a way that eliminates the opportunity to choose *pasta* tomorrow.

This is an example of what is called *sophisticated choice when a change in tastes is anticipated*. References to the literature in which this sort of model is developed are given in the next section, but it is hard to resist giving an early example. Odysseus, when sailing past the Isle of the Sirens, wishing to hear their song but at the same time wishing not to give in to their spell, had himself lashed to the mast. This behavior, which showed a clear preference for smaller opportunity sets, cannot be explained by the standard model of dynamic choice in economics.

Uncertain future tastes and maintained flexibility

Our second example concerns a case in which three meals are possible, $X = \{chicken, fish, pasta\}$, and a consumer whose immediate preferences over M are given by

$$\begin{aligned}\{chicken, fish, pasta\} &\succ \{chicken, fish\} \sim \{chicken, pasta\} \\ &\sim \{fish, pasta\} \succ \{chicken\} \\ &\succ \{fish\} \succ \{pasta\}.\end{aligned}$$

Again we have static (strict) preferences that are asymmetric and negatively transitive. And again we find (♣) violated; $\{chicken\} \succ \{fish\}$, and so if (♣) were to hold, we would require that $\{chicken\} \sim \{chicken, fish\}$. Indeed, if (♣) holds, then some single-element set would necessarily be indifferent to the menu containing all the options. And that certainly doesn't hold here.

The explanation offered by our consumer in this case is again a simple story. She is uncertain what her tastes will be when she sits down to dine tomorrow. Perhaps she will like *chicken* most of all, then *fish*, and then *pasta*. But perhaps her preferences will be *fish*, then *chicken*, and then *pasta*. And perhaps they will be *pasta*, then *chicken*, and then *fish*. She believes that each of these is as likely as any other and she evaluates a menu as follows: Given the menu and her (then) preferences, she gets three utils if her first choice is available, two utils if her second choice is the best available, and only one if all she can have is her third choice. If you compute "expected utils" by this scheme, you will come to the preferences above.

The reader is entitled to object that we are introducing uncertainty into the story surreptitiously. Had we mentioned this uncertainty about future preferences at the start, we wouldn't have expected (♣) to be necessary and sufficient for the standard model. This objection is accepted, but

then let us turn the question around: Which preference relations $\succ$ on M can we explain by the "standard model" where we allow the endogenous incorporation of this sort of uncertainty?

Proposition 4.2. *Assume X is finite. A (strict) preference relation $\succ$ on M is asymmetric and negatively transitive and, for the associated weak preference and indifference relations $\succeq$ and $\sim$, satisfies*

$$m \supseteq m' \text{ implies } m \succeq m', \text{ and} \tag{$\star$}$$

$$m \supseteq m' \text{ and } m \sim m' \text{ implies } m \cup m'' \sim m' \cup m'' \text{ for all } m'', \tag{$\star\star$}$$

if and only if there is a finite set S and a function $u : S \times X \rightarrow R$ such that

$$m \succ m' \text{ if and only if } \sum_{s \in S} \max_{x \in m} u(s,x) > \sum_{s \in S} \max_{x \in m'} u(s,x).$$

We won't attempt to prove this here; it is here only for purposes of illustration and doesn't reappear later in the book.[2] But let us interpret it. We start with preferences $\succ$ on menus that, when you think of menus as the basic objects, satisfy the standard assumptions of chapter 2. In addition, our consumer always (at least weakly) prefers to leave herself with more choices (which is ($\star$)). And if more choices are of no value in some situation, adding more options "to both sides" doesn't make those original extra options valuable. (This is a rough description of ($\star\star$).) Then (and only then) can we justify this consumer's preferences over menus by a model with preferences over meals that depend on some implicit "states of future preferences" and where preferences for menus are obtained by adding up the utility of the "best" meal the menu provides in each state.

4.3. Bibliographic notes and discussion

To repeat from before, the standard approach to dynamic choice consists of reducing dynamic choice to a problem of static choice of an optimal strategy under the presumptions that (1) the consumer at each point in time has coherent preferences over overall outcomes; (2) she believes at each point in time that these preferences will not shift with time or circumstances; (3) she is smart enough to work out the consequences of her

[2] For a proof, see Kreps (1979).

choices, so she can find an optimal strategy according to the preferences assumed in (1); and (4) the second presumption is correct in the sense that the consumer subsequently carries out the optimal strategy she is assumed to find. This approach is ingrained in almost all microeconomic theory, so much so that it is often employed in textbooks without any discussion whatsoever. This is something of a retreat from the careful consideration given to these issues in seminal work on the theory of choice; see, for example, the discussion and justification of the standard approach given by Savage (1972, 15-17).

It is probably safe to assume that, for at least some consumers, tastes and desires shift as time passes; (4) is likely to be false. Grant for the moment that a consumer for whom this is so has coherent preferences at each point in time. That is, grant (1). And grant that she can work out the consequences of her actions, or (3). Granting (1) and (3), one can wonder whether she acts under the "false" presumption of (2) or not. If she assumes (2) and acts accordingly, then her static behavior at each point in time will look sensible according to the standard approach to dynamic choice, but her dynamic actions will be inconsistent. On the other hand, if she is sophisticated enough to deny (2), and if her current tastes govern her actions, she may act as does our weak-willed dieter, so that her static choice does not conform to the standard model. These sorts of issues are discussed and attacked generally in the literature on changing tastes. A sampler of this literature (taking various positions on the value of such models) would include Strotz (1955-56), Peleg and Yaari (1973), Hammond (1976), Stigler and Becker (1977), and Thaler (1980).

Suppose that we rejected assumption (3) of the standard approach — that individuals are able to foresee all the consequences of their actions. How then can we possibly model dynamic choice behavior? Consider the following general scheme: (a) Model choice at each given point of time according to the basic models of choice of chapters 2 and 3, with strict preferences that are asymmetric and negatively transitive, and so on. But model choice at a given point in time as choice over things that are chosen *at that time only* so that, for example, one incorporates in the description of an item chosen the opportunities that current choice leaves for later choice. When doing this, (b) pay attention to how the individual's bounded rationality might affect what static choices she does make, especially if she is aware of her own limitations. And then, (c) consider how choice behavior at different points in time might or might not fit together.

Our second nonstandard example is somewhat of this character. We can imagine that our consumer, choosing a menu, can't conceive of all the consequences of her choice of menu. But still, at the level of selecting a menu, she has "standard preferences." That is, she satisfies (a). Because of her inability to conceive of all the consequences of her choice of menu, she generally prefers to leave herself with more flexibility. That is, her preferences over menus satisfy (*) of proposition 4.2, as an example of what we intend for (b). Then if her static preferences conform as well to (**), we have from the proposition a fairly nice (and quite standard looking) representation of her preferences over menus, a representation that was originally suggested by Koopmans (1964). And we could imagine considering how her static preferences at different points of time might evolve.

If we admit the possibility that the consumer is limitedly rational in the sense the she cannot foresee all the consequences of her current decisions, this in itself might lead us to reject the approach just sketched on grounds that *even* her static choice behavior might fail to conform to the usual assumptions; her inability to work out strategic plans in a complex, dynamic environment leads her immediate choice by rules of thumb (or serendipity) that cannot be represented by a single numerical index. Even at this level one can pursue interesting models of static (and then dynamic) choice: For example, Bewley (1986) suggests a model where once a consumer has chosen a strategic plan of action, she will deviate from that plan (as unexpected options arise) only if some alternative seems substantially better than the status quo plan. Bewley formalizes this with a model where static weak preferences are not complete and with an "inertia" assumption that gives precedence to the status quo, and he considers how this would affect dynamic choice behavior.

Having mentioned these possible alternatives to the standard approach to dynamic choice, we will follow the standard approach until the end of the book. This discussion is meant to accomplish two things: To make you aware of the strong assumptions that go into the standard approach; and to indicate that interesting alternatives to the standard approach are possible. These alternatives are, at the time that I write this, very far from mainstream microeconomic theory. But when we delve later in the book into the serious deficiences of the techniques of the current mainstream theory, we will return to issues in dynamic choice and to the very strong assumptions of the standard approach.

References

Bewley, T. 1986. "Knightian Decision Theory: Part I." Cowles Foundation discussion paper no. 807, Yale University. Mimeo.

Hammond, P. 1976. "Changing Tastes and Coherent Dynamic Choice." *Review of Economic Studies* 43:159–73.

Koopmans, T. 1964. "On the Flexibility of Future Preferences." In *Human Judgments and Optimality*, M. Shelly and G. Bryan, eds. New York: John Wiley and Sons.

Kreps, D. 1979. "A Representation Theorem for Preference for Flexibility." *Econometrica* 47:565–77.

Peleg, B., and M. Yaari. 1973. "On the Existence of a Consistent Course of Action when Tastes Are Changing." *Review of Economic Studies* 40:391–401.

Savage, L. [1954] 1972. *The Foundations of Statistics*. New York: John Wiley and Sons. Rev. and enl. ed. New York: Dover Publications.

Stigler, G., and G. Becker. 1977. "De Gustibus Non Est Disputandum." *American Economic Review* 67:76–90.

Strotz, R. 1955–1956. "Myopia and Inconsistency in Dynamic Utility Maximization." *Review of Economic Studies* 23:165–80.

Thaler, R. 1980. "Toward a Positive Theory of Consumer Choice." *Journal of Economic Behavior and Organization* 1:39–60.

4.4. Problems

■ 1. (a) Consider the first example, in which the consumer must decide how much artichoke and broccoli to consume today and tomorrow and how much to save (or borrow) today, where there is no uncertainty. Suppose the consumer's preferences are given by the utility function

$$U(a_0, b_0, a_1, b_1) = \ln(a_0) + .9\ln(b_0) + .8\ln(a_1) + .7\ln(b_1).$$

Solve this consumer's dynamic consumption–savings problem, as a function of prices p_0^a, p_0^b, p_1^a, and p_1^b, the gross return r on loans to/from the bank, and the consumer's incomes Y_0 and Y_1. Recall that we assume that the consumer can borrow or lend at gross return r.

(b) Suppose we assumed that the consumer could only lend money at rate r but cannot borrow at all. What is the solution in this case?

■ 2. Suppose in the problem described in problem 1(a) that the consumer is able to store vegetables from today to tomorrow. Assume there is no spoilage or anything like that — vegetable put in the refrigerator today and taken out for consumption tomorrow is just as good as vegetable bought fresh tomorrow. What is the solution to the consumer's dynamic consumption-savings problem?

■ 3. *Part (a) tests your knowledge of finite horizon dynamic programming. If you cannot do this problem using the techniques of dynamic programming, consult the first two sections of appendix 2. (The analysis of this problem is given there in detail, so try the problem before looking at the appendix.) Part (b) tests your knowledge of infinite horizon dynamic programming. This is a good deal more difficult, and you will only need the skills required to solve this for some optional material in later chapters. But if you are curious, this problem is worked out in detail in the second two sections of appendix 2.*

(a) A consumer lives for three periods, denoted by $t = 0, 1, 2$, and consumes artichoke and broccoli on each day. We let a_t be the amount of artichoke consumed on day t and b_t be the amount of broccoli. This consumer has preferences over lotteries on consumption bundles (where a consumption bundle is a six-tuple $(a_0, b_0, a_1, b_1, a_2, b_2)$), which satisfy the von Neumann-Morgenstern axioms and are represented by the von Neumann-Morgenstern utility function

$$U(a_0, b_0, a_1, b_1, a_2, b_2) = (a_0 b_0)^{.25} + .9(a_1 b_1)^{.25} + .8(a_2 b_2)^{.25}.$$

This consumer can buy artichoke and broccoli in the marketplace each day. Because both vegetables spoil rapidly, what he buys on any given day is his consumption for that day. (He is constrained to consume nonnegative amounts of each vegetable.) The price of artichoke is \$1 per unit on each and every day. The price of broccoli is more complex: It begins as \$1 per unit of broccoli at $t = 0$. But at $t = 1$, it is either \$1.10 or \$.90, each of these being equally likely. And at $t = 2$, the price of broccoli is again random and depends on the price the day before: If its price was \$1.10 at $t = 1$, then it is either \$1.20 or \$1.00, with each equally likely. If its price was \$.90 at $t = 1$, then at $t = 2$ it is either \$.98 or \$.80, each equally likely. At date $t = 0$, the consumer has no information (beyond what is given above) about subsequent prices of broccoli. At date $t = 1$, he knows the current price of broccoli and no more. At date $t = 2$, he knows the current price.

This consumer has \$300 to spend on artichoke and broccoli over the three days, which he can divide any way he wishes. Any money he doesn't spend on a given day sits in his pocket where it earns zero interest.

(b) Imagine that our consumer eats at times $t = 0, 1, 2, \ldots$. At each date, he consumes both artichoke and broccoli. His utility function, defined for an infinite stream of consumption $(a_0, b_0, a_1, b_1, \ldots)$ is

$$U(a_0, b_0, a_1, b_1, \ldots) = \sum_{t=0}^{\infty} (.95)^t (a_t b_t)^{.25}.$$

This consumer starts out with wealth \$1,000, which he can spend on vegetable or can deposit in his local bank. Any funds deposited at the bank earn interest at a rate of 2% per period, so that \$1 deposited at time t turns into \$1.02 at time $t+1$.

The price of artichoke is fixed at \$1. The price of broccoli is random. It begins at \$1. Then, in each period, it either increases or decreases. Being very specific, if the price of broccoli at date t is p_t,

$$p_{t+1} = \begin{cases} 1.1p_t & \text{with probability .5, and} \\ .9p_t & \text{with probability .5.} \end{cases}$$

The consumer wishes to manage his initial wealth \$1,000 in a way that maximizes his expected utility. How should he do this?

■ 4. Prove proposition 4.1.

■ 5. Consider the consumer in the first example of this chapter, where we specialize to the case in which $p_0^a = p_0^b = p_1^a = p_1^b = r = 1$, and $Y_0 = Y_1 = 3$. Suppose that this consumer is a standard sort of consumer (in microeconomic theory), with preferences given by the utility function

$$U(a_0, b_0, a_1, b_1) = \ln(a_0) + \ln(b_0) + \ln(a_1) + \ln(b_1).$$

(a) What is the solution to this consumer's consumption-savings problem, under the assumption that she is a completely normal consumer?

But this is not a normal consumer. She is quite abnormal, in that she is susceptible to becoming addicted to broccoli! Specifically, if she eats (a_0, b_0) today, then tomorrow when she goes to purchase vegetables, she chooses according to the utility function

$$U^1(a_1, b_1; a_0, b_0) = \ln(a_1) + b_0 \ln(b_1).$$

Note well what is going on here; the more broccoli she consumes today, the more she values broccoli consumption tomorrow.

(b) Assume that this consumer seeks to consume and save today in a fashion that will maximize her U-utility over the consumption bundle x that she chooses dynamically, and she realizes that tomorrow she will spend her resources in a way that maximizes her U^1-utility. What does she (optimally) consume and save today, and what is the overall outcome that she achieves?

chapter five

Social choice and efficiency

So far we have discussed the choice behavior of a single consumer, both in general and in market settings. For the rest of this book, we are interested in what happens when many consumers are choosing simultaneously. Each consumer will come with the sorts of preferences we described in chapters 2 and 3, and each attempts to choose in an optimal fashion given those preferences. It will typically happen that the diverse preferences of consumers will be in conflict with each other. (For example, each consumer typically wants "more" at the expense of the consumption of his fellow consumers.) We will be looking at various *institutional arrangements* by which those conflicts are meant to be resolved. But before moving to institutional arrangements, we look briefly at the abstract *theory of social choice.*

The theory of social choice is concerned with the selection of some *social outcome* that affects a number of individuals, when those individuals have diverse and conflicting preferences. We try to characterize desirable properties of the outcome that is to be selected, properties that usually involve (and blend) notions of *efficiency* and *equity*. In some cases, those properties are insufficient to fix on a single social outcome, but instead restrict attention to some subset of the feasible outcomes. Adding more properties can sometimes pin down a single social outcome that has all the desired attributes. And in some contexts we find that there is no way to satisfy simultaneously all the properties thought to be desirable.

We will not do justice to this very broad topic here. In this chapter we will acquaint you with a few of the basic notions, with emphasis on the concept of efficiency, and we will give a smattering of the classic results. But we only scratch the surface.

5.1. The problem

We begin by formulating the problem in a very general setting. A finite number of individual consumers, indexed by $i = 1, 2, \ldots, I$, make up a society. Each member x of a set X of social outcomes describes how every member of this society is treated. Each consumer has preferences over the possible social outcomes. We assume that these preferences can be represented numerically, and we let $V_i : X \to R$ represent i's preferences.

We are interested, ultimately, in answering the question: For each subset $X' \subset X$ of *feasible social outcomes*, which social outcome(s) should be selected? Following our general approach to choice, along the way we will be making binary comparisons — trying to say when one outcome is better than another.[a]

Utility imputations

Fixing specific numerical representations for the preferences of each consumer, to each social outcome x there corresponds a vector of utilities $v = (V_1(x), V_2(x), \ldots, V_I(x)) \in R^I$ for the I consumers. We call such a vector of utility levels a *utility imputation*. Given a set X' of feasible social outcomes, there is a corresponding set of feasible utility imputations. For each social outcome x, we write $V(x)$ for the I-dimensional vector $(V_1(x), V_2(x), \ldots, V_I(x))$, and for a set X' of feasible social outcomes, we will write $V(X')$ for the set of *feasible utility imputations*:

$$V(X') = \{v \in R^I \ : \ v = V(x) \text{ for some } x \in X'\}.$$

Pictures of feasible utility imputations for the case of two consumers will be used below, so we give a couple of examples in figure 5.1. In 5.1(a) the

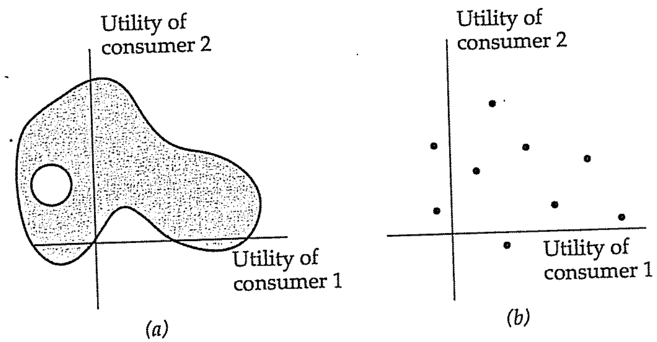

Figure 5.1. Feasible utility imputations for two-consumer societies.

[a] Be a bit sceptical here. Just because individual choice is typically modeled as being driven by such binary comparisons is no reason to suppose that when it comes to social choice this approach ought to be taken. We will discuss this point.

shaded area gives one typical picture, while the finite collection of points in figure 5.1(b) corresponds to a case in which the set X' is finite. Of course, these pictures are all relative to specific numerical representations of the preferences of individual consumers; as we change those representations, we make corresponding changes in pictures such as those in figure 5.1.

Exchange of commodities

The general problem just described may be a bit too abstract to think about, so let us sketch a more concrete setting. Imagine K commodities; write Z for the positive orthant in R^K; and assume that each consumer i consumes some bundle $z \in Z$. Exactly as in chapter 2, imagine that consumer i has preferences for his own levels of consumption, given by a utility function $U_i : Z \to R$.

A social outcome, in this case, is a vector $x = (z_1, \ldots, z_I) \in X = Z^I$, which specifies the bundle z_1 that consumer 1 consumes; the bundle z_2 that consumer 2 consumes, and so on. Note that each z_i is itself a (nonnegative) K-dimensional vector.

The preferences of consumer i over social outcomes x in this case are represented by the function V_i, with domain X, defined by

$$V_i(x) = U_i(z_i), \text{ where } x = (z_1, \ldots, z_I).$$

Note well, in this specialization the preferences of consumer i depend by assumption only on what consumer i consumes; consumer i's preferences do not depend on what is consumed by anyone else. In our general formulation of the problem this assumption is not imposed, and it may seem to you a pretty silly assumption (unless you have taken economics courses before and so are properly socialized into the discipline). But this is an assumption that will play a critical role in developments in chapter 6.

As for the set of feasible social outcomes in this specification, it is typical to imagine that society has at its disposal a given stack of consumption goods, called the *social endowment* and represented by an element $e \in Z$, which can be divided among the I consumers. So the set of feasible social outcomes, in this context called the set of *feasible allocations* (of society's resources), is

$$Z' = \{x = (z_1, \ldots, z_I) \in Z^I \ : \ z_1 + \ldots + z_I \leq e\}.$$

Note that the sum of what is allocated to the consumers is required to be *less or equal to* the social endowment; if one or more of the goods is noxious and we can't dispose freely of this good, then we might wish to replace the inequality with an equality.

The Edgeworth box

For the case of two consumers and two goods, a standard picture of this situation is drawn, known as an *Edgeworth box*. We label the two consumers 1 and 2 and the two goods a and b, and we write $x = ((z_{1a}, z_{1b}), (z_{2a}, z_{2b}))$. Consider figure 5.2. In figure 5.2(a), we represent the preferences of consumer 1 by the usual indifference curve diagram. Note the heavy dot; this is meant to be the location of the social endowment $e = (e_a, e_b)$.

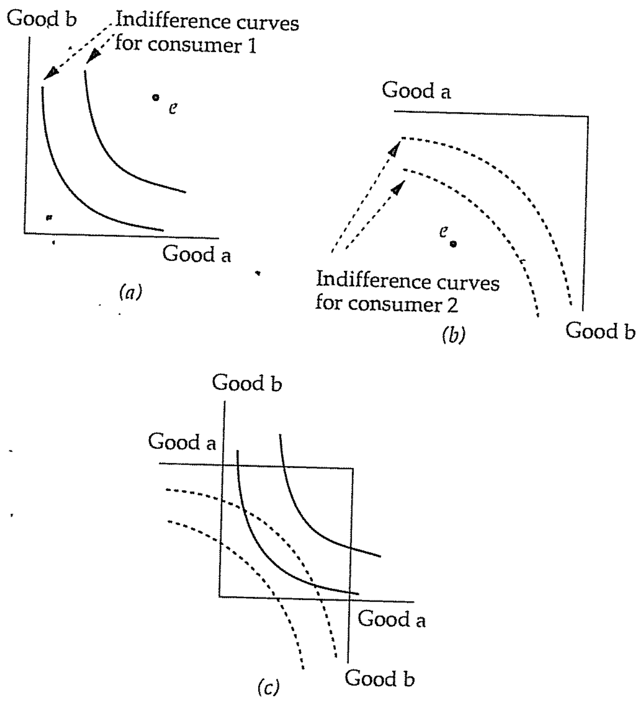

Figure 5.2. Edgeworth boxes.
Figures 5.2(a) and (b) give the two sets of data that make up an *Edgeworth box:* the social endowment and the indifference curves of the two parties, in (a) for consumer 1 and in (b) for 2. In figure 5.2(c) the indifference curves are superimposed.

In figure 5.2(b) the preferences of consumer 2 are represented by the usual indifference curve diagram, except that we have rotated the usual picture by 180 degrees and have put the origin "up" at the level of e in figure 5.2(a), so the social endowment e in this picture is down at the level of the origin in figure 5.2(a).

Given e, what are the feasible allocations? They are all allocations x such that $z_{1a} + z_{2a} \leq e_a$ and $z_{1b} + z_{2b} \leq e_b$. Since (according to figure 5.2) our consumers always prefer more of either good to less, let us restrict attention to allocations where none of the social endowment is wasted — where the two inequalities above are equalities. Then given z_{1a} and z_{1b} (and holding e fixed), we know that $z_{2a} = e_a - z_{1a}$ and $z_{2b} = e_b - z_{1b}$.

In figure 5.2(c), we have all this recorded very neatly and compactly. First, note that we have superimposed figures 5.2(a) and (b), locating the origin of consumer 2's coordinate system at the point e in consumer 1's coordinate system so that e in 2's coordinates is the origin in 1's coordinate system. Then the box formed by the two sets of axes represents all the feasible allocations of e (assuming no wastage of the social endowment); each point (z_{1a}, z_{1b}) in 1's coordinate system is at the same time the point $(e_a - z_{1a}, e_b - z_{1b})$ in 2's coordinate system.

5.2. Pareto efficiency and optimality: Definitions

Return to the general setting of the first part of section 5.1. That is, there is a set X of social outcomes and I consumers whose preferences over the social outcomes are represented by functions V_i. We begin with pairwise comparisons of two social outcomes x and x'.

Definitions. *Outcome x is said to be* **Pareto superior** *to outcome x' if $V_i(x) \geq V_i(x')$ for every consumer i, with a strict inequality for at least one i. Outcome x is said to be* **strictly Pareto superior** *to x' if $V_i(x) > V_i(x')$ for all i.*

Or, in words, one outcome is Pareto superior to another if no consumer finds the first any worse than the second and at least one consumer finds the first strictly better. And the first is strictly Pareto superior to the second if every consumer strictly prefers the first. In such cases, we also say that the second outcome is (strictly) Pareto *inferior* to the first.

It should be carefully noted that Pareto superiority is a *partial ordering* of social outcomes, in general. That is, we might have two social outcomes x and x' such that some consumers strictly prefer x to x' and others strictly prefer x' to x. In such cases, we would say that x and x' are Pareto *incomparable*.

It is easy to see Pareto superiority and inferiority in pictures of utility imputations for two consumers such as figure 5.1. The outcome x is Pareto superior to x' if the corresponding utility imputation $V(x)$ is above and/or to the right of $V(x')$. And x is strictly Pareto superior to x' if $V(x)$ is above *and* to the right of $V(x')$.

Definitions. *Given a set X' of feasible social outcomes, an outcome $x \in X'$ is said to be* ***Pareto efficient*** *or Pareto optimal in X' if no other feasible outcome $x' \in X'$ is Pareto superior to x. The subset of Pareto efficient outcomes in X' is called the* ***Pareto frontier*** *of X'.*

In the definition of Pareto efficiency, a fine point arises concerning whether Pareto inferiority or strict Pareto inferiority should be the criterion by which a point is disqualified. That is, suppose we have a point x which is Pareto dominated by some other feasible point x', but which is not strictly Pareto inferior to any feasible point. Should we say that x' is Pareto efficient? In the definition we have given, we do not. But in other treatments of this subject, such a point would be included. This is a place where you have to watch carefully the definitions that are made.

Pareto efficiency is easy to see in pictures of utility imputations. Consider the sets of feasible utility imputations in figure 5.3. These are just the same as in figure 5.1. In both 5.3(a) and (b) we show a utility imputation marked v', which is not Pareto efficient, and another marked v, which

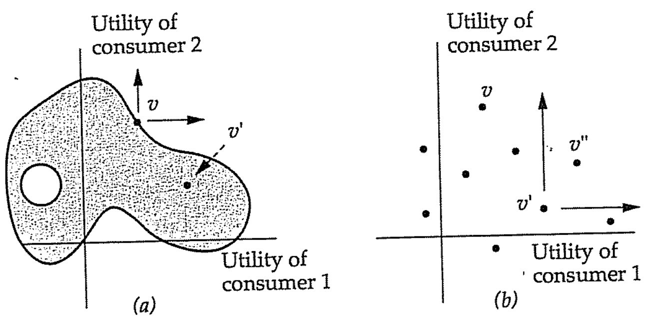

Figure 5.3. Pareto efficiency in the space of utility imputations.
In each figure, the point marked v is Pareto efficient and the point marked v' is not.

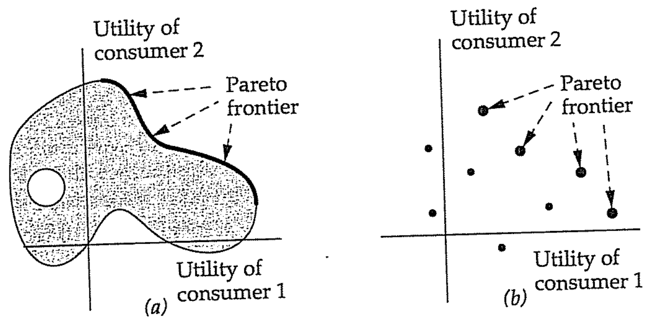

Figure 5.4. Pareto frontiers in the space of utility imputations.
In (a), the Pareto frontier is the heavy curve. In (b), points on the Pareto frontier are the heavier dots.

is.[1] In each case there is a simple test. From any utility imputation, say v in 5.3(a), draw a "positive orthant" with origin at v. If no feasible utility imputation falls in this orthant (including its edges), then v is Pareto efficient. In figure 5.3(b) we do the same thing for v', and we see, by virtue of the point v'', which is Pareto superior to v', that v' is not Pareto efficient. In figure 5.4 we draw in (with a heavy curve in [a], and with heavier dots in [b]), the two Pareto frontiers.

Pareto efficiency and the Edgeworth box

To illustrate these notions, consider the Edgeworth box depicted in figure 5.5(a). Consider in particular the points x and x' marked there. From the point x', it is possible to find points that give more utility to each of the two consumers — the set of points for which this is true is the set of points in the shaded lens-shaped area. On the other hand, the point x, which lies at a point of tangency of the two consumer's indifference curves, is Pareto efficient. To make one consumer better off (staying along or above the consumer's indifference curve through x), you must go into a region that is below the indifference curve of the other consumer through x. Thus x is Pareto efficient and x' is not.

We draw the corresponding pictures of utility imputations in figure 5.5(b). We show both $v' = V(x')$ and $v = V(x)$. Note that v is on the Pareto frontier and v' is not. (Test your understanding of this picture by answer-

[1] We will work simultaneously with social states and their utility imputations. And we will abuse the language and use terms such as *Pareto efficiency, Pareto superiority,* and *Pareto frontier* both for social states and for their utility imputations.

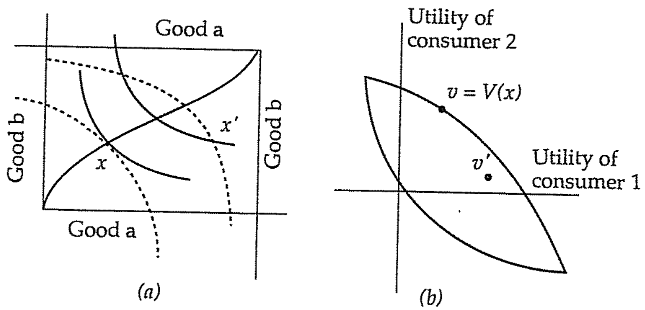

Figure 5.5. Pareto efficiency and Edgeworth boxes.
In (a), we show a Pareto efficient point x, an inefficient point x', and (the heavy line) the Pareto frontier or contract curve. The points v and v' in utility imputation space corresponding to x and x' are shown in (b).

ing: Where in figure 5.5[b] do you find utility imputations corresponding to the shaded lens shaped area in figure 5.5[a]?)

Finally, returning to figure 5.5(a), note the heavy line passing through x. This is the set of points at which the indifference curves of the two consumers are tangent; hence this is the Pareto frontier for this Edgeworth box. When dealing with Edgeworth boxes, the Pareto frontier is often called the *contract curve*.

One variation on the picture in 5.5(a) should be mentioned. We have characterized the contract curve as the set of points in the Edgeworth box for which the two indifference curves through the points are tangent. You should try to draw a picture where this characterization doesn't work, because the point in question is against a boundary of the box, corresponding to a nonnegativity constraint for the consumption of some good by some consumer. In general, the contract curve is defined to be those points that are Pareto efficient in the box; the "tangent indifference curves" characterization is only used informally.

5.3. Benevolent social dictators and social welfare functionals

Imagine that you, an outsider (not one of the consumers), have been appointed dictator for a given society; it is up to you to select a social outcome from among those feasible. We will suppose that you wish to be

a benevolent dictator, and we explore what might be said about how you should choose.

An obvious first question is whether we should expect that your choice can be rationalized by some (asymmetric and negatively transitive) preference relation $\succ$ on the set X of social outcomes. To see why this might not be such a reasonable thing, suppose there are three individuals in this society and two feasible outcomes: $X' = \{x, x'\}$. Suppose that two of the consumers strictly prefer x to x' and one strictly prefers x' to x. You might worry about the relative strengths of these strict preferences, but such interpersonal comparisons may be difficult to make, and so you might reasonably suppose that you should select x. But now imagine that a third option is available, x''. The first two consumers rank $x \succ x' \succ x''$ while the third ranks $x' \succ x'' \succ x$. Since everyone prefers x' to x'', it seems unreasonable that you would select x''.[2] But is it now reasonable to suppose that you will take x over x'? The outcome x' is no worse than the second choice of each individual, whereas x is the third consumer's worst alternative. It certainly isn't clear that x' should be chosen in this case, but neither is it clear that x should be chosen; it is certainly less clear that x should be chosen than it was when only x and x' were available.[3]

Nonetheless, if your choice behavior (as benevolent dictator) can be rationalized by a preference relation, a choice of x in the first instance necessarily implies a choice of x (assuming x'' isn't chosen) in the second; this is an easy consequence of Houthakker's axiom of revealed preference.

Objections concerning our discussion of this example are easy. Perhaps you find x the obvious choice in the second case, and (what I personally find more compelling) perhaps x is not so obvious a choice in the first instance; as a benevolent social dictator, you (perhaps) feel that you must somehow consider intensity of preferences. Nonetheless, we proceed on the basis of the following assumption, although you may wonder if this assumption is at all sensible in this context.

Assumption 1. *The social dictator's choice behavior can be rationalized by an asymmetric and negatively transitive strict preference relation $\succ$ on X. Moreover, $\succ$ is sufficiently well behaved so that it admits a numerical representation $V^* : X \to R$.*

[2] We'll formalize this restriction on your choice behavior in just a moment; for now we are only dealing with an example.

[3] Your intuition might be that x should be prefered to x' in both cases because x is preferred to x' in both cases by a majority of society. We will discuss majority rule below.

The last part of assumption 1 follows immediately if X is a finite set. But if X is infinite, we may need something more, such as continuity of $\succ$.[b]

To this we add two assumptions that relate your preferences as social dictator to those of the members of the society.

Assumption 2. *If x is Pareto superior to x', then $x \succ x'$ or, equivalently, $V^*(x) > V^*(x')$.*

Assumption 3. *If $V_i(x) = V_i(x')$ for all $i = 1, 2, \ldots, I$, then $V^*(x) = V^*(x')$.*

Assumption 2 is virtually an "axiom" of benevolence, *if* you are willing to accept that members of society are entitled to their own tastes. Given this, how could one argue with the assertion that if one social outcome is Pareto superior to a second — if everyone in society at least weakly prefers the first, and some strictly prefer it — then the dictator, if she is benevolent, should prefer the first? We could imagine a slightly weaker assumption, where it is necessary that x is strictly Pareto superior to x' before we conclude $x \succ x'$, but assumption 2 as stated seems pretty reasonable.

Assumption 3, in a sense, pushes assumption 2 to a "limit."[c] If everyone is indifferent between x and x', then what business does a benevolent social dictator have in choosing between them? One might retort: If every consumer is indifferent between x and x', then no one will mind if the dictator chooses according to whim or her own tastes. But we will proceed assuming this as well.

Proposition 5.1. *Fix numerical representations of the consumers' preferences V_i. Then the social dictator's preferences satisfy assumptions 1, 2, and 3 if and only if the social dictator's preferences are represented by a function $V^* : X \to R$ that takes the form*

$$V^*(x) = W(V(x))$$

for some function $W : R^I \to R$ that is strictly increasing on the range of the vector function $V = (V_1, \ldots, V_I) : X \to R^I$.[4]

[b] We won't spend time on continuity of the preferences of the social dictator, but you might think about whether this is reasonable if the preferences of the consumers in this society are not continuous.

[c] Suppose that social outcomes come from some convex subset of a finite dimensional Euclidean space, and for each x and $\epsilon > 0$, there is some $\hat{x}$ that is Pareto superior to x and within ϵ of x. This would hold, for example, if social outcomes are consumption allocations, individual's preferences depend only on what they consume, and at least one consumer is locally insatiable. Suppose, moreover, that the dictator's preferences are continuous. Then assumption 3 follows from assumption 2 (and continuity) in precisely the sense of a limit.

[4] Recall that $V(x)$ is shorthand for the vector $(V_1(x), \ldots, V_I(x))$.

The bit about "strictly increasing..." is less ferocious than it may seem. The idea is that if we have x and x' from X such that $V_i(x) \geq V_i(x')$ for all i, with a strict inequality for at least one i, then $W(V(x)) > W(V(x'))$.

Proving one direction of this proposition is easy. Suppose that for some given W with the postulated properties we define V^* by $V^*(x) = W(V(x))$ and then we define the dictator's preferences from V^*. Showing that these social preferences satisfy assumptions 1, 2, and 3 is a simple matter of comparing definitions. The other direction (going from the assumptions to the existence of such a W) is a bit harder, but only a bit. Assumption 1 guarantees that some V^* represents $\succ$. For each vector $r \in R^I$ such that $r = V(x)$ for some $x \in X$, define $W(r)$ to be $V^*(x)$. Then W is well-defined by assumption 3, and it is strictly increasing on the range of V by assumption 2. One can extend W to all of R^I in any way you please. One might wish W to be extended to be increasing or even strictly increasing on all of R^I, and the mathematically adept reader might find it interesting to consider whether either or both of these wishes can be fulfilled in general.

Note well that the function W defined on R^I depends crucially on the particular V_i that are chosen to represent the preferences of individual consumers. We could, for example, replace V_1 with a function V_1' given by $V_1'(x) = (V_1(x) + 1000)^3$ (which is a strictly increasing transformation), and then we would have to change how W responds to its first argument. Still, the notion is simple. Fixing representations of our consumer's preferences, every "reasonable" benevolent social dictator has preferences given by a strictly increasing function W on the space of utility imputations, and every such function W defines a "reasonable" benevolent social dictator, where the definition of reasonable is that the dictator conforms to assumptions 1, 2, and 3.

Functions W of this sort are sometimes referred to as *social welfare functionals*. Standard examples are

(a) $W(r_1, \ldots, r_I) = \sum_{i=1}^I r_i$, the so-called *utilitarian* social welfare functional

(b) $W(r_1, \ldots, r_I) = \sum_{i=1}^I \alpha_i r_i$ for a set of strictly positive weights $\{\alpha_i\}$, the so-called *weighted utilitarian* social welfare functional. This class of functions is also referred to as the class of *Bergsonian social welfare functionals*. They will reappear later.

In example (a), all members of society are given equal weight.[d] In example (b), we may possibly weight consumers differently. We might agree, for example, that it would be sensible to give the utility of professors ten

[d] Of course, this statement is not quite sensible, since this is all relative to some fixed numerical representation of consumer's preferences.

times the weight given to the utility of undergraduates, who in turn get ten times the weight given to the utility of graduate students.[5]

As an alternative way to weight utilities, you might consider making the weight given to a consumer depend on the relative utility the consumer receives.

(c) Let $\alpha_1 \geq \alpha_2 \geq \ldots \geq \alpha_I \geq 0$ be a nonincreasing sequence of positive weights, and for any vector $(r_1, \ldots, r_I)$, let $r_{[i]}$ be the ith smallest number in the set $\{r_1, \ldots, r_I\}$. (That is, if $I = 4$ and $r = (4,5,4,3)$, then we have $r_{[1]} = 3, r_{[2]} = r_{[3]} = 4$, and $r_{[4]} = 5$.) And set $W(r_1, \ldots, r_I) = \sum_{i=1}^I \alpha_i r_{[i]}$. This gives relatively more weight to the "worse off" members of society.

(d) To take an extreme case of (c), let $W(r_1, \cdots, r_I) = \min\{r_i : i = 1, \ldots, I\}$. That is, the worst off member of the society gets all the weight. This isn't quite going to give a social welfare functional as we've defined the term, since it isn't strictly increasing. (It gives rise to a benevolent social dictator for whom assumption 2 is replaced by the weaker assumption: If x strictly Pareto dominates x', then $x \succ x'$.) This (almost-) social welfare functional is sometimes called a *Rawlsian social welfare functional*.

The range of possible choices of a social welfare functional

What are we to make of these social welfare functionals? Put another way, what conclusions can we draw if we believe that social choice is determined by some preference ordering that satisfies assumptions 1, 2, and 3?

Proposition 5.2. *Suppose that choice out of any set X' of feasible social states is made by maximization of a function $W(V_1(x), \ldots, V_I(x))$, where W is strictly increasing in each argument. Then the chosen social state will always be Pareto efficient (in X').*

There is almost nothing to this; it is virtually a matter of comparing definitions. So we can rephrase the question: Are there any Pareto efficient outcomes that are not selected by some suitably chosen social welfare functional? For all practical purposes the answer to this question is no. In fact, we can, under certain conditions, get a virtually negative answer if we restrict attention to Bergsonian social welfare functionals.

To see how this works, we need the mathematical notion of the *convex hull* of a set. For any set of points A of R^K (for any integer K), the convex hull of A is the set of all points $z \in R^K$ such that $z = \sum_{n=1}^N \alpha_n z_n$ for some integer N, some selection of points $z_n \in A$, and some selection of scalars

[5] You may decide how much weight is appropriate for assistant professors.

$\alpha_n > 0$ such that $\sum_{n=1}^N \alpha_n = 1$. Don't be put off by the symbols; the convex hull of any set is just the smallest convex set that contains the original set. It is obtained by "filling in" any holes or concave pieces in the set.[6]

Proposition 5.3. *Suppose x^* is Pareto efficient in some set X' and the utility imputation corresponding to x^*, or $V(x^*)$, is Pareto efficient in the convex hull of $V(X')$.[7] Then for some selection of **nonnegative** weights $\{\alpha_i\}_{i=1,\ldots,I}$, at least one of which is positive, x^* maximizes $\sum_{i=1}^I \alpha_i V_i(x)$ over the set X'.*

(We will sketch the proof in a bit.)

Note two things about this proposition: first, this ugly condition about $V(x^*)$ being Pareto efficient in the convex hull of $V(X')$. Since x^* is Pareto efficient in X', we know that $V(x^*)$ is Pareto efficient in $V(X')$ itself. So, for example, if we know that $V(X')$ is a convex set, then this added condition is unnecessary. It is a little too much to hope that $V(X')$ will be convex, but still we can eliminate the ugly condition by some quite standard assumptions.

Proposition 5.4. *Suppose X' is a convex set and each $V_i : X \to R$ is concave. Then if x^* is Pareto efficient in X', the utility imputation $V(x^*)$ is Pareto efficient in the convex hull of $V(X')$.*

(The proof is left as a homework exercise.) Consider, for example, the specific application of division of some endowment of commodities among I consumers, where each consumer cares only about how much he consumes. Then X' is convex, and all we need is an assumption that the utility functions of the consumers are concave. We discussed at some length the reasonableness of such an assumption on consumer's utility, and we do not repeat that discussion here. But this assumption is met in many applications.[e]

The second remark to make about proposition 5.3 is that the weights in the sum of utilities are only nonnegative; we don't guarantee that they are strictly positive. If one or more of the weights is zero, then we don't have a social welfare functional as we've defined the term, because the

[6] If you've never seen this notion before, get someone who has to draw a few pictures for you.

[7] Recall that $V(X') = \{v \in R^I : v = V(x) \text{ for some } x \in X'\}$.

[e] What if the ugly condition is not met? Can one still guarantee that every Pareto efficient point is selected for some legitimate social welfare functional, albeit not a Bergsonian social welfare functional? The answer is virtually yes, where one waffles a bit for the same reason that we permit zero weights in proposition 5.3. Untangling this statement is left to the more mathematically inclined reader.

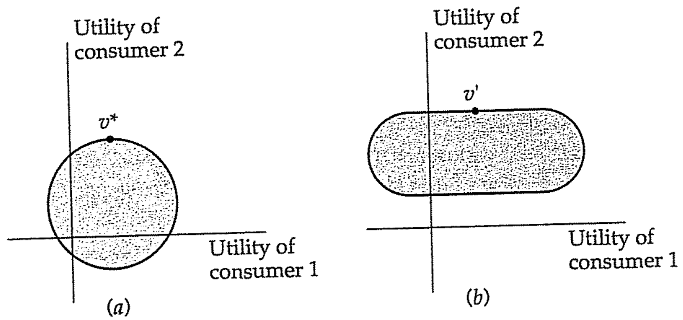

Figure 5.6. Why zero weights are needed in the proof of proposition 5.3.

function is not strictly increasing in the utility of consumers who have zero weight in the sum. Allowing for zero weight is necessary if we are to get all the Pareto efficient points. Consider, in this regard, figure 5.6(a). We have here the set $V(X')$ for a two-person society, and we imagine that $V(X')$ is a circle. Since the slope of the Pareto frontier at the point marked v^* is zero, if we put any (positive) weight on consumer 1's utility in a weighted sum, we will not choose v^* as a maximizing point. But if we allow zero weights, and if $V(X')$ is as in 5.6(b), then points such as v' might be selected as maximizing social outcomes. In terms of our definitions, such points are not strictly Pareto dominated by any other (as long as one of the weights is strictly positive, we could never select a point that is strictly Pareto dominated), but they are not Pareto efficient either. One can either give up guarantees that points such as v^* are obtained as maxima, or allow the possibility that points such as v' are; both these "solutions" to the problem of zero weights can be found in the literature.

Now we give a proof of proposition 5.3. This will be in small type and so is optional material, but all readers are urged to persevere. We are about to introduce the separating hyperplane theorem, a very important tool in economic theory.

We begin by stating the separating hyperplane theorem. Suppose A and B are nonintersecting convex sets in R^2. By drawing pictures, you should be able to convince yourself that it is always possible to draw a line between them, with all the points from A lying on one side of the line and all the points from B lying on the other. If you are really clever in your pictures, you will be able to draw a situation in which points from one or both of the two sets must lie on the line, so we will permit this. The mathematical

result that formalizes your ability always to draw such a line or, in higher dimensions, to draw the appropriate analogue to a line, is

The Separating Hyperplane Theorem. *If A and B are convex sets in R^K that don't intersect, then there is a vector $\alpha = (\alpha_1, \ldots, \alpha_K) \in R^K$, not identically zero, and a scalar β, such that $\alpha \cdot a \leq \beta$ for all $a \in A$ and $\alpha \cdot b \geq \beta$ for all $b \in B$, where $\cdot$ means the dot or inner product.*

With this it is easy to prove proposition 5.3. Take any Pareto efficient point x^* that satisfies the hypothesis of the proposition. Consider the following two convex sets in R^I. Let $A =$ the convex hull of $V(X')$; and let $B = \{v \in R^I : v > V(x^*)\}$. In the definition of B, the strict inequality (which is between two I-dimensional vectors) should be read v is greater or equal to $V(x^*)$ in every component and strictly greater in at least one. Because x^* is Pareto efficient in the convex hull of $V(X')$ and B is the set of utility vectors that Pareto dominate the utility imputation associated with x^*, A and B do not intersect. The set A is convex by definition; it shouldn't take you too long to prove that B is convex. Hence there exists a vector $\alpha = (\alpha_1, \ldots, \alpha_I)$ and a scalar β such that $\alpha \cdot a \leq \beta$ for all $a \in A$ and $\alpha \cdot b \geq \beta$ for all $b \in B$. Since $V(x^*)$ is "almost" in B, $\alpha \cdot V(x^*) = \beta$. (To be precise, use continuity.) Hence $V(x^*)$ maximizes the function $\alpha \cdot v$ over v in A, hence over $v \in V(X')$. And hence x^* maximizes the function $\sum_{i=1}^{I} \alpha_i V_i(x)$ over $x \in X'$. We are done as soon as we show that each $\alpha_i \geq 0$. For this we use our definition of the set B. Suppose that $\alpha_i < 0$ for some i. For each integer N, let

$$v^N = (V_1(x^*), \ldots, V_{i-1}(x^*), V_i(x^*) + N, V_{i+1}(x^*), \ldots, V_I(x^*)).$$

That is, we look at the utility imputation resulting from x^*, and we give consumer i an increase of N in his utility. Then $v^N \in B$ by definition. And if $\alpha_i < 0$, then by making N very large, we can make $\alpha \cdot v^N$ as small as we wish, in particular smaller than β, which would contradict the defining property of α and β.

If you get lost in this mathematical proof, simply consider the picture in figure 5.7. We've drawn the set $V(X')$ here as a finite collection of points, presumably resulting from a finite set X'. We then shade in both the convex hull of $V(X')$ and the set of utility vectors that Pareto dominate the utility imputation $v^* = V(x^*)$ for some $x^* \in X'$. We can separate these two by a line that necessarily passes through $V(x^*)$. Using this line and lines parallel to it to define "iso-social welfare" lines, we see that for some linear social welfare functional, x^* is socially optimal. Since the separating line has to stay outside the interior of the set B, it can't have positive slope. Note also the point v', which is Pareto efficient in $V(X')$ but *not* in the convex hull of $V(X')$. We can't, therefore, obtain v' as the choice of a Bergsonian social welfare functional. Finally, draw the corresponding picture for the point v^* in figure 5.6. This should show you again why we sometimes need zero weights.

Apparently, then, assumptions 1, 2, and 3 don't do a lot to pin down social choice. They do tell us how to restrict social choice to the Pareto

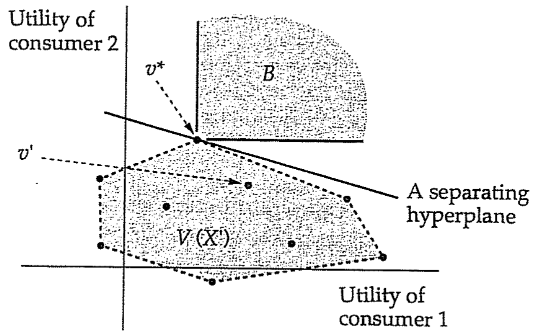

Figure 5.7. Separating feasible utility vectors from Pareto dominating vectors in the proof of proposition 5.4.

frontier, but (following propositions 5.3 and 5.4) in many cases of interest even Bergsonian social welfare functions allow us to get almost the entire Pareto frontier as "socially optimal." So how should we proceed?

We might try to say more about how social choice should be determined in general from arrays of consumer preferences. That is, we might deal not with a single array of consumer preferences but instead with a family of them, discussing how social preferences should change with changes in the array of consumer preferences. This approach is discussed in section 5.5 with the most famous result along these lines, *Arrow's possibility theorem.*

5.4. Characterizing efficient social outcomes

But before doing this, we illustrate how social welfare functionals can be used as a tool of analysis. Their usefulness analytically springs from precisely their weakness in normative choice theory: They can be used to find the entire Pareto frontier as the solution to a parametric optimization problem. That is, suppose we have a social choice problem in which (for simplicity) proposition 5.4 applies. Then we know that every point on the Pareto frontier is the solution to maximizing social welfare using a Bergsonian social welfare functional for some selection of weights. As we vary the weights, we trace the entire Pareto frontier. It might be interesting to see what conditions must hold at an efficient outcome in a particular setting, which thus comes down to the conditions that must hold at solutions of maximizing a weighted sum of individuals' utilities.

To illustrate this technique, we develop three examples that will be used to some extent in the rest of the book.

The production and allocation of private goods

Our first example takes place in the context of production and allocation of commodities. We imagine a situation with K commodities and I individuals. We write Z for the positive orthant in R^K, and we write $x = (z_1, z_2, \ldots, z_I)$ for a social outcome — an allocation of commodities to the individuals. Each z_i is an element of Z, a nonnegative K-dimensional vector, and we write z_{ik} for the amount of good k allocated to individual i.

We restrict attention for now to the case in which these are *private goods,* meaning that the preferences of individual i concerning the social outcome x depend only on i's own bundle of commodities. More precisely, if $V_i : X \to R$ represents i's preferences over X, then V_i can be written as $V_i(z_1, \ldots, z_I) = U_i(z_i)$ for some function $U_i : Z \to R$. We assume that U_i is concave and continuously differentiable; the usual remarks about these assumptions are taken as read. We also assume that none of the commodities are noxious, so that each U_i is nondecreasing.

So far everything is as in section 5.1. But now we add one complication. Instead of supposing that a social endowment bundle is to be distributed, we imagine that this society is endowed at the outset with a supply of some $K+1$ good that no one wishes to consume but which can be transformed into various bundles of the K goods individuals wish to consume.[f] We assume that this society has at its disposal a production or transformation technology that is characterized by a function $\phi : Z \to [0, \infty)$ with the following interpretation: For each bundle $z \in Z$, the amount of the $K+1$ good needed to produce z is $\phi(z)$. We assume that society has at its disposal an amount e_0 of this $K+1$ good, so the social choice problem is to (a) choose a bundle from Z to produce from society's supply e_0 of the $K+1$ good and then (b) divide the bundle produced among the I individuals. That is, the space of feasible social outcomes X' is

$$X' = \{x \in X : x = (z_1, \ldots, z_I),\ \phi(\sum_{i=1}^{I} z_i) \leq e_0\}.$$

Let us be clear: $\sum_{i=1}^{I} z_i$ is the vector sum of the commodity bundles allocated to the individuals; it is a K-dimensional vector. So the constraint reads that it must be feasible to produce from e_0 what is to be allocated.

[f] This $K+1$ good is simply an expository device. Readers who know a bit about modeling technologies will see that we could dispense with it.

We assume that the function ϕ is strictly increasing (it always takes more to make more), continuously differentiable, and quasi-convex. Differentiability of ϕ is assumed for ease of exposition,[g] but quasi-convexity of ϕ is essential for what follows. Quasi-convexity basically says that if z and z' can both be produced from e_0 units of the $K+1$ good, then so can any convex combination of z and z'.[8] The reason for the assumption is so that we obtain: *The set X' defined above is convex, hence proposition 5.4 applies.* We leave this to the reader to prove. We go on to harvest the consequences, which are that every efficient social outcome can be found as the solution of the following maximization problem:

$$\text{Maximize } \sum_{i=1}^{I} \alpha_i U_i(z_i) \text{ subject to } \phi(\sum_{i=1}^{I} z_i) \leq e_0,$$

for nonnegative weights (α_i).

To keep matters simple, we will assume that the weights α_i are all strictly positive, that solutions to the problem above are characterized by the first-order conditions, that at those solutions relevant multipliers and partial derivatives are strictly positive, and that the only constraint that binds is the social feasibility constraint, or $\phi(\sum_{i=1}^{I} z_i) \leq e_0$. The first of these three assumptions can, at the cost of some analysis, be dispensed with. The second is fairly innocuous (and is completely innocuous if the function ϕ is convex). The third takes some work, but for most applications it will hold.[h] But the fourth assumption is quite strong. Essentially we are characterizing efficient allocations where every consumer is allocated a positive amount of each of the K consumption goods. The diligent reader may wish to redo our analysis without this assumption.

The characterization is now easy to derive. Letting λ be the multiplier on the social feasibility constraint, the first-order condition for z_{ik}, the amount of good k allocated to consumer i is

$$\alpha_i \frac{\partial U_i}{\partial z_{ik}} = \lambda \frac{\partial \phi}{\partial z_k},$$

where partial derivatives are, of course, evaluated at the optimal solution. We should be careful about one thing in this first-order condition. When

[g] The mathematically well-versed reader can try to do without it.

[8] For more on this assumption, which is essentially an assumption that the production technology is convex, see chapter 7.

[h] It involves an implicit assumption that each consumer's utility rises in increases in each good, at least for some levels of the good's consumption.

reading $\partial\phi/\partial z_k$ on the right-hand side, remember that ϕ is a function of K variables; we mean here the partial derivative of ϕ taken with respect to the total social production of commodity k.

Take any two goods k and k', and take the two first-order conditions for z_{ik} and $z_{ik'}$. With all the assumptions we have made, we can divide one by the other, cancel the α_is on one side and the λs on the other, and obtain

$$\frac{\partial U_i}{\partial z_{ik}} \Big/ \frac{\partial U_i}{\partial z_{ik'}} = \frac{\partial \phi}{\partial z_k} \Big/ \frac{\partial \phi}{\partial z_{k'}}.$$

The ratio on the left-hand side is called the *marginal rate of substitution for individual i of good k for k'*, whereas the ratio on the right-hand side is called the *marginal rate of technical substitution of k for k'*, and we have the result; at an efficient social outcome these ratios should be equal for each consumer for each pair of goods. Moreover, since the right-hand side is independent of the consumer i, we see that the marginal rates of substitution of each pair of goods should be equal for each pair of consumers.

These characterizations of efficient social production and allocation are easy enough to explain. First, suppose that for two consumers, i and i', their marginal rates of substitution of k for k' were different; suppose the marginal rate for i was larger. Then we could take a bit of k' away from i, compensating her with just enough k so she is indifferent, and give the k' to i' in return for the k we gave to i. The ratio of k to k' that makes i indifferent will, because of the presumed difference in the ratios, strictly benefit i'. Hence we didn't have a socially efficient allocation. Next, suppose that for consumer i the marginal rate of substitution of k for k' is not the same as the marginal rate of technological substitution. Suppose her ratio is larger. Then we can make a bit less k' and a bit more k in a ratio that is technologically feasible and give the extra k to i in exchange for the k' we need to make the extra. Given our assumptions on her rate of marginal substitution relative to the rate of technological substitution, this makes her strictly better off, and we were not efficient in terms of our choice of production plan.[i]

[i] Note that it has to be possible to take k from the second consumer and k' from the first to make these variations. This is where our assumption that nonnegativity constraints don't bind come into play, and it is where we would have to modify these simple characterizations if nonnegativity constraints did bind.

The production and allocation of private and public goods

If you have had a course in intermediate microeconomics, you probably already knew all that, at least at some level. But now we consider a variation that you may not have seen before. We suppose that of our K commodities, the first is a *public good.* A public good is one where the consumption of the good by one individual in no way prevents others from consuming the good or diminishes their enjoyment of it. Classical examples of public goods include clean air and national defence. Parks and highways have some of the character of a public good, but they don't quite qualify — the more people consume Yellowstone Park or the Pennsylvania Turnpike, the less utility each takes from it.[j]

We formalize this situation as follows. Now a social outcome is a $1 + I(K - 1)$ vector,

$$x = (y, (z_{12}, \ldots, z_{1K}), \ldots, (z_{I2}, \ldots, z_{IK})),$$

where y is the amount of the public good, $z_1 = (z_{12}, \ldots, z_{1K})$ is the amount of the private goods consumed by individual 1, and so on. The preferences of individual i over social states are given by a utility function U_i defined on the space of vectors $(y, z_1) = (y, (z_{12}, \ldots, z_{1K}))$. Otherwise, the formulation of the technological possibilities for this society remains the same, so now the space of feasible social outcomes is

$$X' = \{x = (y, z_1, \ldots, z_I) \in R^{1+I(K-1)} : \phi(y, \sum_{i=1}^{I} z_i) \leq e_0\}.$$

Convexity of X' is unaffected, and if we make assumptions similar to those made in the previous subsection, to characterize efficient social outcomes in this setting we have to solve the problem

$$\text{maximize } \sum_{i=1}^{I} \alpha_i U_i(y, z_i) \text{ subject to } \phi(y, \sum_{i=1}^{I} z_i) \leq e_0.$$

[j] Public goods are extreme examples of goods with externalities, which we discuss in the next chapter. A useful exercise, once you've read the discussion given in the next chapter on externalities, is to characterize the efficient allocation of goods in the style of this section in cases with general externalities.

First-order equations on the z_{ik} for $i = 1, \ldots, I$ and $k = 2, \ldots, K$ are just as before, but for the public good we get the first-order condition

$$\sum_{i=1}^{I} \alpha_i \frac{\partial U_i}{\partial y} = \lambda \frac{\partial \phi}{\partial y}.$$

Now consider any one of the private goods, say good #2. By manipulating the first-order condition for z_{i2}, we obtain

$$\alpha_i = \lambda \frac{\partial \phi}{\partial z_2} \Big/ \frac{\partial U_i}{\partial z_{i2}}.$$

Substituting for α_i in the first-order condition for y gives

$$\sum_{i=1}^{I} \left[\lambda \frac{\partial \phi}{\partial z_2} \Big/ \frac{\partial U_i}{\partial z_{i2}} \right] \frac{\partial U_i}{\partial y} = \lambda \frac{\partial \phi}{\partial y},$$

and dividing through by $\lambda(\partial \phi / \partial z_2)$ on both sides, we get

$$\sum_{i=1}^{I} \left[\frac{\partial U_i}{\partial y} \Big/ \frac{\partial U_i}{\partial z_{i2}} \right] = \frac{\partial \phi}{\partial y} \Big/ \frac{\partial \phi}{\partial z_2}.$$

Or, in words, the marginal rate of technological substitution of the public good for any of the private goods (the right-hand side) should equal the *sum* of the marginal rates of substitutions of the public good for the private good summed over consumers.

Syndicate theory

Our third illustration of this technique, *syndicate theory* (Wilson, 1968), comes from the economics of uncertainty. There are I consumers who jointly own a gamble that pays off different amounts of money in N different states of nature. The N states of nature are written $\{s_1, \ldots, s_N\}$, and in state s_n the gamble pays off $\$Y_n$. A "social outcome" consists of a sharing rule for dividing the payoff of the gamble in each state. We write y_{in} for the amount of money received by consumer i in state s_n, so the set of all feasible social outcomes is the set of all vectors $(y_{in})_{i=1,\ldots,I;n=1,\ldots,N} \in R^{IN}$ such that for each n, $\sum_{i=1}^{I} y_{in} = Y_n$. The last constraint is the "adding up" constraint; the consumers can't divide among themselves more than they have to divide. (Thus, in the notation of the general formulation, all vectors (y_{in}) that meet this constraint make up the set X' of feasible social outcomes.) Note that we don't restrict the y_{in} to be nonnegative; we will permit consumers in some states to *contribute* funds.

Each consumer (or member of the syndicate) cares only about what he receives. Each evaluates his part of any division $(y_{in})_{n=1,\ldots,N}$ using an expected utility calculation with probabilities $\{\pi_n\}$ and with utility function u_i. That is, the utility to i of the sharing rule (y_{in}) is given by

$$\sum_{n=1}^{N} \pi_n u_i(y_{in}).$$

Note carefully: We are assuming that the different consumers all have the same probability distribution over the N states. You may wish to consider how what follows will change if we imagine that different consumers have different subjective assessments concerning the likelihoods of the various states. We are also assuming that the individual consumers evaluate their shares in this gamble without regard to any other risky assets they might hold; this (presumably) makes sense only if this risk is independent of any other risks they might face. We assume that the consumers are all risk averse (including risk neutrality as a limit case); that is, the functions u_i are all concave. And for technical reasons we assume that the functions u_i are all continuously differentiable.

Now we pose the question in which we are interested. *Which sharing rules are Pareto efficient?*

The set of feasible sharing rules is clearly convex. And the utility functions of the individuals viewed as functions on the sharing rules are concave because each of the utility functions u_i is concave.[k] So proposition 5.4 applies; and then according to proposition 5.3, we can find all the efficient division rules by maximizing various weighted averages of the consumers' expected utilities.

That is, if we let $(\alpha_i)_{i=1,\ldots,I}$ be a vector of nonnegative weights, then the solution to

$$\max \sum_{i=1}^{I} \alpha_i \left[\sum_{n=1}^{N} \pi_n u_i(y_{in}) \right] \text{ subject to } \sum_{i=1}^{I} y_{in} = Y_n \text{ for each } n$$

is a Pareto efficient sharing rule, and as we vary the weights α_i, we get all the Pareto efficient sharing rules.

[k] The technically astute reader will see that the concavity of u_i is not quite the same as concavity of the "V_i" functions, but it quickly implies concavity of the "V_i." If this is gibberish, don't worry about it.

What happens if some of the weights are zero? Suppose $\alpha_i = 0$. Then we put no weight on the expected utility of consumer i. And since we haven't restricted shares to be nonnegative, the "solution" to the maximization problem just given is to make y_{in} extremely large negative, distributing this "tax" on i among those consumers that have positive weight. More precisely, the maximization problem given has no solution at all.[l] So we can restrict attention to strictly positive weights.[m]

Now let us rewrite the objective function in the maximization problem by interchanging the order of summations. That is,

$$\sum_{i=1}^{I} \alpha_i \left[\sum_{n=1}^{N} \pi_n u_i(y_{in}) \right] = \sum_{n=1}^{N} \pi_n \left[\sum_{i=1}^{I} \alpha_i u_i(y_{in}) \right],$$

so we can rewrite the problem above as

$$\max \sum_{n=1}^{N} \pi_n \left[\sum_{i=1}^{I} \alpha_i u_i(y_{in}) \right] \text{ subject to } \sum_{i=1}^{I} y_{in} = Y_n \text{ for each } n.$$

If you stare hard at this reformulation of the problem, you will see that we solve it by looking at each state separately; if we solve separately, for each state n,

$$\max \sum_{i=1}^{I} \alpha_i u_i(y_{in}) \text{ subject to } \sum_{i=1}^{I} y_{in} = Y_n,$$

then we have the solution to the grand problem. Let us record this and other "facts" we encounter as we go along:

Fact 1. *Efficient sharing among members of the syndicate is done on a state-by-state basis. The probabilities assessed for the various states play no role in this sharing. All that matters to the sharing rule in a particular state is the total*

[l] If u_i is not defined for negative arguments, then trying to figure out what the maximization problem means is an interesting mathematical puzzle. We'll avoid this puzzle by restricting attention to strictly positive weights and by assuming that the problem given has a solution constrained only by the adding up constraint within the domain of definition of the u_i.

[m] Even if the weights are all strictly positive, we cannot guarantee that the maximization problem has a solution. We will proceed assuming that there is a solution, and then, since we have not constrained the y_{in}, this solution is given by simultaneous solution of the first-order conditions. But this assumption is unwarranted for some parameterizations, and we leave it to the careful reader to put in place the necessary caveats.

payoff in that state (and the weights given to the consumers, which are held fixed in any efficient sharing rule). That is, in an efficient sharing rule, one doesn't reward Peter in one state and compensate Paul in another.[n]

So how do we share in state s_n? Letting μ_n be the multiplier on the adding up constraint for state s_n, we get as first-order conditions

$$\alpha_i u_i'(y_{in}) = \mu_n \text{ for } i = 1, \ldots, I.$$

Assume for the time being that each u_i is strictly concave (so that u_i' is a strictly decreasing function). Then for each fixed value of the multiplier μ_n, the equation $\alpha_i u_i'(y) = \mu_n$ has a unique solution in y.[9 o] Let us write that solution as a function $y_i(\mu_n)$ of the multiplier μ_n. Because u_i' is strictly decreasing, the value of $y_i(\mu_n)$ is strictly decreasing in μ_n. Because we assumed that u_i is continuously differentiable, the solution is continuous in μ_n. Hence the function $\sum_{i=1}^I y_i(\mu_n)$ is strictly decreasing and continuous in μ_n. At the value of μ_n for which the sum equals Y_n, we have the unique solution to the first-order conditions for state s_n.

Fact 2. *If all the consumers are strictly risk averse (that is, if all the u_i are strictly concave), the first-order conditions have a unique solution for each state of nature. There is a unique efficient sharing rule* (for the fixed weights).

Consider next how the shares of consumer i vary across various states. Consider two states s_n and $s_{n'}$ such that $Y_n > Y_{n'}$. The solution for state n will be where $\sum_{i=1}^I y_i(\mu_n) = Y_n$. The solution for state n' will be where $\sum_{i=1}^I y_i(\mu_{n'}) = Y_{n'}$. Since the functions $y_i(\cdot)$ are strictly decreasing, $\mu_n < \mu_{n'}$. Hence the share for each consumer i in state n, $y_{in} = y_i(\mu_n)$, will be strictly larger than his share in state n'.

Fact 3. *If all the consumers are strictly risk averse, then the share of any consumer in a state is strictly increasing in the overall payoff in that state.*

So far we've considered efficient division when all the consumers are strictly risk averse. Now let us consider what happens when one of the consumers is risk neutral and the others are (strictly) risk averse. For our

[n] The reader who is thinking about how this would change if the consumers assessed different probabilities for the different states should think hard about this first fact!

[9] The weights α_i are held fixed throughout this discussion. As they shift, we shift the point on the Pareto frontier that we are investigating, a comparative statics exercise that is left for the reader.

[o] We might get $+\infty$ or $-\infty$ as the solution, and the careful reader can worry about what happens then.

risk neutral consumer, say consumer number 1, we have $u_1' \equiv c$ for some constant c. Hence the first-order condition for this consumer in state s_n is $\alpha_1 c = \mu_n$. But the left-hand side of this first-order condition doesn't depend on n. So we conclude that the multipliers μ_n for the various states n must all be the same; they must all be $\alpha_1 c$. Thus the share y_{in} of consumer $i = 2, \ldots, I$ in state n must be the (unique) solution of $\alpha_i u_i'(y) = \alpha_1 c$. Since we have assumed that all the consumers other than number 1 are strictly risk averse, the equation for i has a unique solution. Thus the shares of consumers other than number 1 are constant; consumer 1 bears all the risk.

Fact 4. *If one consumer is risk neutral and the others are strictly risk averse, then the consumers who are risk averse receive, in any efficient division, a payoff that is constant among the various states. The risk neutral consumer bears all the risk; his payoff in state s_n is Y_n plus or minus some constant (that doesn't depend on n).*

This fact will become important in chapter 16.

Now, just to show how far one could take this sort of thing, we move to a very specific parameterization. We assume that each consumer is strictly risk averse and has constant absolute risk aversion, or $u_i(y) = -e^{-\lambda_i y}$ for some $\lambda_i > 0$. Recall that λ_i is called consumer i's coefficient of (absolute) risk aversion. The reciprocal of λ_i, which we write τ_i, is conventionally called i's *coefficient of risk tolerance*; it will be more convenient to work with risk tolerances than with risk aversions, so we rewrite $u_i(y) = -e^{-y/\tau_i}$. It is also conventional to refer to $\sum_{i=1}^I \tau_i$ as *society's risk tolerance;* we will use this term and the symbol T to represent this sum. We have the following result:

Fact 5. *If all the consumers have constant absolute risk tolerance, then the efficient divisions take the form*

$$y_{in} = \frac{\tau_i}{T} Y_n + k_i, \; i = 1, \ldots, I$$

where the constant k_i is a constant that depends on all the parameters of the maximization problem, including the weights (α_i).

We leave this for you to prove in problem 6; some hints are given there. But note how remarkable this is. In this very special case, in any efficient sharing rule consumer i receives a proportional share of the total pie plus a constant payment (which may well be a tax). The size of his proportional share is given by the ratio of his risk tolerance to society's risk tolerance.

We shift around on the Pareto frontier solely by adjusting the constant payments; efficient sharing always involves the same "risky parts" in the efficient sharing rule.

5.5. Social choice rules and Arrow's possibility theorem

Now we return to the problem of normative social choice. In section 5.3, we looked at a single array of consumer preferences, given by an array of numerical rankings $(V_i)_{i=1,\ldots,I}$ on the set of social outcomes X, and we attempted to characterize "reasonable" social preferences. We might get further along in the general program of normative social choice if we expand the scope of our analysis. Specifically, we might consider a *family* of possible arrays of consumer preferences and characterize reasonable social preferences for each array in the family. The point is that this expanded scope allows us to pose conditions concerning how social preferences respond to changes in the array of consumer preferences.

For example, we might pose the following property:

Positive Social Responsiveness. *Suppose that for a given array of consumer preferences the resulting social preferences rank x as better than x'. Suppose that the array of consumer preferences changes so that x falls in no one's ranking and x' rises in no one's ranking. Then the social ranking should continue to rank x better than x'.*

We aren't being very precise in this definition because we haven't set a context for the problem. But the reader can probably glean from this the spirit of what we will try to do.

We will use the following terminology. A map or function from some given family of arrays of consumer preferences to resulting social preferences will be called a *social choice rule.*

A setting and three examples

We begin by looking at this general program in the following setting. Assume we are given a finite set X of social outcomes, a finite collection of consumers indexed by $i = 1, \ldots, I$, and for each consumer, an asymmetric and negatively transitive strict preference relation $\succ_i$ on X. From $\succ_i$ we define weak preference $\succeq_i$ and indifference $\sim_i$ in the usual fashion. We will work directly with the preference relations $\succ_i$; specific numerical representations for these preferences will play no role for the time being. In this setting, an array of consumer preferences is then an I-dimensional vector of preference orders, $(\succ_1, \ldots, \succ_I)$, where each preference order is

asymmetric and negatively transitive. One might put restrictions on the possible arrays of consumer preferences under consideration; we, however, will assume that any vector of asymmetric and negatively transitive orderings of X is permitted.

A *social choice rule* takes an array $(\succ_1, \ldots, \succ_I)$ as input and produces social preferences as output; we assume that the output is in the form of a binary relation $\succ^*$.[p]

Consider the following examples of social choice rules in this setting.

Example 1. The Pareto rule. *In the Pareto rule, $x \succ^* x'$ if and only if x Pareto dominates x'. (Or we could insist on x strictly Pareto dominating x'.)*

The Pareto rule produces social preferences that are not negatively transitive. That is, we might find two outcomes x and x' where x Pareto dominates x', hence $x \succ^* x'$, yet both x and x' are Pareto incomparable with a third x'', hence neither $x'' \succ^* x'$ nor $x \succ^* x''$. You can easily construct specific examples from our discussion of negative transitivity in chapter 2.

Example 2. Majority rule and variations. *For each pair of social outcomes x and x', let $P(x, x')$ be the number of individuals i who express $x \succ_i x'$. In **simple majority rule**, we define $x \succ^* x'$ if $P(x, x') > I/2$. Variations include plurality rule, or $x \succ^* x'$ if $P(x, x') > P(x', x)$; and α-majority rule, for $\alpha > 1/2$, in which $x \succ^* x'$ if $P(x, x') > \alpha I$.*

We could also have α-plurality rules, and so on. And we could have weighted majority rules, where different members of society have different numbers of votes they are allowed to cast.

The problem with majority rule is that preference cycles are possible. Specifically, imagine that we have three outcomes in X, call them x, y, and z, and we have three consumers. The first consumer ranks $x \succ_1 y \succ_1 z$. The second ranks $y \succ_2 z \succ_2 x$. The third ranks $z \succ_3 x \succ_3 y$. Then $x \succ^* y$ by majority rule. (Consumers 1 and 3 feel this way.) And $y \succ^* z$ by majority rule (according to consumers 1 and 2). And $z \succ^* x$ by majority rule (according to consumers 2 and 3). This isn't so good, especially when it comes to choosing from $\{x, y, z\}$.[q]

[p] If we were being formal, we would write $\succ^*$ as a function of $(\succ_1, \ldots, \succ_I)$. We will not indulge in such formal notation; I hope no confusion results in consequence. We could instead think of the output of the social choice rule coming in some other form — say a choice function that doesn't necessarily derive from any binary preference relation. On this point, see the next footnote.

[q] If we allowed the social choice rule to prescribe for an array of consumer preferences a choice rule instead of binary preferences, we might define it as follows: For any pair

*Example 3. **The Borda rule.** Suppose there are N social outcomes. For each consumer i construct a numerical ranking for the N outcomes from $\succ_i$ as follows: If there is some single best social outcome x according to $\succ_i$, let $U_i(x) = N$. More generally, suppose i ranks m social outcomes $x_1, x_2, \ldots, x_m$ as being equally good and all better than anything else. Then set $U_i(x_1) = \ldots = U_i(x_m) = [N + (N-1) + \ldots + (N-m+1)]/m$. In general, the nth best social outcome is given numerical ranking n, where ties average their rankings. Then define $U^*(x) = \sum_{i=1}^{I} U_i(x)$, and define $\succ^*$ to be in accord with U^*.*

Readers who follow American college sports (and, for all I know, others) will know something like this system in the rankings of college football and basketball teams. Sportswriters or coaches rank the top twenty teams, giving 20 points to the best, 19 to the second best, and so on, and then these points are summed to get the "consensus" rankings. (It isn't clear that the rankers are allowed to express ties, and to be the classic Borda rule, the rankers would have to rank all the teams.)

What's wrong with the Borda rule? We leave this for you to discover for homework.

Four properties and Arrow's theorem

Consider the following four properties for a social choice rule in the setting of the previous subsection.

***Property 1.** The social choice rule should prescribe an asymmetric and negatively transitive ordering $\succ^*$ over social outcomes X for every array of individual preferences $(\succ_i)$.*

There are two parts to this property. First, the output of the social choice rule should itself be an asymmetric and negatively transitive order on the social outcomes. As we discussed in section 5.3, this is not entirely non-controversial. Second, the social choice rule is meant to operate no matter what profile of individual preferences is specified. This is sometimes called the *universal domain* property; the domain of definition of the social choice rule should be all conceivable arrays of individual preferences.

$x, y \in X$, let $m(x,y)$ be the number of consumers who strictly prefer y to x. Then given a set X' from which choice must be made, for each $x \in X$, let $M(x, X') = \max_{y \in X'} m(x,y)$. The choice out of X' is defined to be those $x \in X'$ that minimize $M(x, X')$. In words, we choose outcomes against which there is the least "opposition" in any pairwise comparison. For an analysis that begins in roughly this fashion (and for references to antecedents), see Caplin and Nalebuff (1988).

***Property 2. Pareto efficiency.** If $x \succ_i x'$ for all i, then $x \succ^* x'$.*

This is something like assumption 2 from section 5.3, except that we require that x strictly Pareto dominate x' in order to conclude that $x \succ^* x'$.

Properties 1 and 2 concern how the social choice rule works at a given array of consumer preferences. Properties 3 and 4, on the other hand, concern the workings of the rule as we vary the array of consumer preferences.

***Property 3. Independence of irrelevant alternatives.** Take any two x and x' from X. The social ranking of x and x' does not change with changes in the array of consumer rankings of other social outcomes. That is, if $(\succ_i)$ and $(\succ_i')$ are arrays of consumer rankings that satisfy $x \succ_i x'$ if and only if $x \succ_i' x'$ for all i, then the social ranking of x and x' is the same in these two situations.*

This property is meant to capture the notion that all that ought to matter in our social choice rule is the relative rankings of pairs of alternatives: There are no interpersonal comparisons of "intensity" of ranking. To get a sense of this, consider a three-person society with three outcomes. Consider an array of individual preferences where the first two consumers rank $x \succ_i x' \succ_i x''$ and the third ranks $x' \succ_i x \succ_i x''$. In this case, it seems "natural" to expect that $x \succ^* x'$. Now consider an array of individual preferences in which the first two consumers continue to rank $x \succ_i x' \succ_i x''$ and the third ranks $x' \succ_i x'' \succ_i x$. Property 3 would guarantee that society should still prefer x and x'. We argued at the start of section 5.3 this is no longer so obvious *if* we think that since x is consumer 3's worst alternative in the second array we ought to give more consideration to x' as the social choice. But how are we to judge how bad 3's worst outcome is for him (instead of his best outcome) relative to how good 1's and 2's first best outcomes are for them (instead of their second bests)? If we actively forebear from making such comparisons, then property 3 seems sensible.

***Property 4. Absence of a dictator.** No single consumer i is a dictator in the sense that $x \succ_i x'$ implies $x \succ^* x'$ no matter what the preferences of the other consumers are.*

This seems sensible as long as there are more than two consumers; suppose that one consumer prefers x to x' and all the others prefer x' to x. We might be uncomfortable if society then chose x over x'. We'd be even more uncomfortable if society did this for any pair x and x', for a single consumer. Note carefully that someone is a dictator even if he doesn't impose his will in matters of indifference. That is, we allow $x \sim_i x'$ and $x \succ^* x'$ without disqualifying i as a dictator. (See problem 9.)

And now for the punchline:

Arrow's Possibility Theorem. *Suppose there are at least three social outcomes. Then no social choice rule simultaneously satisfies properties 1, 2, 3, and 4.*

The theorem stated is Arrow's theorem 2 (1963, 97). There are in the literature many variations on this basic result, in which the four properties are changed and/or weakened in various ways, sometimes at the cost of changing the result slightly. (A good example of the latter sort of treatment is Wilson [1972] who eliminates the Pareto property, but in consequence replaces the absence of a dictator with something a bit weaker.) In all these variations, the same basic result appears; no social choice rule satisfies four properties of roughly the character of the four properties we've given.

So to keep alive the possibility of a "reasonable" social choice rule, some one (or more) of the four properties will have to go. Doing away with the first part of property 1 removes the whole point of the exercise, unless we find some reasonable way to weaken this part of property 1 without losing its entire content. Getting rid of Pareto efficiency doesn't seem appealing (and doesn't really solve anything; see Wilson [1972]). So we must consider whether we wish to give up having a social choice rule that gives well-behaved social preferences or give up on the universal domain part of property 1, or property 3, or property 4. Since dictators are somewhat out of fashion, the universal domain part of property 1 and property 3 get the most attention in the literature. The references listed in section 5.5 will lead you into this rich literature.

We proceed to sketch the proof of the version of Arrow's theorem given above. (We roughly follow Arrow [1963, 97ff].) Assume we have a social choice rule that satisfies properties 1, 2, and 3. (We will demonstrate that there is a dictator.)

Step 1. For any pair of outcomes x and y, all that matters to the social ranking of x and y for a given array of consumer preferences are the sets of individuals such that $x \succ_i y$, $x \sim_i y$, and $y \succ_i x$. That is, for any two arrays of consumer preferences that have the same partition of I into these three sets, x and y will be ranked similarly socially. This is simply a restatement of the independence of irrelevant alternatives.

Step 2. For any distinct pair of outcomes x and y, we say that a subset $J \subseteq I$ is *decisive* for x over y if, when $x \succ_i y$ for all $i \in J$ and $y \succ_i x$ for all $i \notin J$, $x \succ^* y$. We assert that there exists some pair of outcomes x and y and some single individual i such that $\{i\}$ is decisive for x over y.

The proof of this assertion runs as follows. Look over all pairs of distinct outcomes x and y and all the sets of consumers that are decisive for x over y. Note that by Pareto efficiency I is decisive for every x over every y, so we are looking over a nonempty collection of pairs and sets. Since I is a finite set, some smallest (measured by number of members) set of consumers is decisive for some pair x and y. (We are taking the smallest decisive set, ranging over all pairs and all decisive sets at once.) Let J be this smallest set. Suppose that J has more than one element. Partition J into any two nontrivial subsets, called J' and J''. Pick any third element z. (There is a third element by assumption.) Consider any array of social preferences where x, y, and z are ranked as follows:

For $i \in J'$, $z \succ_i x \succ_i y$
For $i \in J''$, $x \succ_i y \succ_i z$
For $i \notin J$, $y \succ_i z \succ_i x$

Since J is decisive for x over y, at this profile $x \succ^* y$. Since social preference is negatively transitive, either $x \succ^* z$ or $z \succ^* y$ (or both). But if $x \succ^* z$, then J'' is decisive for x over z. And if $z \succ^* y$, J' is decisive for z over x. In either case, we have a pair of outcomes and a decisive set for those outcomes that is strictly smaller than the smallest decisive set, a contradiction. Hence the smallest decisive set must contain a single element.

Step 3. Let i be decisive for some pair x and y. Then i is a dictator.

To show this involves looking at numerous cases.

(1) We will show that for any array of consumer preferences where $x \succ_i z$, it follows that $x \succ^* z$. First consider the case where $z \neq y$. For any array where $x \succ_i z$ rearrange (using universal domain) the relative rankings of y as follows: For i, rank $x \succ_i y \succ_i z$, and for everyone else, rank $y \succ_j x$ and $y \succ_j z$. Since $y \succ_j z$ for all $j \in I$, $y \succ^* z$ by Pareto efficiency. Since i is decisive for x over y, $x \succ^* y$. So by transitivity of $\succ^*$, $x \succ^* z$. This leaves the case $z = y$. But in this case, take any third element w. We just showed that i is decisive for x over w (and more), so repeating the argument above with y replacing z and w replacing y covers $z = y$.

(2) We claim that if $z \succ_i y$ for any z in some array of consumer preferences, then $z \succ^* y$. If $z \neq x$, construct a preference array that preserves the original rankings of z and y and that has $z \succ_i x \succ_i y$ and, for $j \neq i$, $z, y \succ_j x$. By Pareto efficiency, $z \succ^* x$. Because i is decisive for x over y, $x \succ^* y$. Hence $z \succ^* y$. If $z = x$, take any third element w. What was just shown establishes that i is decisive for w over y, hence reapplication of the argument leads to the conclusion that $x \succ^* y$.

(3) We claim that if $z \succ_i x$ for any z in some array of consumer preferences, then $z \succ^* x$. To see this, for any third outcome $w \neq x, y$ set up an array of consumer preferences in which $w \succ_i y \succ_i x$ and, for $j \neq i$ $y \succ_j x \succ_j w$. By Pareto efficiency, $y \succ^* x$. By (2), $w \succ^* y$. Hence i is decisive for w over x and applying (2) again (with x in place of y) leads to the desired conclusion.

(4) Now suppose we have any two outcomes w and z such that $w \succ_i z$. If $w = x$, we conclude from (1) that $w \succ^* z$. If $z = x$, we conclude from (3)

that $w \succ^* z$. While if neither w nor z is x, we can without changing the relative rankings of w and z, arrange matters so that $w \succ_i x \succ_i z$ for i. By (1), $x \succ^* z$. By (3), $w \succ^* x$. So by transitivity $w \succ^* z$.

This finishes step 3 and completes the proof.

Social choice rules with interpersonal comparisons of intensity of preference

One way to try to avoid the negative conclusion of Arrow's theorem is to get into the business of making interpersonal comparisons concerning intensity of preference, at least to some degree. One approach to this concerns what are known as *Nash social choice rules*, which are adaptations of the *Nash bargaining solution* from cooperative game theory (Nash, 1953). We will not give anything like a complete treatment of this subject here; this is not mainstream microeconomic theory. But a few remarks on how the problem is set up may give the diligent reader a sense of at least one direction that can be taken. What follows is loosely transcribed from the treatment given in Kaneko and Nakamura (1979).

The setting in this case begins with a finite set X of social outcomes, as before, and a finite set I of consumers. Three special assumptions are made.

We allow feasible social outcomes to include probability distributions over social outcomes out of X. That is, if X' enumerates the feasible social outcomes, we consider choosing a probability distribution with support X'. We use $P(X')$ to denote the set of probability distributions over X'.

Each individual consumer's preferences over $P(X)$ satisfy the von Neumann-Morgenstern axioms, and so are represented by expected utility. We let u_i denote i's von Neumann-Morgenstern utility function on X.

There is a distinguished social outcome, x_0, such that every other social outcome is strictly preferred by every consumer to x_0.

You can think of x_0 as being "everyone dies," or "no one gets anything to eat."

Having von Neumann-Morgenstern preferences and this distinguished social outcome gives us a "scale" on which intensity of preferences can be measured. Suppose one consumer prefers x to y and another y to x, where neither x nor y is x_0. We can think of measuring which of these two likes their preferred outcome "more" by asking: For the first consumer, what number α satisfies $\alpha\delta_x + (1-\alpha)\delta_{x_0} \sim \delta_y$; for the second consumer, what number β satisfies $\beta\delta_y + (1-\beta)\delta_{x_0} \sim \delta_x$? (Here as in chapter 3, δ_x means the lottery that gives prize x with certainty.) If $\alpha > \beta$, then there is a sense in which it matters more to the second consumer to have y instead of x than it does to the first consumer to have x instead of y. In other words, we can use the *cardinal* aspect of von Neumann-Morgenstern preferences and the distinguished reference point x_0 to calibrate intensity of preference, at least in some sense.

The domain for a social choice rule can now be taken to be all arrays of I von Neumann-Morgenstern utility functions $(u_1, \ldots, u_I)$ on X, where we insist that x_0 is worst for everyone in every utility function. And we can then try to pose properties on a social choice rule with this domain that makes use of the stucture provided. The interested reader can consult Kaneko and Nakamura (1979) to see this carried out in at least one fashion. (For the insatiably curious, they give properties that lead to the conclusion: Social choice out of $P(X')$ should be that lottery p that maximizes $\prod_{i=1}^{I}(u_i(p) - u_i(x_0))$, where $u_i(p)$ is shorthand for the expected utility of p computed with respect to u_i.) All we wish to show here is a way in which interpersonal comparison of "intensity of preference" might be incorporated into a theory; you shouldn't conclude that this is impossible.

The rest of the book

We will not explore abstract social choice theory any further. The books and articles listed below will give you a good entrée into this enormous subject. (We will briefly revisit the subject of social choice in chapter 18 under conditions where we mightn't know the preferences of individuals and so must elicit those preferences.) Instead, the remainder of this book takes for the most part a very different tack. Rather than thinking normatively about desirable properties for social choice, we regard "social choice" as the product of individuals interacting in various institutional environments. We describe those institutions and predict the outcome of individual actions within those institutions. Especially in chapters 6 and 8, we will say a bit in the spirit of this chapter about the efficiency of the "social outcomes" that are so selected. But our approach is that of taking the institutions as given exogenously and describing (to the best of our abilities) what "social choices" will be made.

5.6. Bibliographic notes

The classic reference in the theory of social choice is Arrow (1963). Many other excellent books on the subject that follow Arrow; you might in particular wish to consult Sen (1970). To get a taste of more recent developments, a number of survey articles could be consulted; I recommend Sen (1986) and for a shorter treatment Sen (1988). Either of the later pieces by Sen will provide an exhaustive bibliography.

The analysis of syndicate theory, used here to illustrate how to "find" the Pareto frontier in applications, is drawn from Wilson (1968), who draws out many further consequences of the analysis.

References

Arrow, K. 1963. *Social Choice and Individual Values*, 2d ed. New Haven: Cowles Foundation.